Natural Science

Grade 9

Learner's Book

Natural Science Grade 9 Learner's Book

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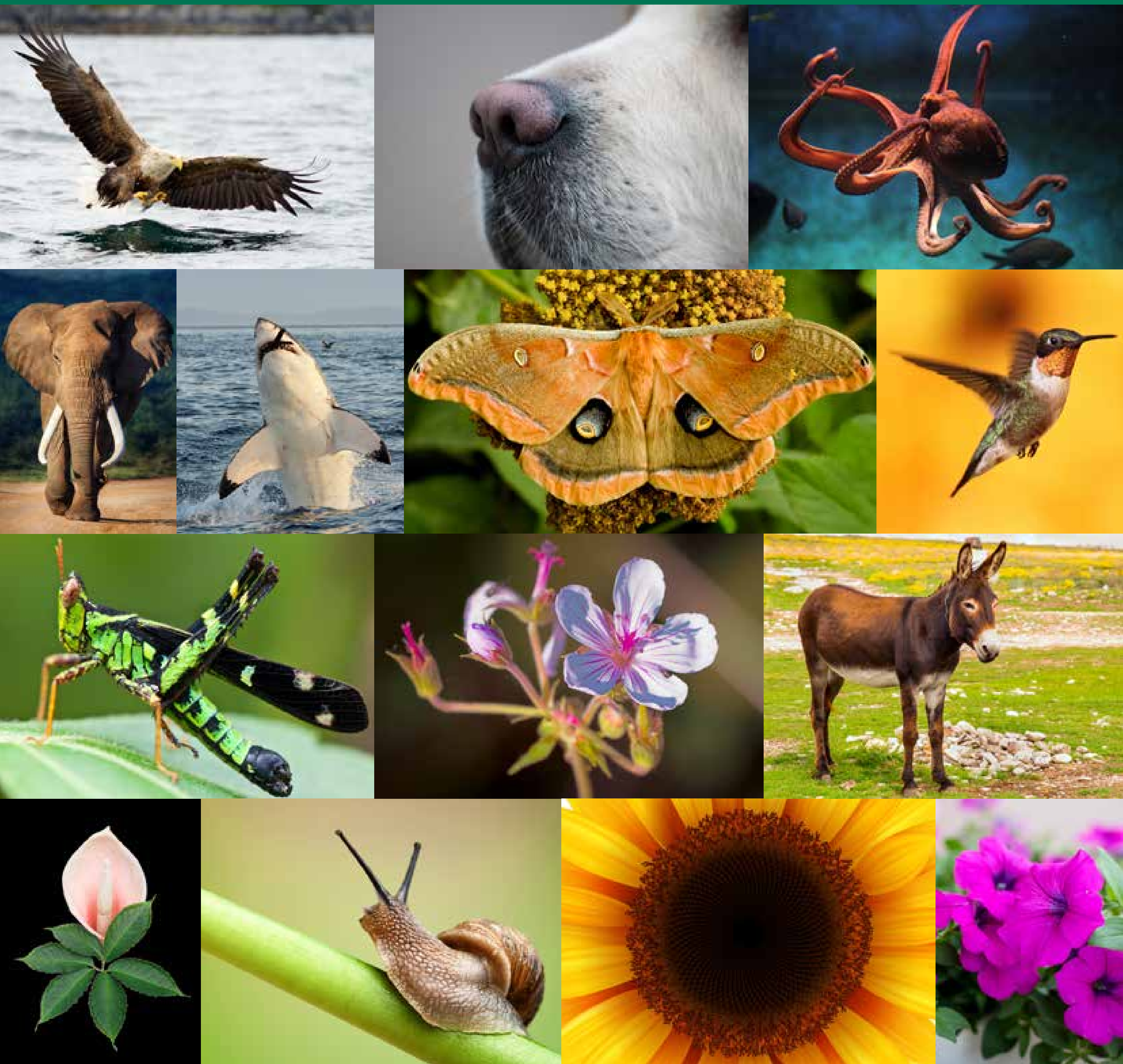
Life and Living



1 Cells as the basic units of life

Key questions

- What are cells?
- Why are cells so small?
- What does it mean to be microscopic?
- Are there different types of cells?
- What is inside a cell and why is it there?
- Are plant and animal cells the same?

In this unit we will learn about the basic units of life which enable all of functions within living organisms – cells.

Did you know?

All the Keywords listed in the boxes in the margin are defined in the Glossary at the end of this strand.

Take note

The Visit boxes in the margins contain links to interesting websites and videos. Simply type the link exactly as it is into the address bar in your browser.

Keywords

- cell
- cytoplasm
- eukaryote
- membrane
- microscope
- microscopic
- nucleus
- organism
- prokaryote

Figure 1.1: Robert Hooke (1635–1703)

1.1 Cell structure

What are cells?

All living organisms, including plants, animals, bacteria and fungi, are made up of cells. Cells are the smallest parts of all living organisms.

There are two main types of organisms based on the structures of their cells. The most important difference in structure is the presence of a nucleus. Cells that contain a nucleus are classified as eukaryotic cells, while those without a nucleus are prokaryotic cells. In this unit we will specifically look at eukaryotic cells that make up organisms such as plants and animals. Examples of organisms with prokaryotic cells are bacteria.

We can say that cells are the basic *structural* and *functional* units of all living organisms. You cannot see most individual cells with the naked eye, because they are too small – you need to use a microscope to be able to see cells. We say cells are microscopic because they can be seen only under a microscope.

Robert Hooke was the first cytologist to identify cells under his microscope in 1665. He decided to call the microscopic shapes that he saw in a slice of cork '*cells*' because the shapes reminded him of the cells (rooms) that the monks in the nearby monastery lived in.

ACTIVITY Brainstorm the Seven Functions of Life

Do you remember the test you learnt about in previous grades to decide whether an organism is living or non-living? Perhaps you had a mnemonic to remember the seven processes, such as 'MRS GREN'.

1. Work in your group and see how many of the seven functions of life you can remember. Write them down in your exercise books.
2. Do you think that an individual cell is living? Explain your answer.

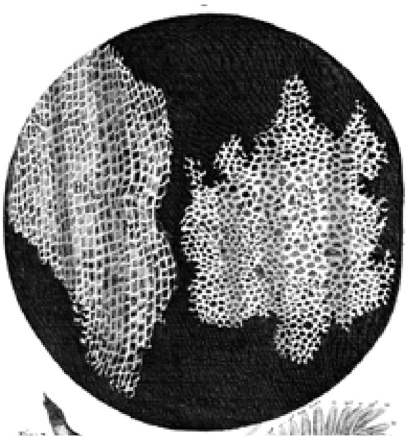


Figure 1.2: Robert Hooke was the first to use the term 'cell' when he studied thin slices of cork with a microscope.

Figure 1.3: Robert Hooke's microscope that he used to view cells for the first time.

Take note

In this section you'll often read the prefix 'cyto'- as in cytoplasm, cytosol or cytoskeleton. Cyto- means 'cell' so if you read 'cytoskeleton' it actually says: 'cell-skeleton'.

Different types of cells

Your body is made up of many different kinds of cells. Cells are **specialised** to perform a specific function. Depending on the function of the cell, it can be specialised by having a different shape or size or may have some components which other cells do not have.

Did you know?

We call the study of cells cytology, 'cyto'- meaning 'cell' and -'logy' meaning 'study', while -'logist' refers to the person who does the studying. The cytoplasm (cytosol and organelles) and the nucleus together are referred to as protoplasm.

Even though there are many different types of cells, there are components of the cell structure which are common to all cells. There are also some structures which most, but not all, cells have. Let us take a look at this in the next section.

Similar components of cells

All cells have some common structures. These are:

- a cell membrane
- cytoplasm; and
- in most eukaryotic cells, a nucleus.

Keywords

- carbon dioxide
- cell membrane
- cellular respiration
- DNA
- hereditary
- inherited
- medium
- mitochondria
- nuclear membrane
- nucleolus
- organelle
- oxygen
- protein
- selectively permeable
- specialised
- species
- vacuoles
- variation

Let's now have a look at the structure of these components of the cell, and some of the other organelles common to cells. An **organelle** is a specialised structure within the cell that performs a function for the cell. Examples of organelles in cells are **vacuoles** and **mitochondria**. Look at the diagram which identifies the different components in a simple animal cell.

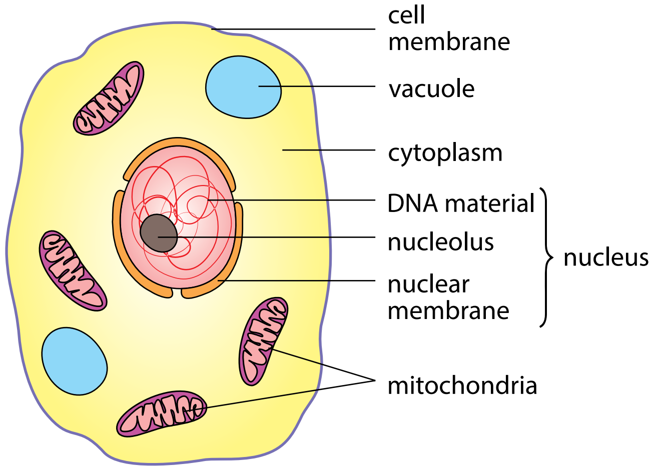


Figure 1.4: A drawing of a typical animal cell

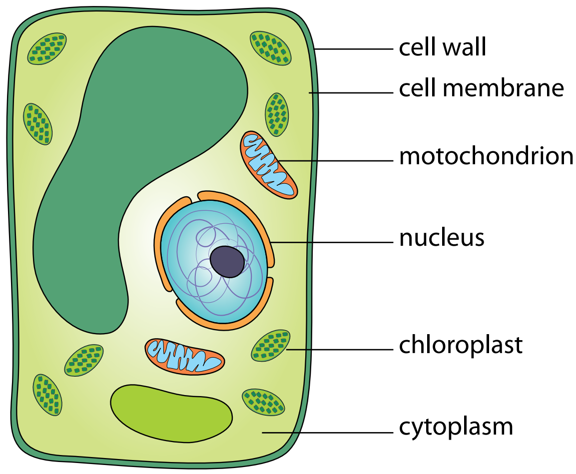


Figure 1.5: A drawing of a typical plant cell

Take note

If something is 'permeable', then it means that substances, such as gases and liquids, can pass through it freely.

Cell membranes

All cells have a cell membrane around them. The cell membrane is a thin layer that encloses the cell's contents and separates the cell from its environment.

Many different substances have to pass in and out of a cell in order for it to function. The cell membrane controls which substances are allowed to enter and leave the cell. The cell membrane is **selectively permeable**. The organelles are also surrounded by membranes.

Cytoplasm

The cytoplasm is made up of a jelly like medium (called cytosol) in which all living parts of the cell are suspended within the cell membrane, but excluding the nucleus. The cytoplasm is made up of the cytosol and the cell organelles. The cytosol is a watery, jelly-like medium made of 70% – 90% water, and is usually colourless. The cytosol suspends organelles and nutrients in the cell, and provides a medium for all the activities in the cell.

The cytosol is a mixture of different soluble substances dissolved in water. These substances within the cytosol include salts, various elements, such as sodium and potassium, and more complex molecules, such as **proteins**.

Did you know?

Identical twins come from the same fertilised egg which splits in two. They have the same DNA. However, they are not exactly the same due to environmental factors that can influence how they develop. Non-identical twins develop from two different eggs and two different sperm.

The cell organelles making up the cytoplasm include mitochondria, chloroplasts and vacuoles. Vacuoles are organelles enclosed by a membrane and contain mostly water with other molecules in solution. The size and number of vacuoles within a cell varies greatly and depends on the type and function of the cell.

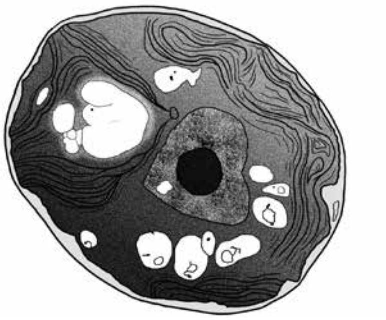


Figure 1.6: This is a micrograph of a plant cell. Can you see the clear, white organelles, which are the vacuoles? The cytoplasm appears very granular in this image.

Nucleus

Plant and animal cells have a **nucleus** inside (but not suspended in) the cytoplasm. It controls all the processes and chemical reactions that take place inside the cell. The nucleus also contains the cell's genetic material which is organised into long **DNA** molecules.

The nucleus is structured as follows:

- A double membrane called the **nuclear membrane** encloses the DNA. This nuclear membrane contains pores (holes) for substances to pass through.
- There is a **nucleolus** inside the nucleus. This is often seen as a darker area within the nucleus.
- The DNA contains information about **inherited** characteristics (**hereditary**), such as whether the person will have blue, brown or green eyes.

Have a look at the micrograph of a nucleus and the diagram of the nucleus.

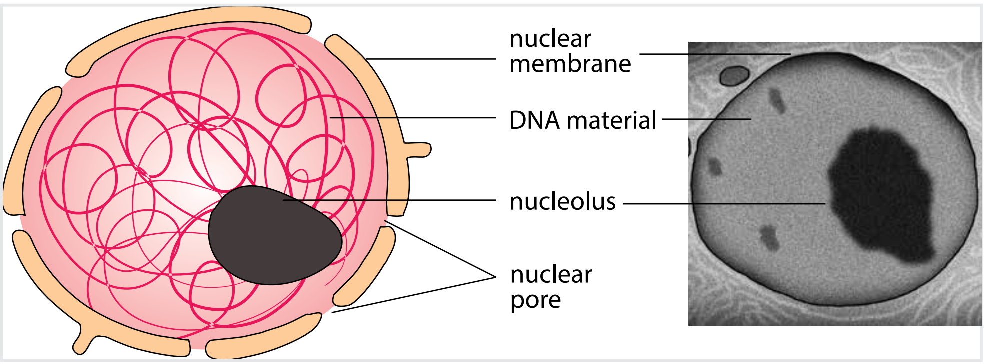


Figure 1.7: Left) A Diagram of a nucleus. Right) A micrograph of a cell nucleus.

DNA is a very important part of all cells and therefore of all life. DNA contains information that encodes all our inherited traits or characteristics. This refers to characteristics which are passed down in families, such as your skin and eye colour, whether you have allergies and also the likelihood of contracting different types of illnesses.

Take note

The difference between eukaryotic and prokaryotic cells is that eukaryotic cells have a nucleus which contains the genetic material surrounded by a membrane. Prokaryotic DNA floats in the cytoplasm without a membrane.

Every organism has unique DNA. The difference in DNA that occurs between individuals is called **variation**. Even the slightest difference in DNA may cause significant variations within **species** and between species. Within species DNA differences or variations can lead to albino animals or the transference of similar illnesses, like sickle cell anaemia.

Figure 1.8: An albino (white) lion lacks pigment due to an alteration in the lion's DNA.

Take note

Singular or plural? Mitochondr*ion* is the singular and mitochondr*ia* is the plural form of the word.

Did you know?

Mitochondria have their own 'mitochondrial DNA' that is completely different from the DNA in the nucleus. What do you think we can we deduce from this fact?

Mitochondria

Mitochondria are organelles enclosed by a double membrane. Cells that are very active would typically have more mitochondria than cells that are less active. The function of the mitochondria is to provide energy for the cells. Which type of cell, do you think, will have more mitochondria: a muscle cell or a bone cell?

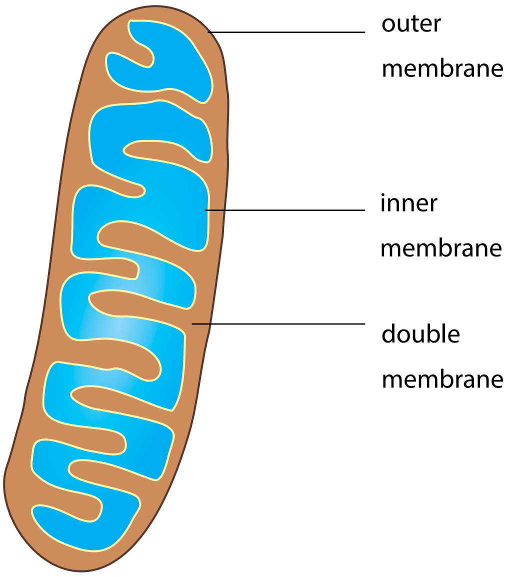


Figure 1.9: Diagram of a mitochondrion

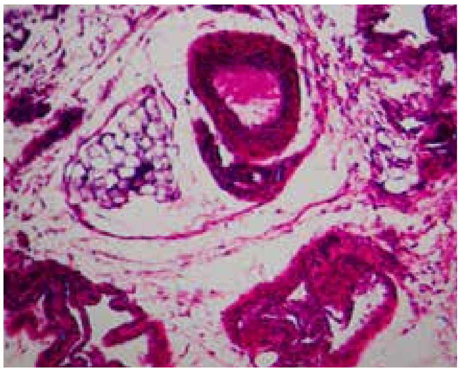


Figure 1.10: A micrograph of muscle tissue in a mouse. Can you see all the darker grey circles? These are mitochondria.

ACTIVITY Summarise what you have learnt

Now that you've studied the internal structure of a cell, let us summarise what we have learnt so far. Copy and complete this table in your exercise book filling in the main function of each of the cell structures.

Cell Structure	Function(s)
Cell membrane	
Cytoplasm	
Nucleus	
Mitochondrion	

1.2 Differentiate between plant and animal cells

Now that we know what the main similarities are between all plant and animal cells, let's see how they are different.

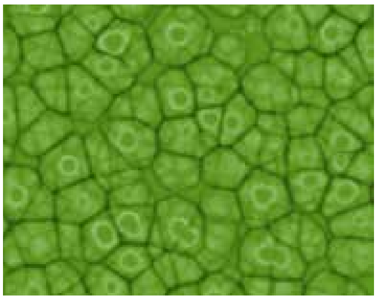
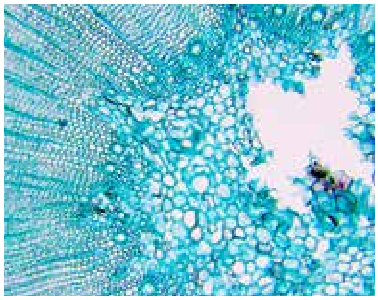
Keywords
- cell wall
- cellulose
- chloroplast
- flaccid
- turgid

Plant cells differ from animal cells

Why do plant and animal cells have differences? Plant and animal cells differ because they have to perform different functions.

ACTIVITY Identify differences between plant and animal cells

1. Study the pictures in the table below. On the left is a picture of plant cells and on the right is a picture of some animal cells, which have been stained blue.
2. Write the differences that you observe between the pictures of the cells in your exercise books.

Plant cells	Animal cells
	

Cell wall

Animal cells often have an irregular shape, whereas plant cells have a much more regular, rigid shape.

Plant cells have an additional layer surrounding the cell on the outside of the cell membrane. This is called the cell wall. The cell wall provides a protective framework for support and stability for the plant cell.

Did you know?

Other organisms also have cell walls, like bacteria or fungi, but in these organisms the cell walls are not the same as plant cell walls. Only plant cells are made of cellulose.

The cell wall is formed from various compounds, the main one being cellulose. Cellulose helps maintain the shape of the plant cell. This allows the plant to remain rigid and upright even if it grows to great heights.

Did you know?

The sea slug, *Elysia chlorotica*, has evolved to take up the chloroplasts from the algae that it eats and incorporate them into its own cells where the chloroplasts function as if in a plant!

Chloroplasts

You might remember learning about photosynthesis in previous grades. Can you still remember why photosynthesis is so important to all life on earth?

Did you notice the green organelles present in plant cells which were not there in the animal cells in the previous activity? These are chloroplasts. Chloroplasts are the only cell organelles that can produce food from the sun's energy. Only plants with chloroplasts are able to photosynthesise because they contain the very important green pigment, chlorophyll. Chlorophyll can absorb radiant energy from the sun and convert this to chemical energy that plants and animals can use.

Vacuoles

Does a plant have a skeleton? Turn to a friend and discuss what could possibly be used in a plant cell as a skeleton. Think for example of a blade of grass, or a long-stemmed rose.

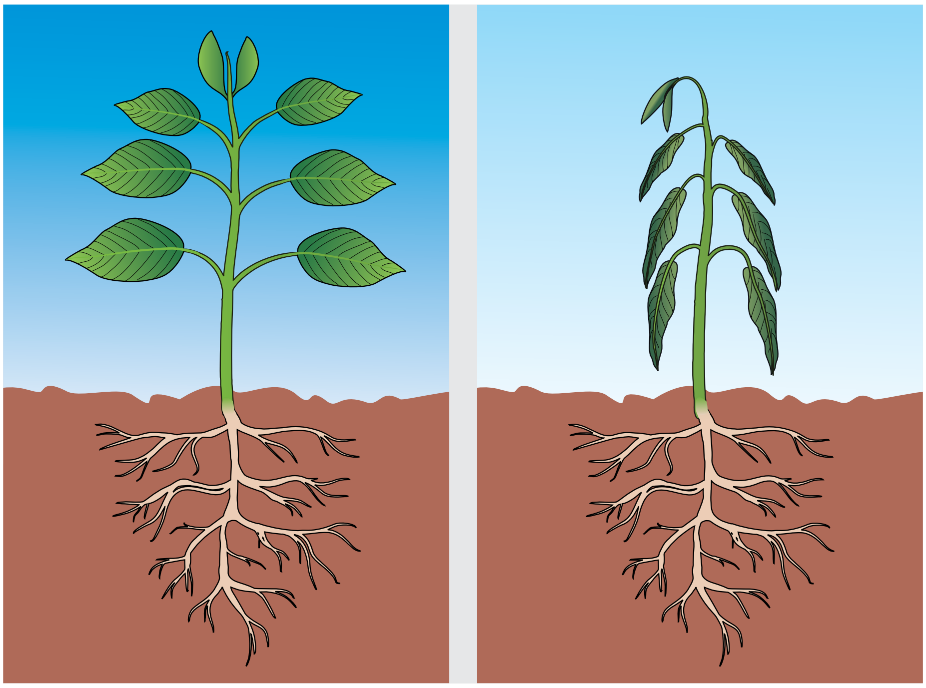


Figure 1.11: Left: A plant with turgid vacuoles is rigid and stands upright. Right: A plant with flaccid vacuoles droops (called wilting).

Vacuoles in plant cells are usually quite large organelles that can occupy as much as 90% of the cell's volume. The liquid in the vacuole, called cell sap, helps to support the plant. The full vacuoles push out against the cell wall and make the cells, and therefore the plant, rigid. We say the cells are turgid in this condition. The opposite to turgid is flaccid.

You can easily see when a plant's vacuoles are full and when they are not. When the vacuoles are full the plant's stem and leaves will be held upright and firm. However, if the leaves and stem are drooping, the vacuoles might have lost a lot of water because the soil is too dry and the cell was forced to use up this water to survive.

Vacuoles are present only in some animal cells and they are typically very small and have a short life-span.

ACTIVITY Comparing plant and animal cells

Study the two diagrams of plant and animal cells below.

1. Draw a table of differences between the two cell types in your exercise books. Give the table a suitable heading.
2. Also provide labels for the different cell structures and organelles.

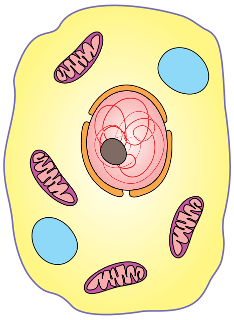


Figure 1.12: A typical animal cell.

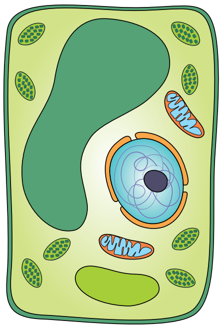


Figure 1.13: A typical plant cell.

ACTIVITY 3D model of a cell

In a 3D cell model, we will be making built models out of materials where we will use other objects to represent the actual parts of the cell.

Instructions

1. You must create a 3D model of a cell.
2. You may use whatever materials or 'media' you choose to create your cell.
3. Your model must clearly show the following:
   a) cell membrane
   b) nucleus with nuclear membrane
   c) cytoplasm
   d) mitochondria
   e) vacuoles
   f) chloroplasts
   f) any other organelles that you might have learnt about.

Requirements for your cell model

- Your model and the examples of the organelles need to show some resemblance to the real organelle that we have learnt about so far.
- Your model needs to be clearly marked with a heading and your name.
- Each organelle needs to be clearly labelled and with each label you need to add a description of the function of that particular organelle.
- You also need to make an accompanying drawing (at least the size of an A4 page) including the labels of the structure of a basic plant and animal cell.
- Your teacher will assess your model according to a rubric.

Keywords

- cover slip
- multicellular
- organisms
- slide
- specimen
- unicellular
- wet mount

1.3 Cells in tissues, organs and systems

Now that you have learnt all about different cells, are you ready to see them for yourself?

Observing cells under a microscope

Have you ever used a microscope before? Microscopes are instruments that are used to look at and study objects that are too small to be seen with the naked eye. Since the days of Hooke's observations, the development of microscopes has come a long way. Today we have incredibly powerful microscopes called electron microscopes which use electrons instead of light to observe very fine detail – even as small as a single column of atoms!

Figure 1.14: A modern electron microscope

Figure 1.15: A basic light microscope

Before we start working with microscopes, let's have a look at the different parts of a basic light microscope and the safety precautions we need to follow when using these pieces of equipment.

A microscope allows you to see detail in specimens that you cannot see with the naked eye. The image you see needs to be:

- well lit with enough light provided to see the specimen
- well focused
- contrasted with its surroundings to clearly see details.

The next image explains the different parts of a light microscope and what they are used for.

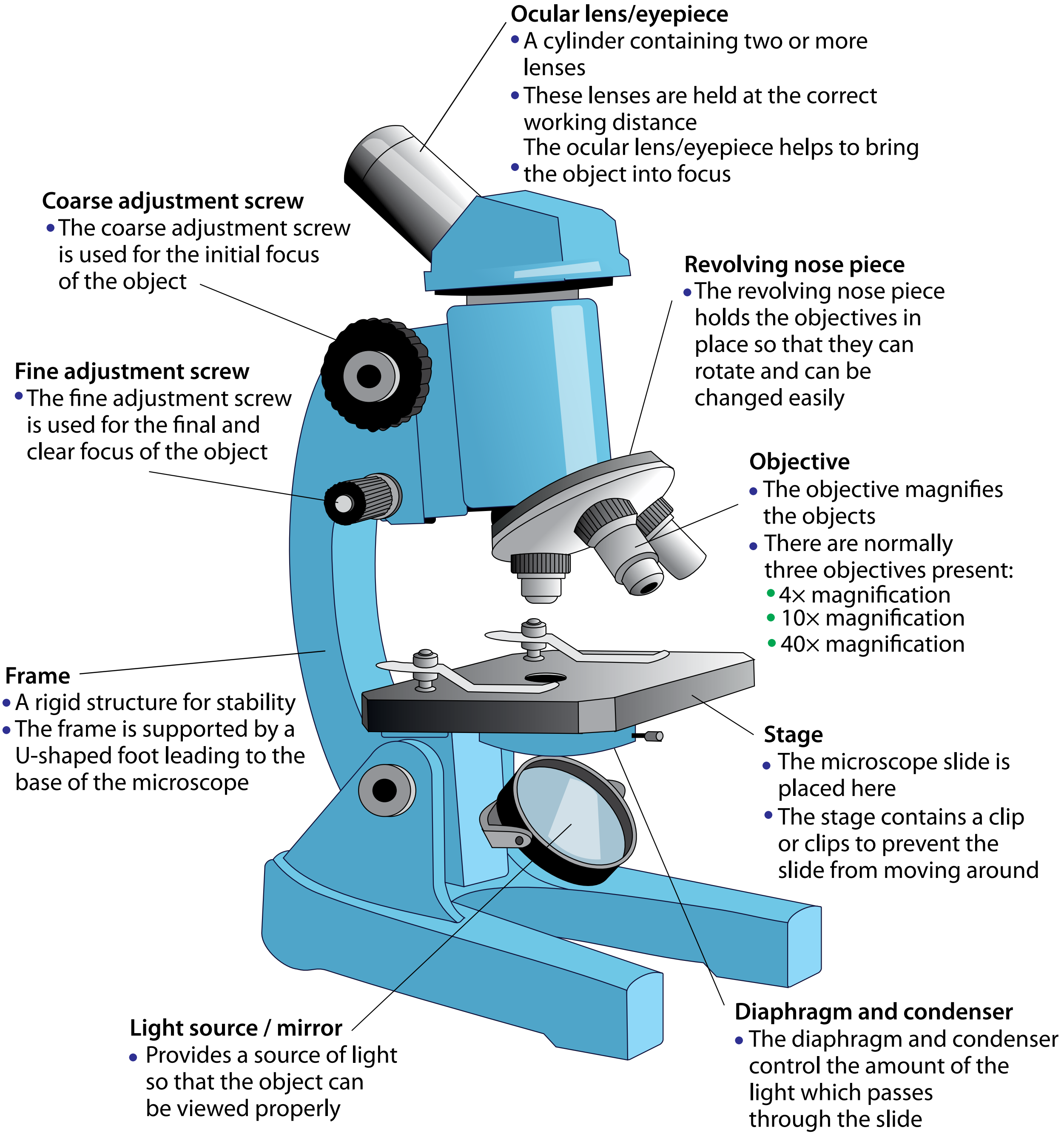


Figure 1.16: Part of the microscope

When you use a microscope, make sure to follow these safety precautions:

1. There is a special way to carry the microscope: one hand supports the base and the other holds the frame of the microscope.
2. Put it down on a stable, horizontal, clear counter.
3. Before using the microscope, clean the lenses with proper lens paper. Do not touch the lenses with your fingers! Make sure the stage and slides are clean.
4. When handling the slides, do not use broken or cracked slides and handle cover slips by the edges.
5. When focusing with the objectives:
   - focus smoothly and slowly
   - be careful with the objectives and do not scratch them.

Did you know?

We can use water, brine (salt water), glycerine or immersion oil for wet mounts.

6. When you are done:
   - always turn the lowest magnification objective into place before storing the microscope
   - make sure that the stage and slides are clean before putting everything away
   - always store the microscope in a box or covered with a dust jacket to avoid dust from settling on the lenses.

To view cells under a microscope, we need to make and prepare something called a **specimen** on a **slide**.

A specimen is a small part or slice, or an example of an organism that we want to examine. When we view a specimen under a microscope it needs to let light pass through the specimen so we can see it. Therefore, we need to prepare the specimen and cut extremely thin slices of less than 0.5 mm. Specimens are then placed on a glass slide.

We can prepare samples or specimens on a slide using these different techniques:

- **wet mount** – good for observing living organisms and is especially used for aquatic samples
- **dry mount** – good for observing hair, feathers, pollen grains or dust
- **smears** are often made of blood or slime that is smeared over the slide and allowed to dry before observing them.
- **stains** are added to wet or dry mounts by dropping colouring chemicals onto the specimens, like iodine solution, methylene blue or crystal violet. We use staining to improve the colour contrasts on the slide.

ACTIVITY Evaluating microscopic images

Instructions

1. Carefully study this image of onion cells that have been stained blue. Evaluate this image in terms of the focus, light and contrast visible in the photo.
2. These same onion cells were viewed under a microscope which had not been adjusted properly and the following photos were taken. Identify what is wrong with the photograph compared to the one above. Complete the table in your exercise books.

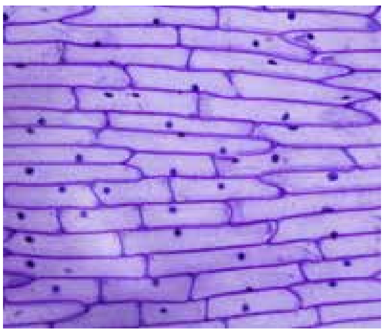
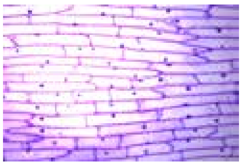
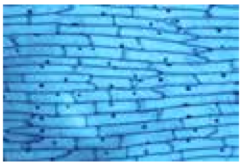


Figure 1.17: Onion cells that have been stained blue.

Image	What is wrong with the image?	How could the image have been adjusted and corrected, using what part of the microscope?
		
		
		

ACTIVITY Making a wet mount with onion and cheek cells

There is a very specific way to prepare slides for viewing under a microscope.

You will use this technique very often in Natural Sciences to study specimens.

Materials

- onion
- scalpel or knife
- dissecting needle
- forceps
- microscope slides
- coverslips
- dropper
- tissue paper or filter paper
- distilled water
- iodine solution
- light microscope

Take note

You will need to work quite quickly as the onion cells will dry out!

Instructions

Step 1: Prepare your microscope and slides as discussed in the safety methods above.

Step 2: Cut the onion into blocks of about 1 cm square with a sharp knife or scalpel.

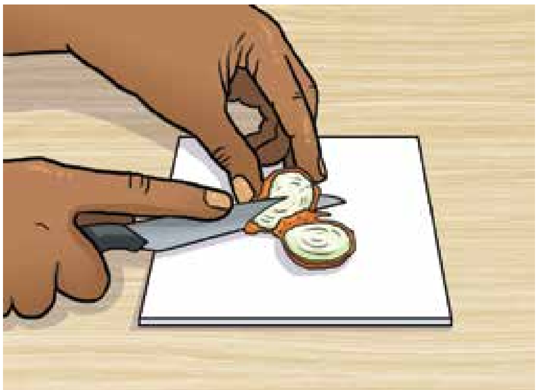


Figure 1.18: Cutting the onion to expose the layers.

Figure 1.19: Carefully pulling the lining off the onion layer.

Take note

If you have accidentally trapped an air bubble, gently press on the middle of the coverslip to get rid of any trapped air using the dissecting needle or drop some extra fluid right next to the edge of the coverslip.

Step 3: Use forceps to pull or peel a small piece of the very thin membrane-like epidermis lining off one of the inner layers of the onion.

Step 4: Place a drop of iodine solution onto the slide.

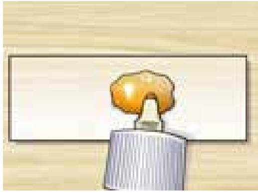


Figure 1.20: Adding iodine solution to the slide.

Step 5: Place the membrane directly in the drop on the slide.

Step 6: Gently lower a coverslip at an angle onto the onion cells. Hold the coverslip up with a dissection needle and gently lower the slip. This prevents air bubbles from getting trapped under the cover slip.

Figure 1.21: Lowering the coverslip onto the specimen.

Step 7: Wipe off excess fluid around the edge of the coverslip with tissue paper or filter paper.

Step 8: Make sure the lowest power objective lens (this is the shortest lens) is in line with the eyepiece. Switch on the lamp or use the mirror to reflect the light onto your stage. Place the prepared slide onto the stage and secure it with the stage clips.

Figure 1.22: The slide secured on the microscope stage.

Step 9: While on the low power, look from the side and lower the objective lens to just above the coverslip. Then look through the eyepiece and use the fine focus to focus your image.

Step 10: Magnify your cells by swapping the objective lens to a higher powered lens. Only use the fine focus adjustment to focus clearly.

Step 11: Make careful drawings of your observations in your exercise books and remember to label what you see. Add a heading including the specimen, the stain used and the magnification.

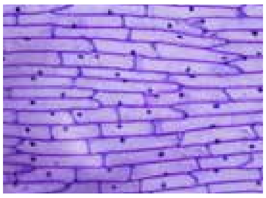


Figure 1.23: Onion cells.

Now that you have prepared slides of onion cell specimens, use a toothpick to scrape the inside of your cheek *gently* to collect cheek cells using the side of the toothpick or ice cream stick. Follow the same instructions as above to prepare the cheek cell specimen and to view it under the microscope. Draw and label the cheek cells that you viewed under the microscope in your exercise books.

Did you see something like this?

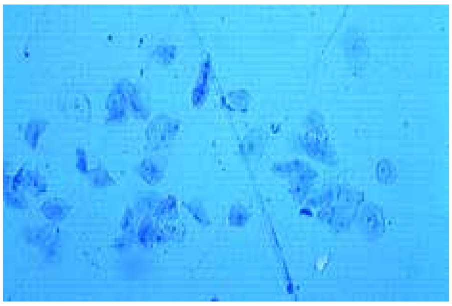


Figure 1.24: Some cheek cells stained with methylene blue.

Question

1. What are some of the differences and similarities you noted between the animal and cheek cells?

ACTIVITY Research the discovery of light and electron microscopes

The invention and improvement of microscopes has led to incredible cellular discoveries (among others) in the last 400 years. Without microscopes, many of the microscopic organisms we know of today would never have been identified!

Instructions

1. You can work individually or in groups for this task.
2. Research the history and discovery of the light and electron microscopes and how they are used today.
3. Design a brochure for the local Science museum where you tell visitors about the history of the development of microscopes.
4. Remember that a brochure must be informative, but not contain too much text.
5. Include some photographs or drawings.

Cells differ in shape and size

We looked at the basic differences between plant and animal cells. However, not all plant cells and not all animal cells are the same. Cells within an organism need to have different shapes and sizes because they fulfil different functions.

Keywords

- differentiation
- stem cell

Look at the photo of the rose. Do you think the cells in the roots, stem, leaves, and petals of the rose all look the same?

The cells in the different parts of the rose all have to perform very specific functions and therefore have different sizes and shapes.

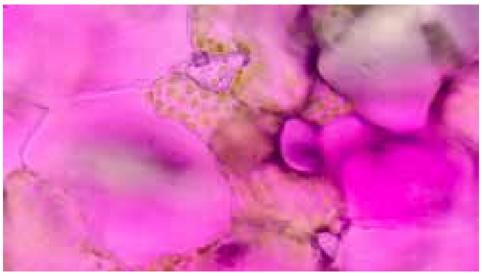


Figure 1.25: The rose's petals are pink due to pigments in the vacuoles of the petal cells which are round.

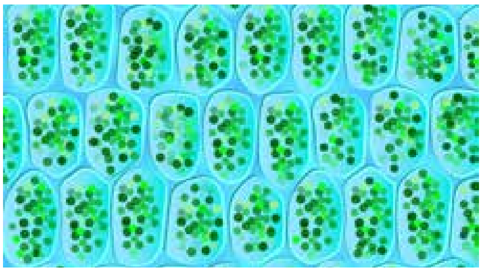


Figure 1.26: Cells in the leaves are full of chloroplasts for photosynthesis. They are long and rectangular in shape.

Your body contains a great number of **specialised** cells, meaning they have different functions. They have differences in their structures allowing them to have different functions. We say they have **differentiated**.

Did you know?

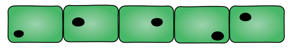
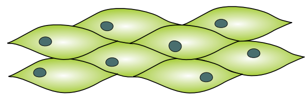
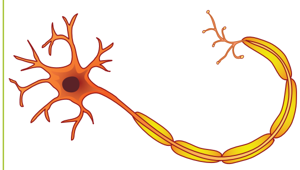
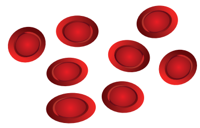
Stem cells are also harvested from the umbilical cord at birth and used for research. There are many ethical concerns regarding stem cell research.

Take note

Microscopic and macroscopic describe whether an organism can be seen with the naked eye, while unicellular and multicellular refer to the number of cells an organism has.

Do you remember we spoke about nerve cells and red blood cells briefly in the beginning of the unit? Some of them are summarised in the following table.

What do you think?

Specialised cell	Structure	Function
Epithelial cells: they are mostly flat		They cover the surface of the body for protection.
Muscle cells: some are long and spindle shaped		Muscle cells can contract and relax allowing for movement within your body.
Nerve cells: they are very long and have branched ends		Nerve cells are specialised to carry messages that coordinate the functions of the body.
Red blood cells: round and biconcave shape		Red blood cells carry oxygen and carbon dioxide throughout the body.

Stem cells

Stem cells are unspecialised cells which can divide and develop into many different types of specialised cells. Stem cells are quite amazing as they can divide and multiply while at the same time keeping their ability to develop into any other type of cell. Embryonic stem cells are the little ball of 50–150 cells that forms 4–5 days after conception. Embryonic stem cells are very special as they can become absolutely any cell in the body, for example, blood cells, nerve cells, muscle cells or brain cells.

For this reason, scientists are using stem cells to conduct research. There are many benefits in doing this, but there are also many controversial and ethical issues surrounding stem cell research.

Are you curious about stem cell research? Find out more and discover the possibilities!

Microscopic and macroscopic organisms

We have just looked at specialised cells within organisms. The organisms that we have discussed, plants and animals, consist of many, many cells. Your body has millions of cells! Did you know that there are some organisms which consist of only a single cell? We have many different specialised cells to perform the different functions within our body whereas in a single-celled organism, all the functions it performs are done in this one cell. We can make a distinction between organisms that are made of one cell (unicellular) and those that are made of many cells (multicellular).

Microscopic organisms

We call one-cell organisms, that can only be seen with the help of a microscope, microscopic organisms. There are many single-celled microscopic organisms.

Have a look at the images.

Figure 1.27: A group of *Escherichia coli* bacteria which are found in the intestines of many animals.

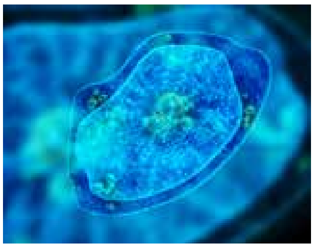


Figure 1.28: An amoeba, which is a single cell organism that lives in water.

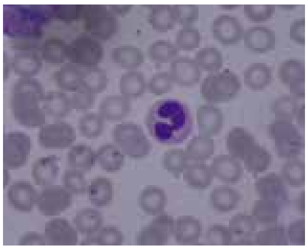


Figure 1.29: Red blood cells showing some which have been infected with malaria (purple dots).

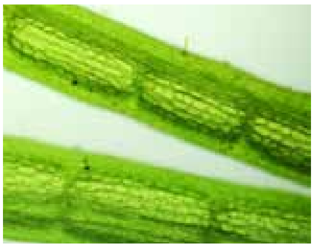


Figure 1.30: A single-celled algae called a desmid.

Macroscopic organisms

In contrast to microscopic single-celled organisms, macroscopic organisms are visible to the naked eye and consist of many cells. Macroscopic organisms can have a few cells working together or trillions of cells that form larger organisms.

Organisation of cells in macroscopic organisms

In microscopic single-celled organisms, the individual cell has to perform all the life processes for that microscopic organism.

So what about the cells in macroscopic organisms that consist of many cells?

We have already learnt about specialised cells in macroscopic organisms, so we know that not all cells perform all the processes – they are specialised to perform a specific function.

Specialised cells that perform a specific function group together to form a tissue. For example, muscle cells will group together to form muscle tissue, epithelial cells will group together to form the skin, and nerve cells will group together to form the brain and nerves.

Groups of tissues that work together form organs. Think of the stomach, for example – it is made of many different specialised cells that form muscle tissue to make it contract and epithelial tissue (made from specialised epithelial cells) which lines the inside of the stomach and produces mucus.

When organs work together we say they form systems or organ systems. There are many different systems in your body where specific organs work closely together to make your body function. Have a look at the following diagram. It shows how cells are organised into tissues in the stomach which form part of the digestive system in a human (the organism).

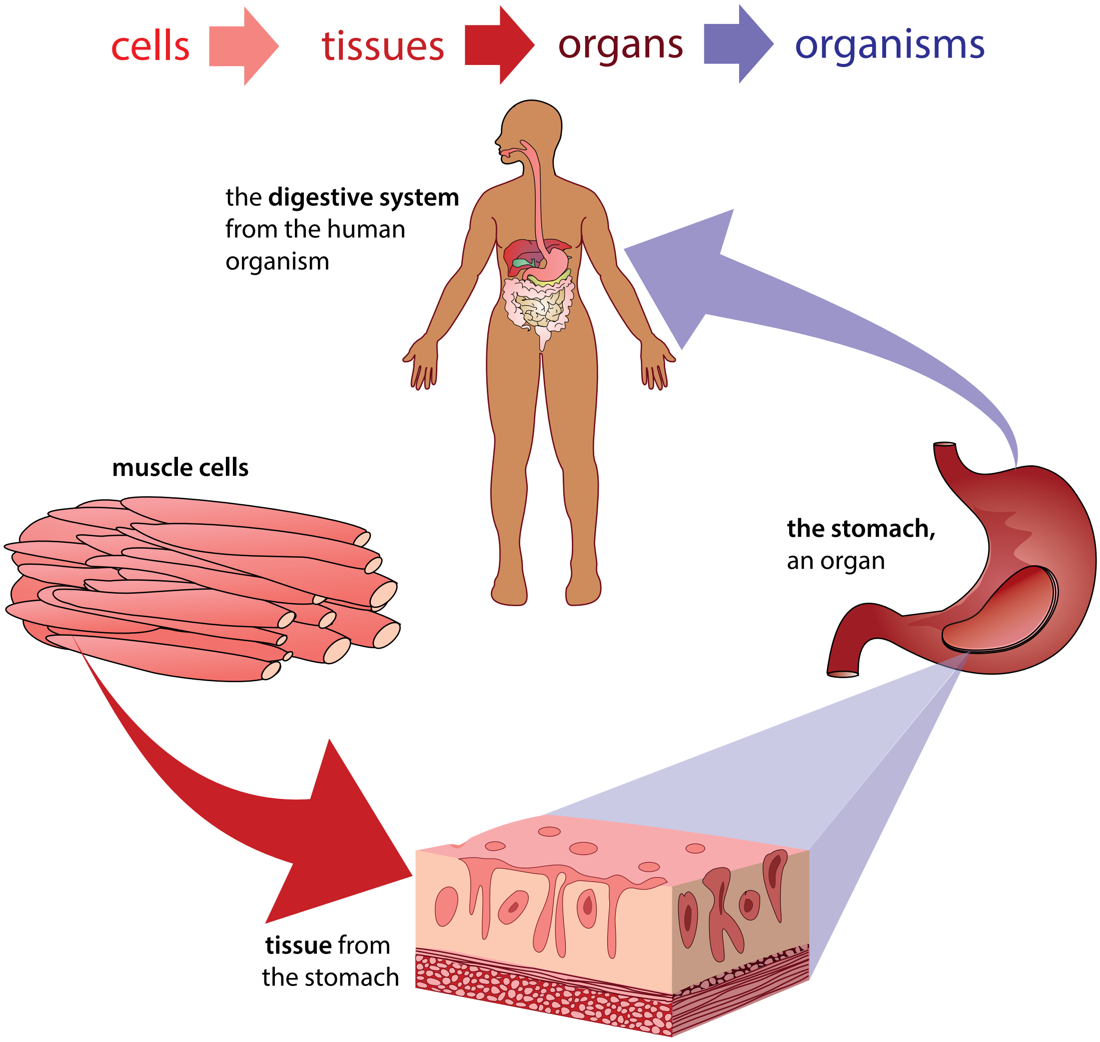


Figure 1.31: How cells are organised into tissues in the stomach to form part of the digestive system.

All the systems work together to form an organism. We will be looking at some of these systems later on in the term.

Summary

In the summary, we first have the 'Key concepts' for this unit. This is a written summary and the information from this unit is summarised using words. We can also create a concept map of this unit, which is a map of how all the concepts (ideas and topics) in this unit fit together and are linked to each other. A concept map gives us a more visual way of summarising information.

Different people like to learn and study in different ways; some people like to make written summaries, while others like to draw their own concept maps when studying and learning. These are useful skills to have, especially for later in high school and after school!

Have a look at the concept map which shows what we have learned about the cell in this unit and how these concepts link together. Can you see how the arrows show the direction in which you must 'read' the concept map?

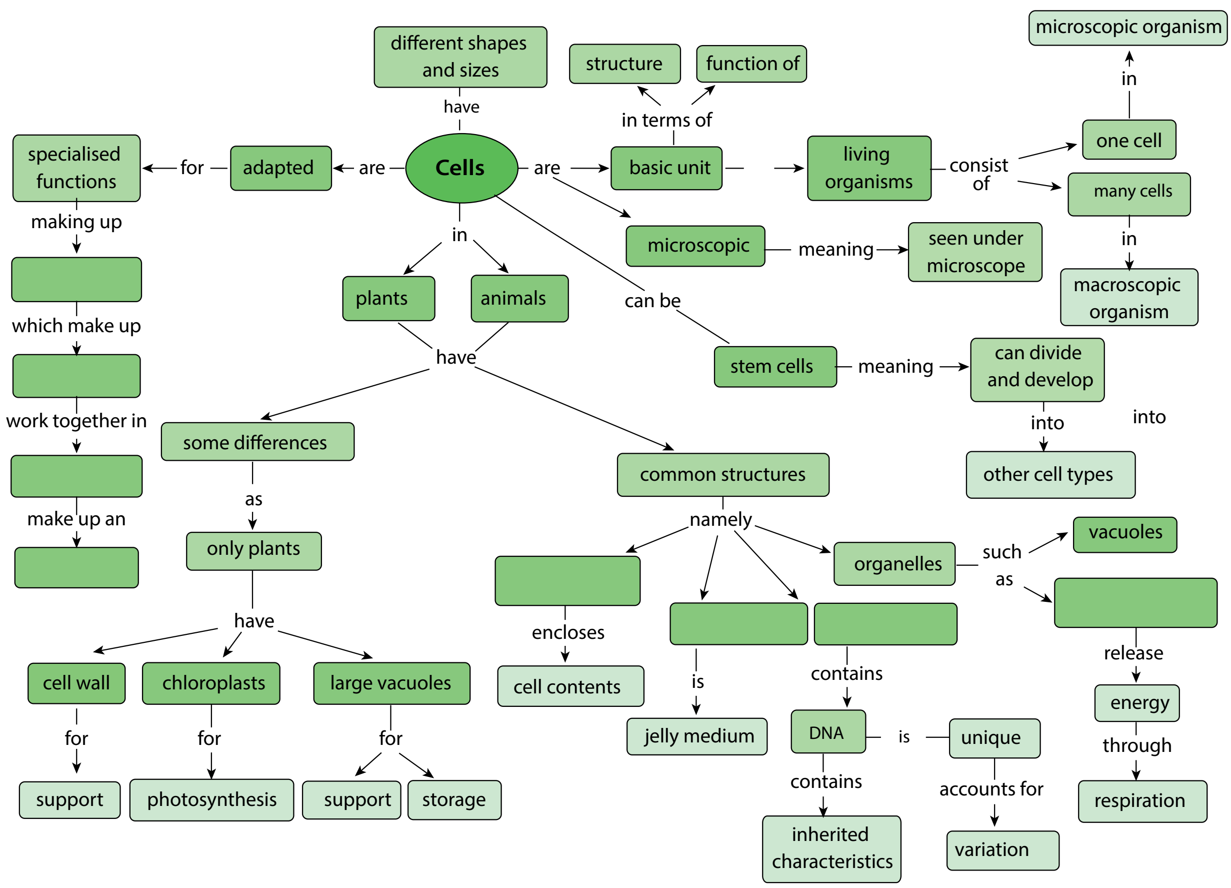
There are some empty spaces in the concept map that you need to fill in. For example, some of the common structures in cells have been left out. You need to look at the concepts linking from these bubbles to work out which structure goes where. For example, what structure in a cell encloses the cell contents? Write the answer in the correct space using the concept map provided by your teacher. On the left-hand side of the concept map there are also empty spaces – can you see that this describes the hierarchy of how cells are organised into tissues, which are organised further into organs, and so on? Fill in each level of organisation into the spaces.

Key concepts

- Cells are the basic structural and functional units of all living organisms.
- Most cells are microscopic and can be seen only under a microscope.
- Plant and animal cells have cell membranes, cytoplasm, a nucleus and organelles such as mitochondria and sometimes vacuoles.
- The cell membrane encloses the contents of the cell and separates it from its environment.
- Cell membranes are selectively permeable, which means they only allow certain substances to pass into and out of the cell.
- The cytoplasm includes the organelles and the cytosol. The cytosol is the jelly-like medium in which many chemical reactions take place in the cell. Everything inside the cell membrane, except the nucleus, is considered the cytoplasm.
- The nucleus in eukaryotic cells is enclosed by a nuclear membrane and contains the DNA.
- DNA contains inherited characteristics of an organism and controls the cell's activities. It is unique to each organism, resulting in variation within a species.
- Cell respiration takes place in the mitochondria.
- Plant cells have a cell wall around the cell membrane that is rigid and provides support and protection of the cell's content.
- Plants have chloroplasts with the pigment chlorophyll to photosynthesise and produce glucose (food).
- Plant cells also have large vacuoles to store water and glucose, and to provide support for the plant
- Vacuoles in animal cells are temporary (or absent) and much smaller.

- Cells come in many different shapes and sizes.
- Stem cells are cells that have the ability to divide and develop into many different cell types.
- Microscopic organisms can only be seen under a microscope. All single-celled organisms, such as bacteria, are microscopic. However, some multicellular organisms such as dust mites are also too small to see with the naked eye.
- Macroscopic organisms consist of many cells and can be seen with the naked eye.
- Specialised cells perform special functions. Specialised cells that work together to perform a specific function form tissue.
- A group of different tissues makes up an organ.
- Organs working together in groups form systems or organ systems.
- Organ systems make up an organism, such as a human.

Concept map



Revision

1. Why would you say cells are considered to be the smallest unit of life? [2]
2. Explain what **selectively permeable** means when referring to the cell membrane. [1]
3. Eukaryotic and prokaryotic cells differ. What is the main difference between these two types of cells? [2]
4. What is the main function of the nucleus and what is the function of the DNA? [2]
5. When a Grade 9 learner labelled one of the cell organelles 'Powerhouse', their teacher marked it wrong. What should the learner rather have written? [1]
6. A plant and an animal cell are similar in some ways yet very different in others. Compare the two types of cells in a paragraph. [10]
7. Make two drawings to show the differences between plant and animal cells using the examples of plant and animal cells you studied under the microscope. Follow the drawing guidelines for making scientific drawings. [10]
8. There are different types of specialised cells and tissues in plants and animals that perform different functions. Match each function to the corresponding tissue. [3]

Smooth muscle tissue	receives and sends messages and helps the body respond to stimuli.
Nerve cell	carry oxygen around the body in mammals
Red blood cells	contracts and enables movement

9. Use words from this box to complete the sentences below. Write the sentences out in full in your exercise books. [4]

Organs Tissues organ systems specialised cells

Macroscopic organisms consist of many different____________ that are made of individual ____________ that work together in a very particular way. These are formed from ____________ that are in turned created when groups _______________ of function together in a specific way.

Total [35 marks]

2 Systems in the human body

Key questions

- How does the body do the things it does, such as breathe, move and think?
- What happens when one of the systems in our bodies does not work properly and has a 'system error'?
- How can you best look after your body?

The human body has been studied by artists and scientists, mechanical engineers and medical practitioners throughout history. The mechanical beauty and operation of each and every part in the human body has fascinated human beings throughout history. Be curious, and get ready to be fascinated!

New word

- integrate

Body systems

The human body consists of several integrated systems for the body to function well.

In the following pages we willl study seven of the main systems in our bodies. At the end of each organ system you will need to make a summary of that organ system to show:

- the main purpose or function of the system in the body;
- the main processes that take place in the system;
- the main components (organs) that make up the system; and
- the main health issues that relate to that particular system.

Therefore pay close attention and make notes as you study each organ system to help you with your summary.

ACTIVITY Research and writing about health issues

Instructions

1. You are going to learn about many of the health issues related to each of the different systems. Choose one of these health issues to research.
2. You will need to:
   a) Consult at least 3 different resources to find out more about that particular health issue.
   b) Suggest ways that this health issue may be prevented (if this is possible).
   c) Suggest treatment for the health issue in question.
3. Present your findings on an A3 poster as part of an oral presentation (of 3–4 minutes) to the class.

2.1 The digestive system

Our cells need **protein**, **carbohydrates**, **fats**, **vitamins** and **minerals** to function. Yet we eat large pieces of food that are too big to pass through the selectively permeable cell membranes. So how does the food we eat eventually get to our cells in a small enough form to be absorbed?

Purpose of the digestive system

Our digestive system is responsible for breaking down the food that we eat into small particles that can be **absorbed** into the bloodstream. They are then **transported** to the cells throughout our body.

The digestive system is made up of the different parts of the **alimentary canal**.

This canal is a long, twisting, pipe-like structure (about 9 metres in total) that starts at the mouth and ends at the anus. Along the way the food is broken down from chunks into molecules small enough to pass through cell membranes and supply energy to cells.

Main processes in the digestive system

There are four main processes that occur in the digestive system at different parts along the alimentary canal. They are:

- **Ingestion**: This occurs when you take food into your body through your mouth by eating or drinking it.
- **Digestion**: This is the process of breaking down large food pieces into particles that are small enough to be absorbed and pass through cell membranes.
- **Absorption:** This occurs when the digested particles move into the cells of the digestive tract (they are absorbed) and move to the bloodstream from where they are carried to all the cells in the body.
- **Egestion**: Any undigested or unwanted particles that travel through the digestive tract are later passed out as faeces. This process is known as egestion.

Components of the digestive system

Have a look at the diagram on the right which gives an overview of the different parts making up the digestive system.

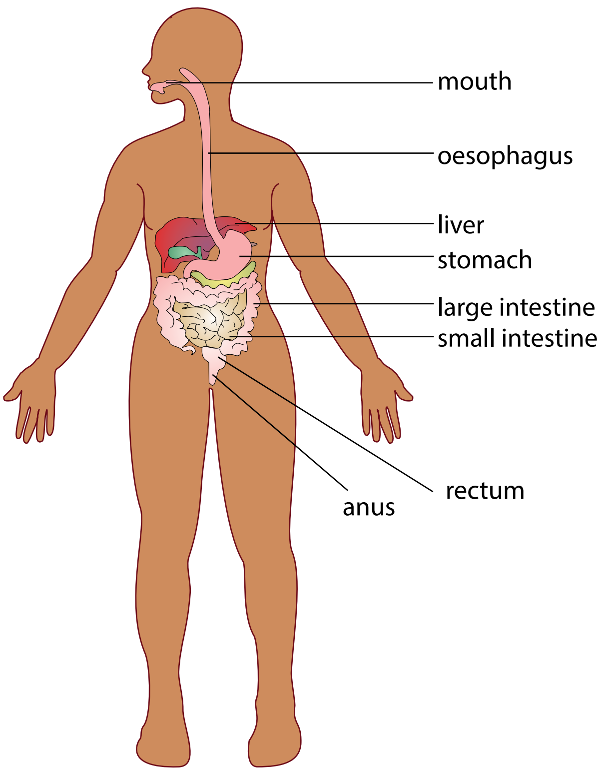


Figure 2.1: Components of the digestive system.

Keywords

- absorption
- alimentary canal
- carbohydrates
- digestion
- dissolve
- fats
- glucose
- minerals
- nutrients
- protein
- transport
- vitamins

Did you know?

Our digestive system is sometimes also referred to as the 'gastrointestinal tract', the 'digestive tract' or the 'alimentary canal'.

Keywords

- anus
- chemical digestion
- egestion
- faeces
- fibre
- ingestion
- jaundice
- metabolic
- saliva
- small intestine
- stomach
- ulcer

The mouth and oesophagus

Digestion starts in the mouth as food is chewed and mixed with saliva. It then travels down the oesophagus when you swallow.

Stomach

The chewed food enters the stomach and is further digested. The stomach has substances called enzymes to help digest the food. The stomach also contracts to break the food down further into a liquid.

Small intestine

Most of the digestion takes place in the small intestine. Absorption of the food particles also takes place in the small intestine.

Large intestine (or colon)

By the time the food reaches the large intestine, most of the nutrients have been absorbed. What is left is water, salts, and indigestible fibre. The water that is left is absorbed in the large intestine.

Rectum and Anus

The remaining substances (called faeces) are passed into the rectum and then out through the anus. This is called egestion.

Did you know?

Bacteria that live in the large intestine produce Vitamin K and some Vitamin B.

ACTIVITY Flow diagram of the digestive system

Flow diagrams are diagrams that show how the different sets of a process fit together in a sequence. They show the direction (flow) by using arrows. These are important tools to help you think about processes in Science.

Instructions

1. Draw a flow diagram to represent the passage of food from the time it is taken into the body to the time it is egested from the body.
2. The blocks must show the main components involved in digestion, listed in order, with arrows in between. Under each component include the main processes that occur at each of these stages.

Take note

There is a difference between egestion and excretion. Egestion occurs when undigested particles are passed out as faeces. Excretion occurs when the body gets rid of metabolic waste formed from chemical reactions taking place in the body.

Health issues involving the digestive system

Common diseases of the digestive system

Ulcers

Sometimes open sores or ulcers develop on the lining of the mouth, oesophagus, stomach or upper portions of the small intestine. Ulcers can be very painful. They are generally caused by bacterial infections and some medications.

Anorexia nervosa

This is one of many eating disorders. People who suffer from this eating disorder have an abnormal fear of gaining weight and therefore starve themselves on purpose. This can lead to many health issues such as bone thinning, kidney damage, heart problems and even death.

Diarrhoea

Someone who passes very frequent, loose, watery stools has diarrhoea. Many diseases cause undigested food to pass through the large intestines too quickly for water to be absorbed and cause diarrhoea.

Do not forget to wash your hands with lots of soap and water!

Liver cirrhosis

This disease slowly replaces healthy liver tissue with scar tissue and eventually prevents the liver from functioning properly. Alcohol abuse and fatty liver caused by obesity and diabetes are the most common causes of liver cirrhosis.

Take note

It is good to know the dangers and health consequences of an unhealthy lifestyle.

2.2 The circulatory system

Did you know that the blood moving throughout your body forms a system? To 'circulate' means to move around, and so we have the circulatory system within the human body which transports blood.

Purpose of the circulatory system

The circulatory system is responsible for transporting blood with oxygen (O_2) from the lungs to cells and then transporting blood with carbon dioxide (CO_2) back to the lungs. It also has to distribute nutrients from the digestive system to the cells in the body and remove waste products to be excreted.

Keywords

- arteries
- blood
- blood vessels
- capillary
- closed blood system
- deoxygenate
- excrete
- gaseous exchange
- heart
- lungs
- network
- oxygenate
- temperature
- veins

Components of the circulatory system

The circulatory system is composed of the heart and a system of blood vessels, including arteries, veins and capillaries.

Heart

The heart is a very strong muscle and pumps blood throughout the body. There are four chambers in the heart that receive and send blood to all parts of the body. The two top chambers are called *atria* (singular = atrium) and the two bottom chambers are called *ventricles*.

Blood vessels

There are various blood vessels which carry the blood throughout the body.

These are:

- arteries
- capillaries
- veins

Did you know?

The first heart transplant in the world was done by Dr Chris Barnard in South Africa in 1967!

Did you know?

On average your heart beats about 100 000 times a day and 30 million times a year. If you live to age 70, your heart would beat an average of 2,5 billion times!

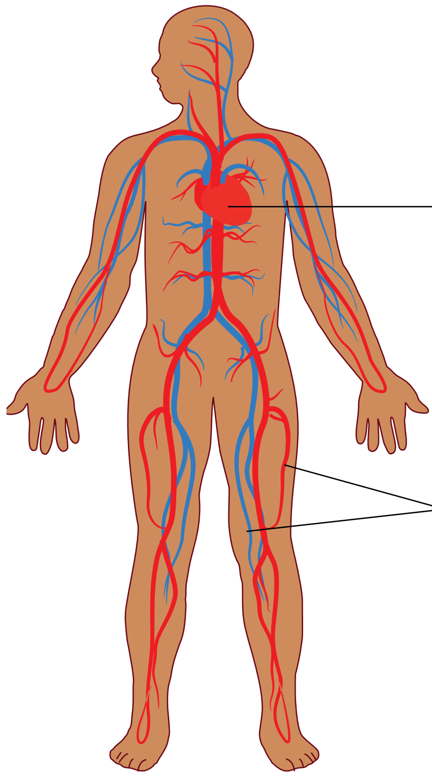


Figure 2.2: The circulatory system is composed of the heart and blood vessels.

Blood

The blood is transported throughout your body and carries various substances.

Substances such as carbon dioxide, nutrients and waste products can be dissolved in the blood liquid (plasma), or else in the red blood cells (like oxygen).

Main processes in the circulatory system

Our circulatory system is actually made up of two systems that function together:

- a short system that circulates blood between the lungs and the heart; and
- a much longer system that circulates blood from the heart throughout the body and back again.

This process occurs as follows:

- Blood is circulated from the heart to the lungs. At the lungs, carbon dioxide (CO_2) leaves the blood and oxygen (O_2) enters the blood. This process is known as **gaseous exchange**. Since the blood now contains more oxygen than carbon dioxide, we say it is **oxygenated**. This oxygenated blood returns to the heart again.
- Once in the heart the oxygenated blood is circulated to deliver the oxygen to all the cells in the body before returning to the heart. At the same times as it delivers oxygen, the blood also collects carbon dioxide from the cells. This blood has more CO_2 than O_2, so it is **deoxygenated** blood. The carbon dioxide is excreted when it returns to the lungs.

This process occurs over and over again throughout your life, thousands of times a day!.

ACTIVITY Chart the circulatory system

Instructions

1. Study the diagram below that explains the circulatory process.

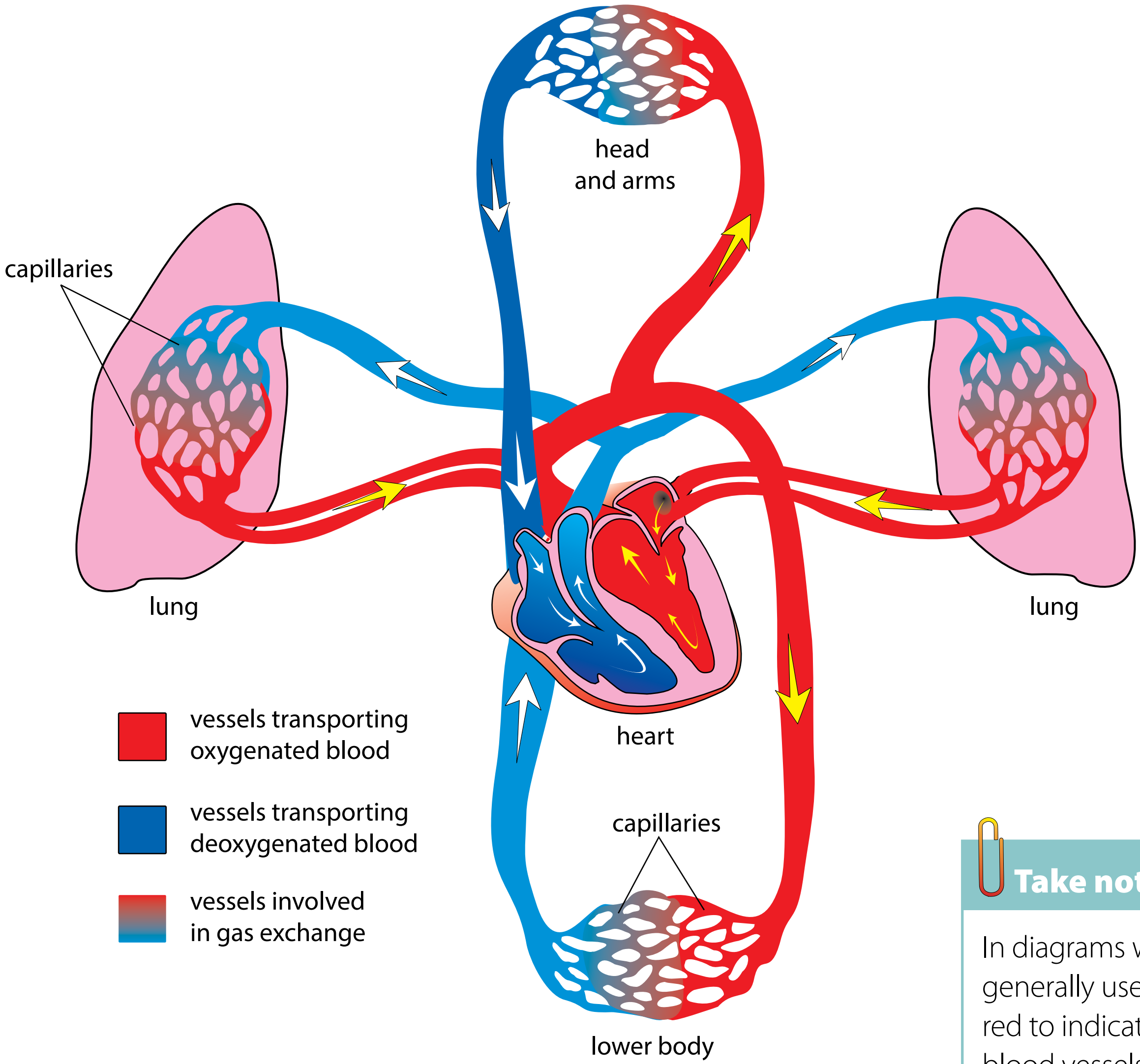


Figure 2.3: The circulatory process

Use the diagram above to draw a circular diagram in your exercise books to show how blood travels through the circulatory system (composed of two systems).

2. Your circular diagram will form a complete circle.
3. Add arrows to show the direction in which the process occurs.

Take note

In diagrams we generally use red to indicate blood vessels that contain oxygenated blood and blue to show blood vessels with deoxygenated blood.

Take note

You can find out lots more online by visiting the links provided in the Visit boxes. Be curious, and discover the possibilities!

Health issues involving the circulatory system

Common diseases of the circulatory system include:

Keywords

- blood pressure
- deprived
- rupture

High blood pressure

This occurs when the force with which the blood pushes against the walls of the blood vessels is too high and can cause damage to the capillaries and several organs.

Heart attacks

Heart attacks occur when a narrowing or blood clot develops in one of the blood vessels that supplies the heart muscle with blood. If the narrowing or blood clot is big enough it can stop the blood flow to the heart muscle and can stop the heart from pumping, which is called a heart attack. The person can die.

Strokes

Strokes occur when cells in your brain are deprived of oxygen. This often occurs as a result of a blockage in the blood vessels leading to the brain, or when one of these vessels rupture (break or burst open).

2.3 The respiratory system

Keywords

- alveoli
- breathing
- bronchi
- cilia
- diaphragm
- exhale
- inhale
- lungs
- mucus
- pharynx
- trachea

Closely linked to the circulatory system is the respiratory system. The circulatory system maintains the circulation of blood in the body, while the respiratory system deals with the exchange of gases in your body.

Purpose of the respiratory system

The respiratory system is responsible for supplying the body's cells with oxygen and for removing carbon dioxide.

Components of the respiratory system

Various organs play a vital role in the respiratory system.

Mouth and nose

Oxygen-rich air enters the body through the mouth and nose where it is warmed.

Trachea (also called the windpipe)

The trachea is a tube that enters the chest and allows air to flow from the mouth into the bronchi and from there into the lungs. It is kept open by cartilage rings.

When dust particles and germs in the air enter the trachea during inhalation, the mucus lining the trachea traps these particles and the **cilia** work together to remove them from the body. When you sneeze or cough you expel the mucus and foreign particles from your body.

Did you know?

Using a patient's own stem cells, surgeons were able to develop and transplant the first human trachea in 2008.

Bronchi

The trachea splits into two air tubes, called bronchi, that connect to each of the lungs. These tubes divide even further into smaller and smaller tubes that connect with the tiny air bags (alveoli) of the lungs.

Lungs

The main organs of the respiratory system are the lungs. The tiny alveoli or air bags in the lung are surrounded by small capillaries where gaseous exchange takes place.

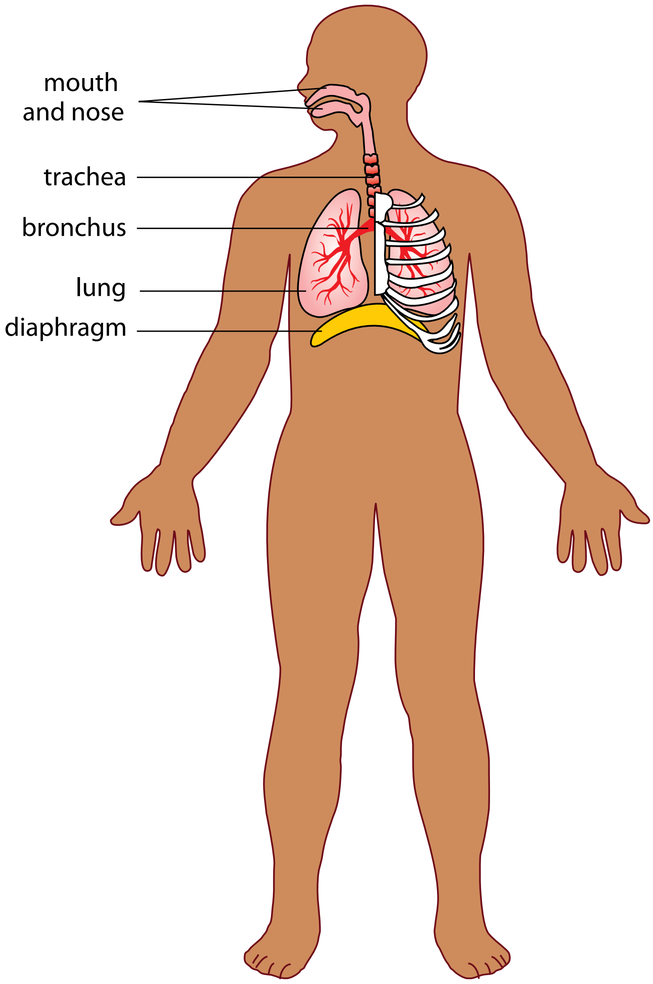


Figure 2.4: A diagram of the structures that make up the respiratory system

Diaphragm

This dome-shaped muscle below the lungs enables you to breathe. When it contracts, it moves downwards and your lungs fill with air. When it relaxes again it moves upwards and forces the air out of your lungs. This is the main muscle used for breathing.

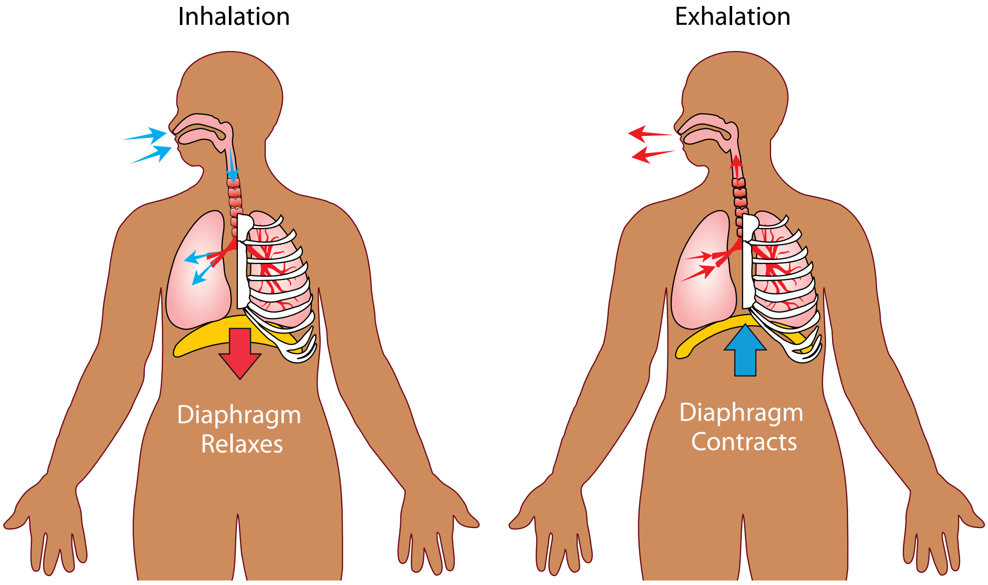


Figure 2.5: This diagram helps us to understand how breathing occurs showing how the diaphragm contracts and relaxes.

New word

- diffusion

Main processes in the respiratory system

Three distinct processes occur in the respiratory system:

- **Breathing** occurs when we take oxygen into the body (lungs) and push carbon dioxide out of the body (lungs). Breathing therefore occurs in two folds:
  - Inhalation – drawing air in
  - Exhalation – pushing air out.
- **Gaseous exchange** takes place at two locations by a process called **diffusion**:
  - in the alveoli, oxygen diffuses into the blood from the lungs and carbon dioxide diffuses from the blood back into the lungs
  - in the body tissues oxygen diffuses from the blood into the cells and carbon dioxide from the cells diffuses into the blood
- **Cellular respiration** occurs within the mitochondria of cells to release the chemical energy in food.

Take note

People often confuse respiration with breathing. Breathing is taking air into the body through the lungs. Respiration or cellular respiration takes place inside the cells to release energy when oxygen is combined with glucose and other nutrients.

Health issues involving the respiratory system

Some common health issues of the respiratory system are:

- **Asthma:** caused by allergies that inflame and narrow the airways
- **Lung cancer:** uncontrolled growth of cells within the lungs which may be caused by smoking and toxic air pollution
- **Bronchitis:** swelling of the lining of the bronchi due to infection which causes coughing and makes it difficult to get air into the lungs
- **Pneumonia:** an infection in the lungs where the alveoli fill with fluid
- **TB (Tuberculosis):** an infectious disease caused by the bacteria, *Mycobacterium*

2.4 The musculoskeletal system

All the movements that your body performs rely on a system of muscles, tendons, ligaments, bones and joints that work together. These are the components of your musculoskeletal system.

Purpose of the musculoskeletal system

Muscle tissue is responsible for producing movement in the body. However, muscles need to be attached to a frame structure to produce movement.

The bones of the skeleton provide a frame for muscles to attach to, so that movement is possible. The skeleton also protects the body, especially the soft, fragile organs such as the heart, lungs and brain.

Keywords

- brittle
- cartilage
- collagen
- contract
- fracture
- frame structure
- joint
- ligament
- mineral salts
- muscle
- tendons

Components of the musculoskeletal system

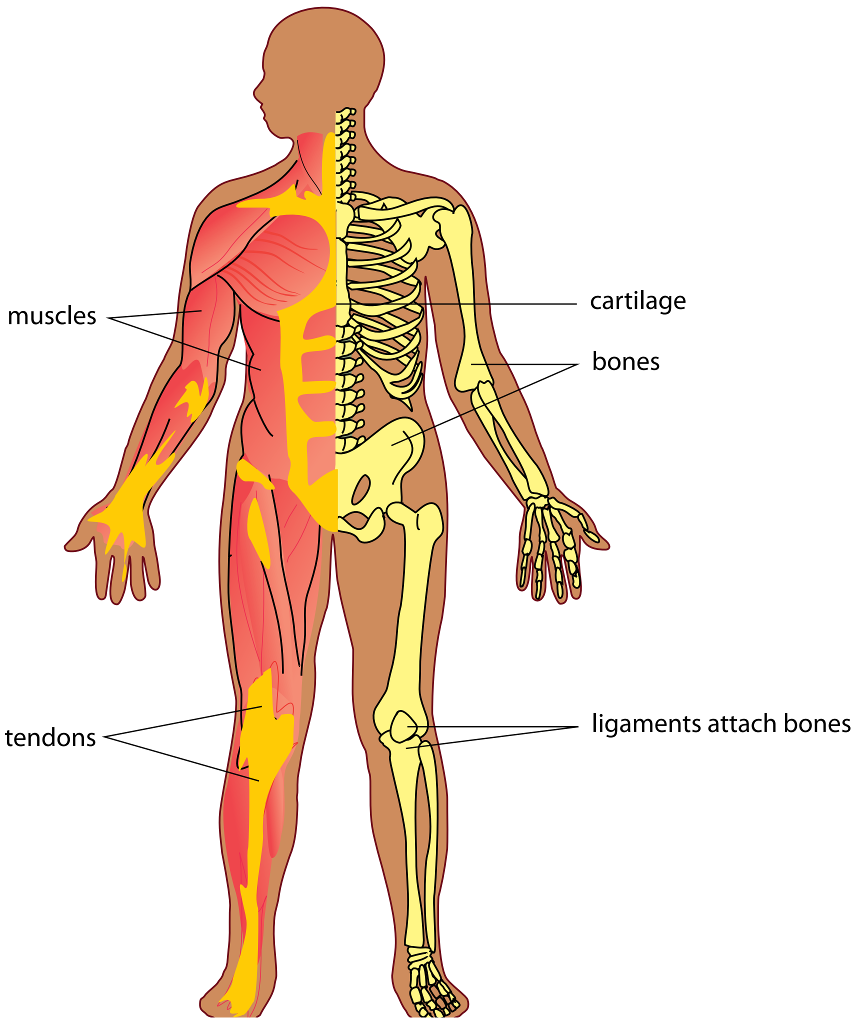


Figure 2.6: The components of the musculoskeletal system help bring about movement.

Did you know?

You use 200 muscles when taking only one step!

The components of the musculoskeletal system include the following:

Muscles

Muscles allow us to move because they are able to contract (become shorter) and relax (become longer).

Bones

Bones provide support and help to form the shape of the body. The place where bones meet is called a joint - think of your knee or elbow joint, or your finger and toe joints.

Cartilage

Cartilage is stiff yet flexible and is found between bones in joints and between the ribs and breastbone (as indicated in the diagram). It also forms the ears, nose and bronchial tubes, and forms discs between the bones of the spinal column.

Tendons

Your muscles attach to the bone with strong cords called tendons. You can feel some of the tendons in your body, for example behind your ankle (called the Achilles tendon).

Ligaments

Ligaments occur between bones at joints and hold bones together within the joint. Ligaments are extremely strong.

Keywords

- bowing
- locomotion
- self-propulsion

Main processes in the musculoskeletal system

We can move our entire bodies from one place to another through **self-propulsion**. This is called **locomotion**. Locomotion is different to movement. Movement is the change in shape, direction, position or size of a part of the body. Animals show movement and locomotion. What about plants? Do you think plants show movement and locomotion?

Locomotion and movement are made possible through the contraction and relaxation of muscles. Muscles are stimulated by nerves to contract.

Did you know?

The muscle that moves the fastest in your body is the muscle that controls your eyelids! There is a reason why we use the phrase 'in the blink of an eye' to mean very fast!

Health issues involving the musculoskeletal system

Common disorders of the musculoskeletal system include:

Rickets

This disorder is caused by a lack of vitamin D, calcium or phosphate which results in soft, weak bones. A typical symptom in children who have rickets is a **bowing** (bending outwards) of the bones of the legs.

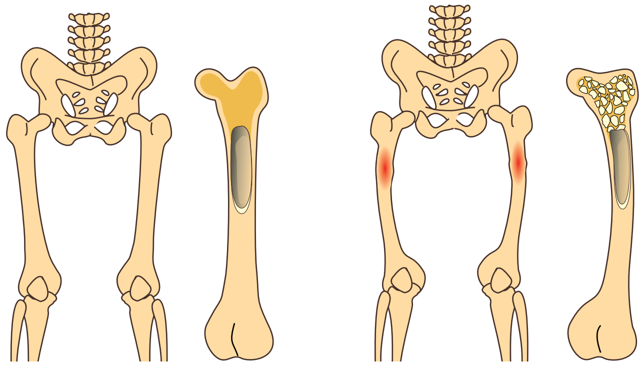


Figure 27: Can you see how the shape of the bones changes when a person has rickets?

Arthritis

This is a condition where the joints in the body become inflamed, painful and swollen. The cartilage between the joints breaks down, causing the bones to rub against each other, which is very painful.

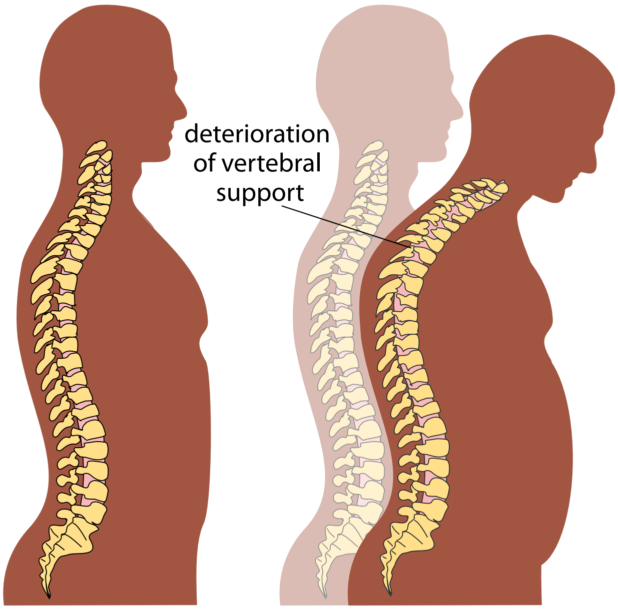


Figure 2.8: As this woman grew older, she developed osteoporosis, causing her vertebral column to crumble and collapse. So she now stoops over.

Osteoporosis

This occurs when the bone tissue becomes brittle, thin and spongy. These fragile bones can break easily, and they start to crumble and collapse. Although osteoporosis is common in older people (especially older women), teenagers and young adults may also develop it.

Did you know?

Babies are born with 305 bones while adults only have 206 bones. As babies grow into adults, many smaller bones fuse together to form bigger bones

2.5 The excretory system

We will now be looking at the excretory system. This is often confused with egestion, which we previously learned about.

Keywords

- bladder
- excretion
- kidney
- metabolic waste products
- metabolise
- toxic
- urea
- ureter
- urethra
- urinate

ACTIVITY Differentiating between excretion and egestion

Do you remember learning about the difference between excretion and egestion? Explain what you understand the difference between these terms are.

1. Egestion is …
2. Excretion is …

Purpose of the excretory system

Our cells use oxygen and nutrients to function and in the process also produce various metabolic waste products including:

- urea: a substance that is formed when protein is broken down in the liver
- carbon dioxide: a by-product of cellular respiration.

The organs of the excretory system are responsible for removing these harmful metabolic waste products from the blood so that they do not build up to high concentrations. But in the process, they have to retain the nutrients and water for the body to function. One of the main functions of the excretory system is to prevent too much or too little water in the body.

Did you know?

The first kidney transplant occurred in 1954.

> **Did you know?**
>
> On average, your kidneys produce 1,5 litres of urine each day.

Components of the urinary system as an example of the excretory system

We already know that the lungs excrete carbon dioxide (CO_2) when you exhale.

Another organ that excretes waste is the skin. When you sweat, your skin excretes excess water, salts and a small percentage of urea. In this section, however, we will focus on the excretory system to remove metabolic waste from our blood in the form of urine.

Organs of a urinary system

To do this, the body uses the urinary system, which consists of four main parts.

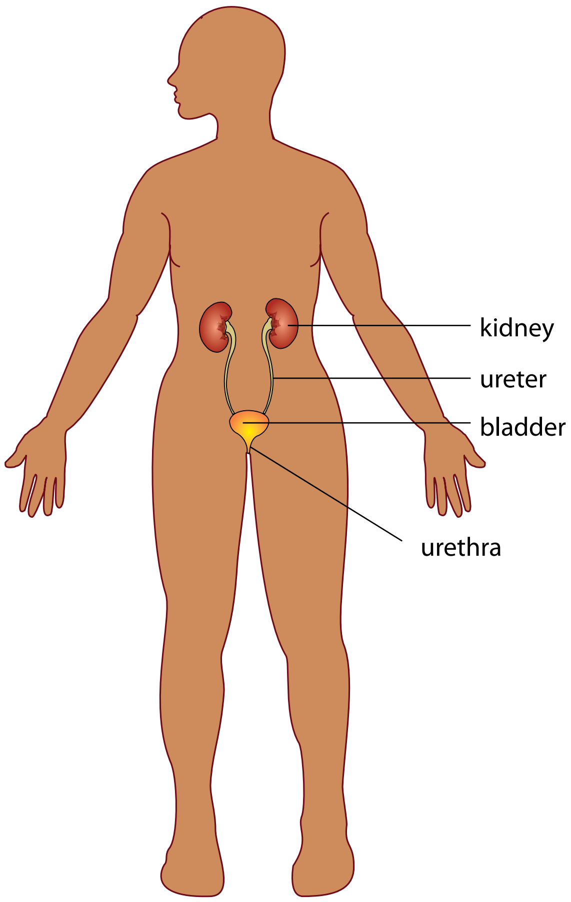


Figure 2.9: The excretory system is responsible for removing metabolic waste products from the blood.

> **Did you know?**
>
> Your kidneys filter about 125 ml of blood every minute! Since you have about 7 to 8 litres of blood in your body, all your blood gets filtered 20 to 25 times per day through your kidneys!

Kidneys

The kidneys filter all the blood in your body to remove urea from the blood. You have two kidneys, each about the size of your fist and bean-shaped. Your kidneys produce urine which is a combination of excess water and waste products.

Ureters

There are two ureters (thin tubes) which connect each kidney with the bladder and carry the urine from the kidney to the bladder.

Bladder

The bladder is a balloon-like organ that stores the urine before excreting it during urination.

Urethra

The urethra is a tube that connects the bladder to the outside of the human body through which the urine is excreted.

Main processes in the excretory system

There are four main processes discussed below.

Filtration

All the blood in the body passes through the kidneys as part of the circulatory system. The kidneys filter the blood to remove unwanted minerals and urea, and also excess water. Some water is removed so that the metabolic waste products can be excreted in solution in the liquid urine.

Absorption

Once the blood is filtered by the kidneys, the substances that the body needs are re-absorbed back into the blood so that they are not lost in the urine.

Diffusion

The substances are transported into and out of the specialised cells of the kidney through the process of diffusion.

Excretion

The kidneys funnel the liquid urine through the ureters to the bladder where it is stored. When the bladder has filled up, it uses muscles to force the urine out of the body through the urethra. This is called excretion.

Keywords

- antibiotic
- infection
- filtration
- absorption
- excretion

Health issues involving the excretory system

Common diseases of the excretory system include:

Kidney failure

When this happens the kidney loses its ability to properly filter and remove metabolic waste which allows this waste to build up in the body. This is very harmful and may be fatal. In such cases the patient needs to undergo very regular kidney dialysis. Dialysis involves using a machine which filters the blood for the patient to remove waste products.

Figure 2.10: A patient is receiving dialysis to filter his blood because the kidneys are not working as they should.

Kidney stones

Kidney stones form when fluid intake is too low, resulting in the concentration of solutes (salts and minerals) in the kidney becoming too high. This can result in a small crystal (stone) forming. The kidney stone may stay in the kidney or move down the ureter to be excreted in the urine. A larger stone may, however, cause severe pain along the urinary

Figure 2.11: A kidney stone, which is about 4,5 mm in diameter

tract and may even get stuck, blocking the flow of urine and causing severe pain or bleeding.

Bladder infection

This is one of the most common infections in women but is quite rare in men. Bacteria can enter the bladder and cause an infection. This causes swelling and pain when urinating.

2.6 The nervous system

Purpose of the nervous system

Our nervous system is a complex network that transmits nerve impulses between different parts of the body. The nerves in our body receive stimuli from inside the body or from the environment (from the ears, eyes, skin or tongue, for instance). These are turned into impulses, which travel to the brain and spinal cord.

Keywords

- auditory
- brain
- conduct
- degenerative
- impulse
- nerve
- neuron
- optic
- stimulus
- transmit
- vision

Components of the nervous system

The nervous system consists of two parts, the central nervous system and the peripheral nervous system.

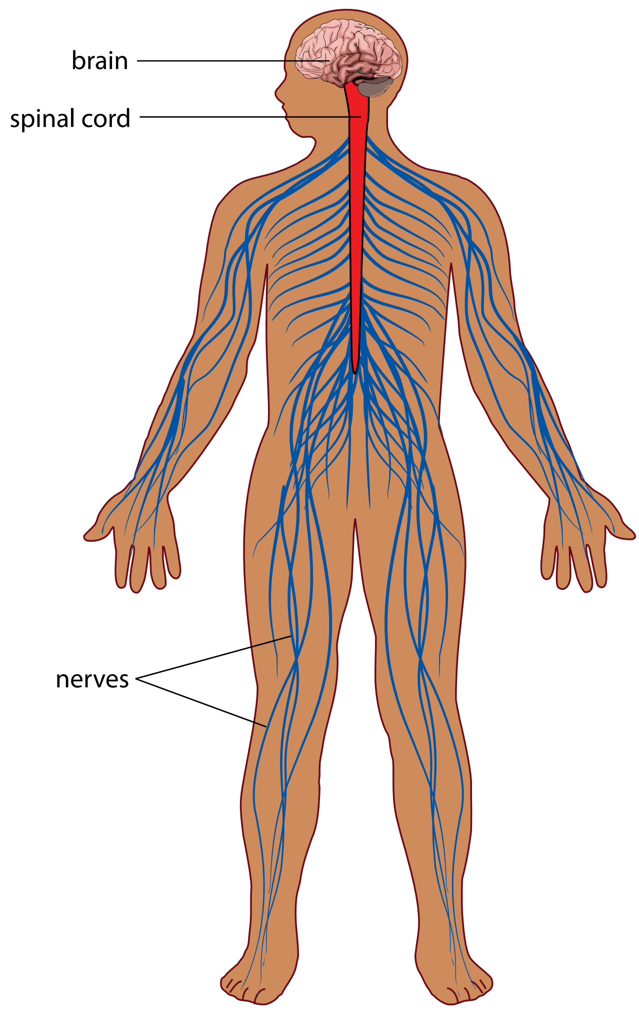


Figure 2.12: Components of the nervous system

Nerves

Nerves are the long fibres which transmit messages from the brain and spinal cord to the rest of the body and back. Each nerve is actually an enclosed bundle of nerve cells, called neurons. The nerves work together to carry messages throughout the body. They make up the nerve tissue in the nervous system.

Take note

'Stimuli' is the plural form of the word 'stimulus'.

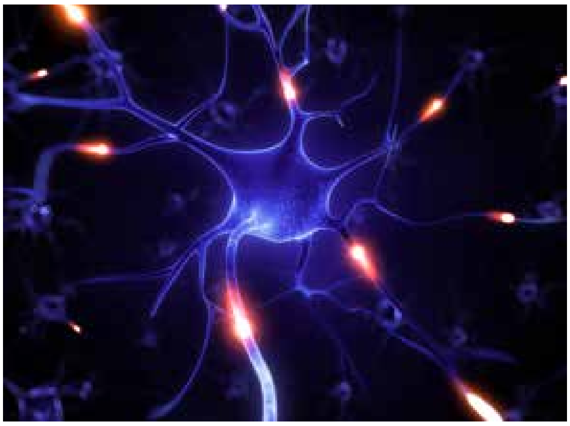


Figure 2.13: This fluorescent image shows nerve cells from a rat brain, which were grown in the laboratory.

Brain

Your brain is located inside your skull. The brain is part of your central nervous system and sends messages to the rest of your body. There are different areas in the brain that have different functions. All these different areas also communicate with each other.

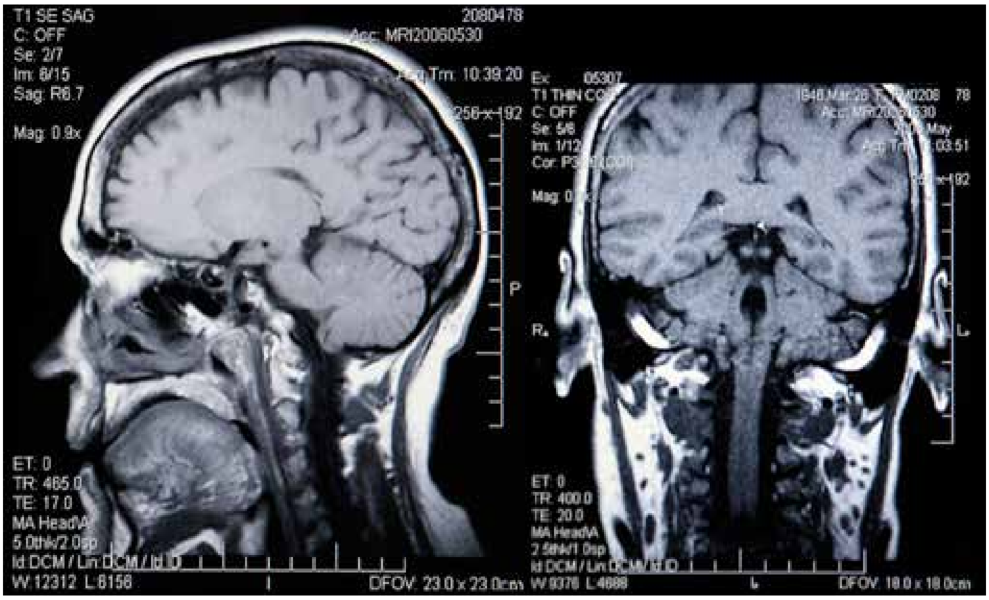


Figure 2.14: A MRI (magnetic resonance imaging) scan of a person's head showing the position of the brain within the skull.

Spinal cord

The spinal cord runs from the brain through your spine, protected by your vertebral column. The spinal cord is a bundle of nervous tissue and other support cells. The brain and the spinal chord are the main organs of the central nervous system.

Take note

'Peripheral' means on the outside. So the peripheral nervous system is outside the central nervous system.

Did you know?

The amount of electricity used by the brain to send and receive messages can power the light in your refrigerator!

Sensory organs

The second part of the nervous system within our bodies is the peripheral nervous system.

The peripheral nervous system connects the central nervous system to the muscles and organs. Various sensory organs are responsible for collecting information and sending it via sensory nerves to the central nervous system.

Our sensory organs are our:
- ears
- nose
- eyes
- skin
- tongue.

Main processes in the nervous system

The nervous system is responsible for the following processes. These are discussed next.

Sending and receiving impulses: Nerve cells in the brain send and receive multiple messages from multiple sources at any given moment. These are transmitted as electrical impulses. The central nervous system interprets these signals and this is how we sense the world around us.

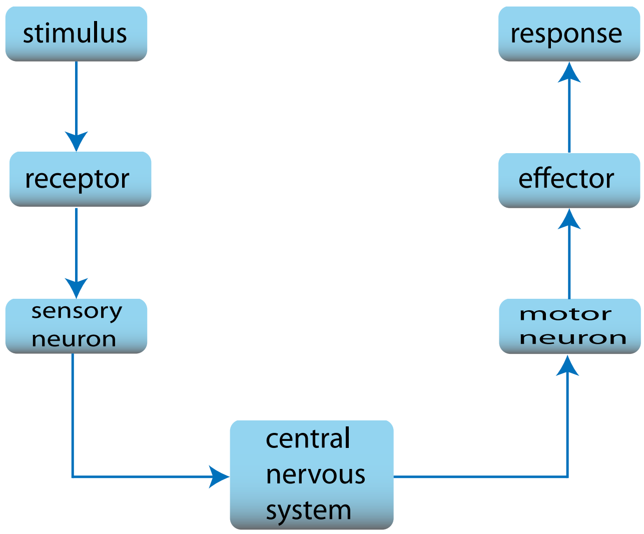


Figure 2.15: Sending and receiving impulses.

Hearing

In the ear, sound waves are transformed into electrical signals that travel along the auditory nerve to the brain. This allows us to understand what we are hearing.

Seeing

Seeing and understanding what you see are complex processes.

Light enters your eye and stimulates specialised cells within your eye. These cells transmit signals to the brain along the optic nerve, where they are interpreted as sight.

Feeling

The skin allows us to feel and experience the world around us through touch. Millions of nerve endings in the skin, called receptors, cover the skin, muscles, bones and joints, as well as internal organs and the circulatory system. These receptors respond to pressure, pain, temperature and movement.

Tasting

Taste buds in your tongue and parts of your mouth can distinguish between the different flavours: sweet, sour, bitter, salty. These receptors work very closely with the receptors in the nose. Together the taste and odour of food is sent to the brain where it is processed and interpreted.

Smelling

Nerve cells in the lining of your nose respond to molecules in the air. They send messages to the brain which interprets the smell accordingly and recognises any one of about 10 000 different smells!

Regulating temperature

An important part of the nervous system is to maintain a balance within the human body. This includes regulating our body temperature. Our bodies need to be kept at about 37 °C to work effectively. If the body is too hot the brain may try to cool the body through increased sweating. If you are very cold, your body will start to shiver to generate heat energy. These responses to changes in body temperature are controlled by your nervous system.

Did you know?

Nerve impulses travel to and from the brain at 274 km per hour.

Did you know?

Your brain makes up less than 2% of your total body weight, yet it uses 20% of your body's energy!

Health issues involving the nervous system

Trauma and injuries to brain and spinal cord

Any damage to the brain or spinal cord can have devastating effects on the human body. For example, people who break their necks in an accident, often damage their spinal cord. This prevents the brain from sending and receiving messages to the body and the person can become paralysed.

Stroke

If blood flow to the brain is stopped, brain cells begin to die, even after just a few minutes without blood or oxygen. This can lead to a stroke, where a part of the brain function is lost.

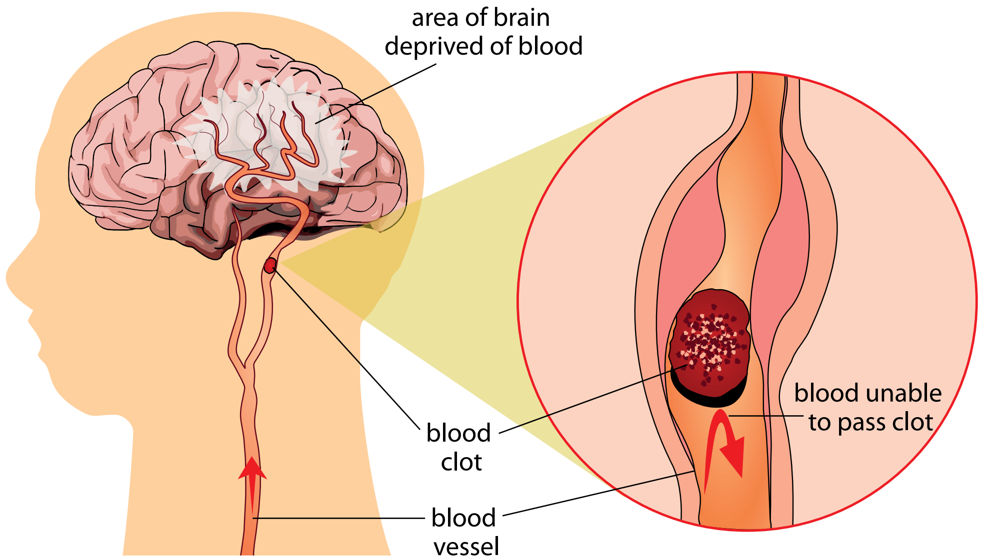


Figure 2.16: When the brain is deprived of blood and oxygen due to a blood clot, the person may suffer from a stroke.

Degenerative disorders

There are several problems associated with the nervous system that cause a gradual loss of function over time (degenerative). These conditions include Alzheimer's Disease, Parkinson's Disease and Multiple Sclerosis.

Mental health problems

Examples include depression, anxiety disorder and personality disorders.

Sensory organ problems

We have discussed the various sensory organs that are associated with the nervous system. These organs can also have problems, such as:

- deafness
- blindness
- short sightedness.

Effects of drug and alcohol on the brain: Different types of drugs target different areas in the brain and it is mostly the brain's reaction(s) that make people want to take drugs and/or alcohol.

Alcohol and drug abuse can cause irreversible brain damage, a loss of memory, decreased learning capability, an increased risk of strokes and heart attacks, and a variety of emotional and mental health problems.

Keywords

- ejaculation
- fertilisation
- ovary
- ovulation
- ovum
- penis
- puberty
- reproduction
- scrotum
- sperm
- testes
- uterus
- vagina
- womb

2.7 The reproductive system

Purpose of the reproductive system

In humans, as in other eukaryotic organisms, the main purpose of the reproductive system is to produce sex cells to ensure the continuation of the species.

Components of the reproductive system

We will be looking at the reproductive organs in more detail in the next unit.

For now, let's get an overview of the main components in the reproductive system.

The male and female reproductive organs differ:

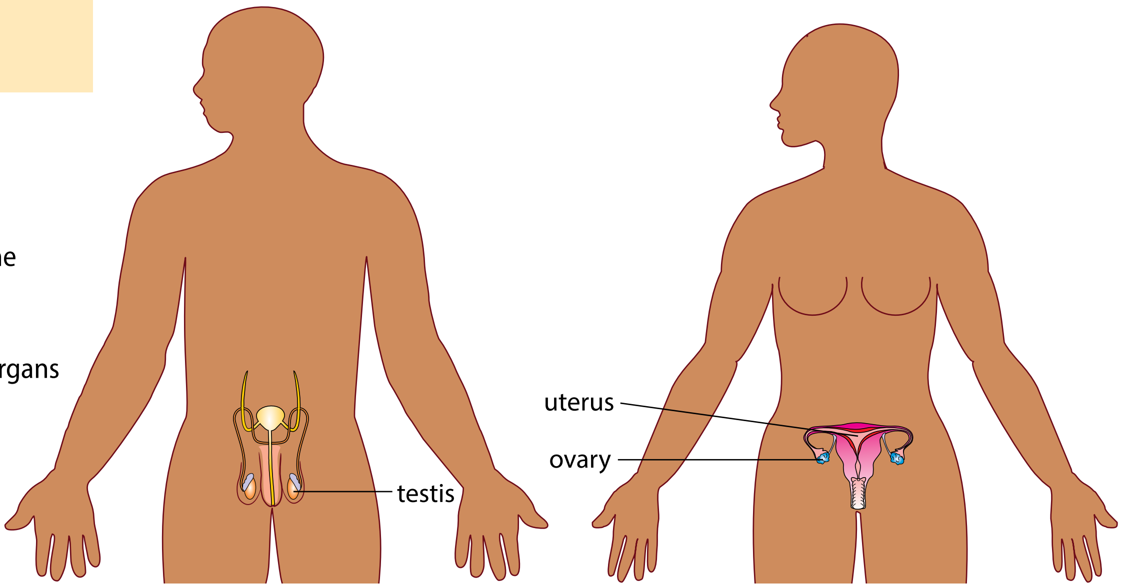


Figure 2.17: The male (left) and female (right) reproductive organs

Ovaries

The ovaries are located inside the female's body in the lower abdomen and produce mature egg cells (ova).

Uterus

The uterus (also known as the womb) has a thick lining and muscular wall. This is where a fertilised egg will implant and develop during pregnancy.

Testes

The sex organs in males are located in the scrotum, a pouch of skin that hangs between the legs. During puberty the testes start to produce sperm cells.

> **Did you know?**
>
> The largest cell in the human body is the ovum (egg cell) and the smallest cell is the sperm cell

Main processes in the reproductive system

During sexual reproduction, the egg and sperm have to combine to form a new individual. Let's do an activity to find out about the main processes in the reproductive system.

ACTIVITY Defining the main processes involved in reproduction

Instructions

- Below is a list of the main processes involved in the reproductive system.
- Look up each term, either in your dictionary or on the internet, and write a brief description in your exercise books.
- The first three have been done for you.

Growth

Growth is the increase in size and mass of an organism as it develops over time.

Cell division

Cell division is the process when a parent cell divides into two daughter cells. In the reproductive system, cell division occurs within the ovaries and testes to produce gametes (sperm and egg cells).

Maturation

Maturation is the process of becoming mature. In humans, this refers to puberty where sexual organs mature so that they are able to reproduce.

1. Copulation
2. Ejaculation
3. Ovulation
4. Menstruation
5. Fertilisation
6. Implantation

> **Did you know?**
>
> Some of the strongest muscles in the female body are found in the uterus! Can you think of reasons why this is?

Health issues involving the reproductive system

Infertility

About 10% of heterosexual couples have problems falling pregnant and may even be completely unable to reproduce sexually. This is known as infertility, and it affects both men and women.

Foetal Alcohol Syndrome

When a pregnant mother drinks alcohol during her pregnancy, the alcohol may cause serious birth defects in the unborn baby. This will affect the child throughout their entire life and in most cases cannot be reversed.

Sexually Transmitted Diseases (STDs)

Many life-threatening diseases such as HIV/AIDS, syphilis and gonorrhoea can be transferred during sexual intercourse.

Summary

Key concepts

- There are many complex systems functioning in our bodies.
- Each system has very specific organs and tissues that are key components in making the system function optimally.
- Different processes take place that are dependent on the key components in each system.
- There are various health issues that affect the systems of the body and that can often be prevented with a healthy lifestyle and wise (informed) life choices.

Concept map

Revision

1. What does digestion mean? [4]
2. List the four main processes involved in the digestive system. [4]
3. Describe the different components of the digestive system and their function. [15]
4. Diarrhoea is can be very dangerous in babies.
   a) Why do you think this is so? [1]
   b) How can diarrhoea be prevented? [3]
5. Distinguish between inhalation and exhalation. [2]
6. Is carbon dioxide in your body excreted or egested? Explain why you say so. [3]
7. Draw a simple diagram to show how blood is circulated around the body in a closed system. [10]
8. What is the difference between breathing and respiration? [5]
9. Give two parts of your musculoskeletal system you use when you have to climb stairs. [3]
10. What are the functions of the bones in the skeleton? [2]
11. Drugs and alcohol have various negative effects on the body. List at least three. [3]
12. Explain why it is so dangerous for a pregnant woman to drink alcohol during pregnancy. [2]

Total [71 marks]

3 Human reproduction

Key questions

- What is puberty, and what does it mean when we 'reach puberty'?
- Why do we all go through puberty at different times and rates?
- What changes take place inside our bodies during puberty?
- What do our reproductive organs look like when they are mature?
- How does reproduction occur?
- What is menstruation and why does it occur once a month?
- How does a baby grow inside a woman's uterus?
- Are there ways to prevent pregnancy and the transmission of STDs?

At this stage in your life, your body is probably going through all sorts of changes as it grows, develops and matures. In this unit we will learn more about these changes and why they occur.

3.1 Purpose and puberty

The purpose of reproduction

You have previously learnt that reproduction is one of the seven life processes, and like all organisms, humans need to reproduce to ensure the survival of the species.

ACTIVITY Reflecting on population growth

Have a look at the website link provided in the visit box about our 'Breathing Earth'. This will give you an idea about how our population is growing.

In 2011 the world's population grew to 7 billion people, one billion more than in 1999. Medical advances and increases in agricultural production (food) allow more and more people to live longer lives. In ancient times, countries such as India, Rome and Greece saw a large population as a source of power. The Romans even made laws about how many babies a couple could have and punished those who did not follow the rules. Yet Confucius (551–478 BC) thought that too many people was a problem, as there wouldn't be enough food to feed everyone, leading to war and famine and various other problems. Today in China this philosophy still applies and couples are allowed only one baby and are heavily taxed if they have more than one.

South Africa's population grew by 15,5%, or almost 7-million people in the space of 10 years to reach a total of 51,7-million in 2011. This is according to the country's latest national census, which took place in 2011. The previous census took place in 2001.

Keywords

- birth control
- conception
- fertilisation
- implantation
- population growth rate
- puberty
- sexual intercourse
- sperm

Questions

1. List any possible reasons why you think South Africa would want to have a large population.
2. What are some advantages and disadvantages to the country in which the number of children per couple is limited so that the population growth is limited?
3. Predict what possible long-term problems may arise if the population in South Africa continues to grow at the fast rate at which it is currently growing.
4. Have a look at the following diagram which shows the percentage growth of a country's population in a year. The different colours give an indication of the growth rate, as shown in the key. For example, countries which are colour coded yellow have an annual growth rate of 3%. This means their population increases by 3% each year. Answer the questions which follow.

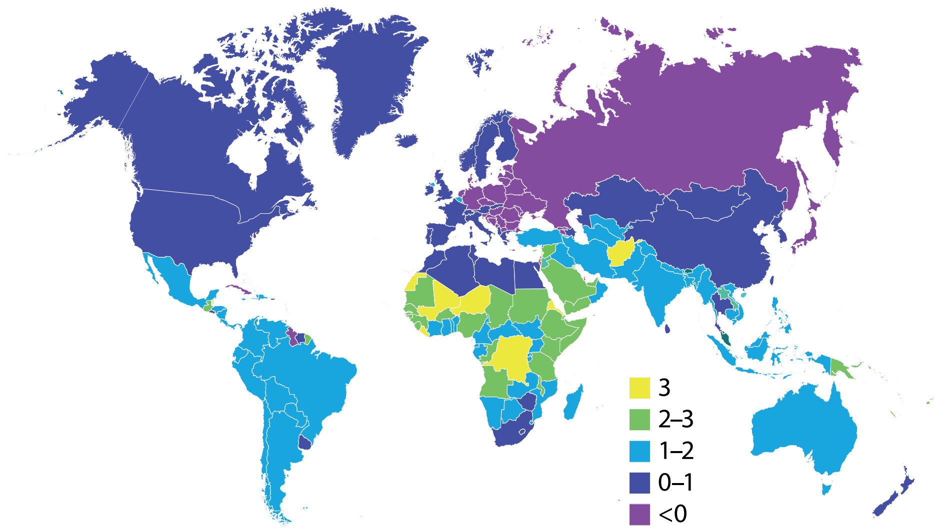


Figure 3.1: Percentage growth rate of each country.

5. Which continent would you say has the largest percentage growth rate each year? Justify your answer.
6. Many countries in Europe are coloured light purple in the diagram. What does this mean?
7. Various population control methods are put in place around the world: contraceptives to stop women from falling pregnant, abortion clinics, large tax incentives to convince people not to want more children, and others.
8. What is your opinion about population control methods, and do you think they should be allowed in modern society?

Keywords

- hormone
- menstrual cycle
- oestrogen
- penis
- testes
- testosterone
- uterus

What is the purpose of puberty?

The human body is geared towards reproduction to ensure the survival of the species. Men have to produce sperm and ensure that they come into contact with a female egg cell. Women have to produce (and store) egg cells that can be fertilised by a male sperm cell.

Children's bodies and sexual organs are not mature and cannot yet perform the reproductive function. Puberty is the time when a child's body develops and changes. The sexual organs mature to enable the body to produce sex cells. These sex cells are called gametes.

How does puberty just 'start'?

- Puberty is the stage in the life cycle of humans when we become capable of sexual reproduction. Girls and boys do not, generally, go through puberty at exactly the same time. So how does puberty 'start'?
- Many of the complex actions that take place in our bodies are controlled by chemical messengers called hormones. Hormones are produced by different glands in our bodies. The pituitary gland is an important gland which controls most of the body's hormones and hormonal activities. It is about the size of a pea and located at the base of the braiån.
- Puberty is brought on when the pituitary gland releases specific hormones into the bloodstream. These hormones then travel to the immature sex organs and signal the hormones in these to be released.
- In girls, the ovaries are stimulated by hormones released by the pituitary gland to release the hormone oestrogen. In males, the testes are stimulated to release the hormone testosterone. These hormones initiate all the bodily changes that you experience during puberty.

What changes during puberty?

The main purpose of puberty is for the sexual organs to mature. However, the hormones which are released from the reproductive organs also start a number of other changes in the human body. We call these secondary sexual characteristics.

Puberty brings about the following secondary changes in females:

- **Breasts** start to develop that may be used for breastfeeding a baby after childbirth.
- **Pubic hair** starts to grow at the onset of puberty. Underarm hair also starts to grow.
- **Menstruation** occurs in girls in a monthly cycle once they reach puberty.
- **Body shape** also changes due to the rising levels of oestrogen in the body.
- **Body odour and acne** develop as more oil is secreted and the smell of sweat in the body changes.

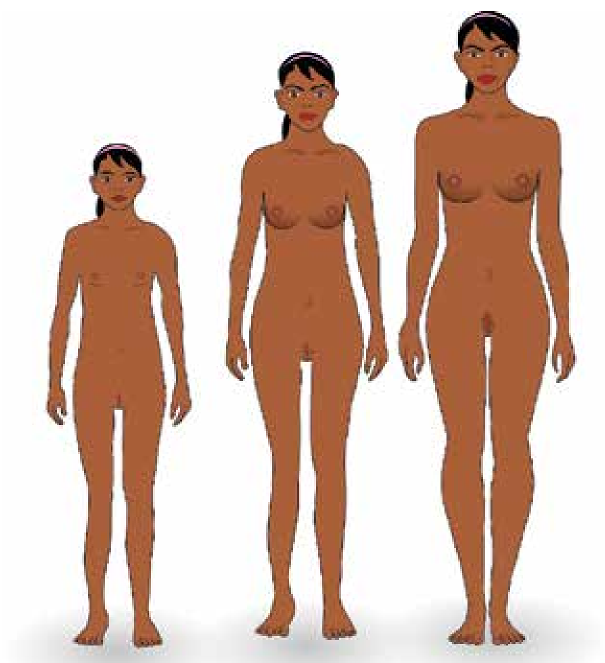


Figure 3.2: Puberty changes in females

At the start of puberty boys are, on average, 2 cm shorter than girls, yet adult men are approximately 13 cm taller than adult women. Puberty brings about the following secondary changes in males' bodies:

Take note

'Testes' is plural and 'testis' is singular.

- Testicle and penis size increases.
- Hair starts to grow on the pubic areas, the limbs, chest and the face.
- Voice becomes deeper as the larynx (voice box in your throat) grows.
- Body shape changes occur as the skeletal muscle and bones increase in size and shape.
- Body odour and acne start to develop, as with females.

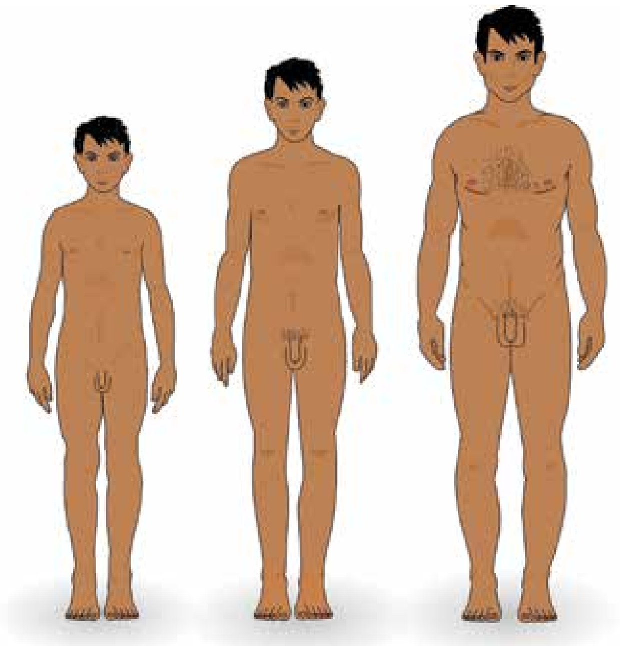


Figure 3.3: Puberty changes in males

Let's take a look at the reproductive organs.

Keywords

- erection
- foreskin
- scrotum
- semen
- sperm duct (*vas deferens*)
- vagina

3.2 Reproductive organs

Let's take a closer look at the male and female reproductive organs to see how they are structured and what functions they perform.

ACTIVITY Identify the role of the male and female bodies in reproduction

In your exercise books, explain what you think the role of the male and female bodies are in reproduction.

1. The male body has to …
2. The female body has to …

Did you know?

Semen contains sperm cells, dissolved nutrients and enzymes that nourish and protect the sperm inside the woman's body. Every millilitre of semen can contain up to 100 million sperm cells!

Male reproductive organs

The male reproductive organs include:

Testes and scrotum

Males are born with their two testes hanging outside their bodies. The testes in young boys do not produce sperm. During puberty the two testes release testosterone which then triggers the production of sperm. The two testes are each contained in a pouch of skin

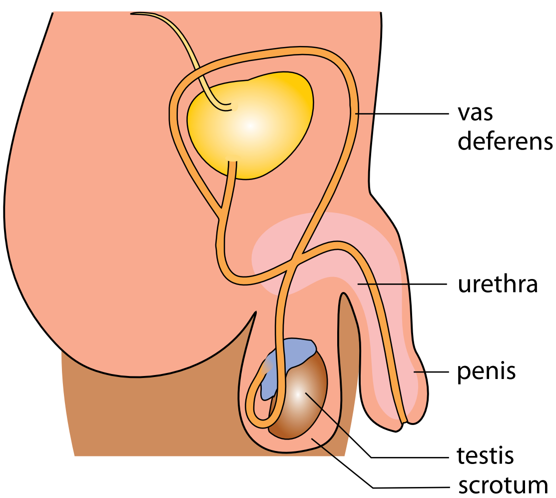


Figure 3.4: The male reproductive organs

called the scrotum. The scrotum ensures that the testes are kept at a constant temperature of 35°C which is the temperature at which sperm is produced.

Sperm duct (*vas deferens*)

Different tubes (ducts) carry the semen from the testes to the penis. The sperm duct carries the sperm from the testes to the urethra in the penis.

The penis

The penis is the external sex organ. The head is often covered by a loose fold of skin called the foreskin. The penis needs to be erect (stiff and hard) to be able to go into the vagina to deliver the sperm to the cervix during ejaculation.

Urethra

The semen moves through the urethra to the outside during ejaculation. The urine passes through the urethra during urination, but the semen and urine do not move through the urethra at the same time.

Take note

Some cultures have the foreskin removed, called circumcision. This is done when the boy is a baby or at puberty.

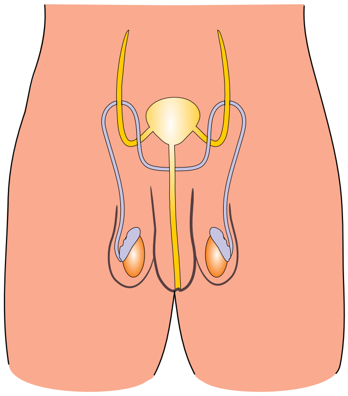


Figure 3.5

ACTIVITY Identify structure and function

1. Study the diagram of the male reproductive system. Label each part using its correct scientific name.
2. In the table, identify the function(s) of the male reproductive organs mentioned.
3. Copy the table below in your exercise books. In the last column, suggest how you think the structure of the organ is adapted to perform the function most effectively.

Reproductive organ	Function	Adaptation
Penis		
Testes and scrotum		

Keywords

- cervix
- Fallopian tube (oviduct)
- ovary
- ovulation
- womb

Female reproductive organs

The female reproductive organs include:

Vagina

The vagina is a tube that connects the uterus with the outside of the body. During intercourse the vagina acts as a canal for the penis to fit into to deliver sperm. Once a month, during menstruation, the menstrual blood leaves the

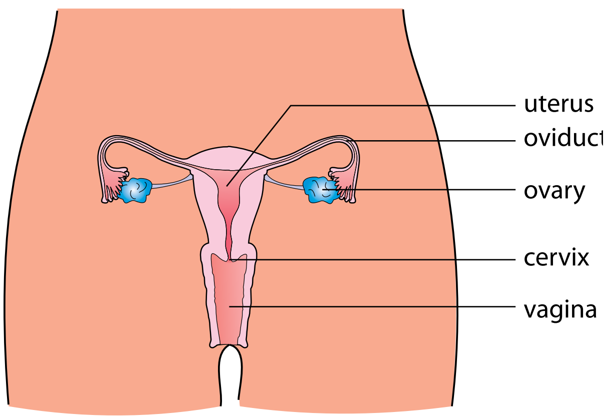


Figure 3.6: The female reproductive organs

Did you know?

Women are born with thousands of immature egg cells in each ovary, but very few ever reach maturity.

body through the vagina. The vagina is also the birth canal during childbirth, when it stretches to allow the baby to pass through.

Uterus

The uterus is hollow, with extremely strong muscular walls that can carry and protect a baby. Two oviducts (Fallopian tubes) at the top of the uterus connect it to the ovaries. The bottom neck of the uterus is called the cervix, which is tightly closed to protect the inside of the uterus.

Take note

The plural of ovum is ova.

Ovaries

There are two ovaries on either side of the uterus. They produce oestrogen and contain the ova. Each month the ovaries take turns to produce a mature ovum. This is called ovulation.

Oviducts (Fallopian tubes)

The uterus and ovaries are connected through a pair of muscular tubes called the oviducts or Fallopian tubes. The mature ovum travels into these tubes to the uterus. Fertilisation occurs in the oviduct.

ACTIVITY Comparing the reproductive organs

1. Explain how the structures of the vagina, cervix and uterus are specially adapted to fulfil their functions.
2. Provide at least 2 reasons why the uterus needs to have strong muscular walls.
3. Compare the position and functions of the ovaries with those of the testes. Create a table to show these differences.

Keywords

- contraception
- contraction
- embryo
- gestation (pregnancy)
- labour
- menopause
- surrogacy
- umbilical cord
- zygote
- foetus

3.3 Stages of reproduction

We have already mentioned most of the processes that take place during reproduction. These processes occur in stages. Let's first have a look at the female reproductive cycle.

The reproductive cycle

The female reproductive cycle repeats every 28–30 days to release an egg cell to be fertilised if sperm are present. This cycle will repeat for many years from puberty to **menopause** (when the reproductive cycle comes to an end). The processes that occur will differ depending on whether the ovum is fertilised or not. After ovulation, if fertilisation does not occur, the reproductive organs 'reset' through menstruation to start the process again.

Ovulation

Once a month, one of the ovaries releases one mature ovum into the oviduct. This process is called ovulation. At the same time the uterus wall thickens and develops extra blood vessels. This is in preparation for the *possible* implantation of a fertilised egg.

Menstruation

When there is no fertilised egg cell (**zygote**) to implant in the uterus, the thick layer of blood and tissue is no longer needed. It is passed out through the vagina during menstruation. The entire process is called the menstrual cycle and it normally repeats every 28–30 days.

Fertilisation

During sexual intercourse the erect male penis enters the female vagina. This is called **copulation**.

The male penis deposits sperm into the female vagina through **ejaculation**.

There can be millions of sperm cells in one ejaculation, but only one will be able to penetrate the outer membrane of the ovum.

After ejaculation into the vagina, the sperm swim into the cervix and through the uterus to the oviducts. Once inside the oviducts, the sperm swim to meet the mature egg that was released from the ovaries and is now travelling towards the uterus.

One sperm cell burrows into the surface of the ovum. Only the sperm's head enters, the tail stays outside. As soon as one has penetrated the outer layer, the surface of the ovum changes and no more sperm will be allowed to enter.

This process is called **fertilisation** and it takes place in the outer part of the oviduct, and not in the uterus or vagina.

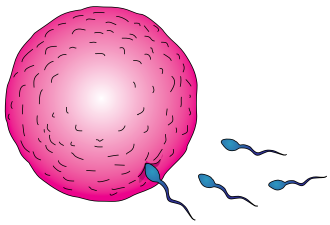


Figure 3.7: Once the sperm on the right has entered the outer layer of the ovum, no more sperm will be able to penetrate.

> **Take note**
>
> There are times in the menstruation cycle near to ovulation when there is a high chance of becoming pregnant. All girls need to be aware of where they are in their own menstrual cycle as each cycle differs slightly.

ACTIVITY Comparing fertilisation and menstruation

Instructions

1. Use the following diagram to compare what happens when an egg is fertilised compared to when it is not fertilised. You can use coloured pens if you have them.
2. Use labels and arrows to illustrate on the left-hand side what happens to the ovum if it is fertilised by a sperm cell.
3. Use arrows and labels to illustrate on the right-hand side what happens if the ovum is not fertilised and the woman subsequently menstruates.

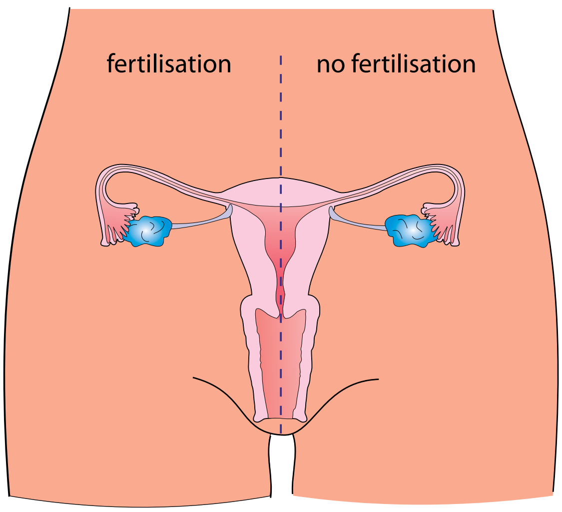


Figure 3.8

ACTIVITY Flow diagram of the pathway of sperm

Instructions

1. Use a flow chart to track the progress of a sperm cell from the testes in the male body to the ovum in the female body.
2. Draw your flow diagram in your exercise books. Remember to draw arrows.
3. Use the following terms in your flow chart, in the correct order.
   a) Cervix
   b) Uterus
   c) Urethra
   d) Penis
   e) Testes
   f) Sperm duct
   g) Oviduct/Fallopian tube
   h) Vagina

Pregnancy and birth

In humans, pregnancy is about 40 weeks (9 months). We call this the gestation period.

Pregnancy begins the moment the female egg cell is fertilised by the male sperm cell. This is then called a zygote. The zygote will then start to divide and grow as it moves down the oviduct. It will then implant in the uterus lining, where it will continue to grow. The fertilised egg is now called an embryo and undergoes cell division over and over again. This forms a cluster of cells with the different cells differentiating to become the specialised cells, tissues and organs that make up the human body.

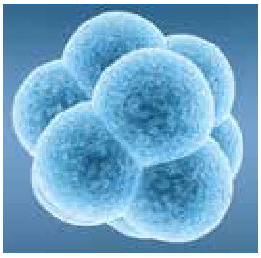


Figure 3.9: An 8-cell human embryo on day 3 after fertilisation.

Where the embryo implants into the spongy, blood-vessel rich lining of the uterus, some of the cluster of cells that formed after fertilisation form the placenta. The placenta is partly formed by the mother and partly by the embryo. The embryo develops an umbilical cord to attach itself to the placenta. The embryo can receive food and oxygen and remove its wastes through the umbilical cord and placenta.

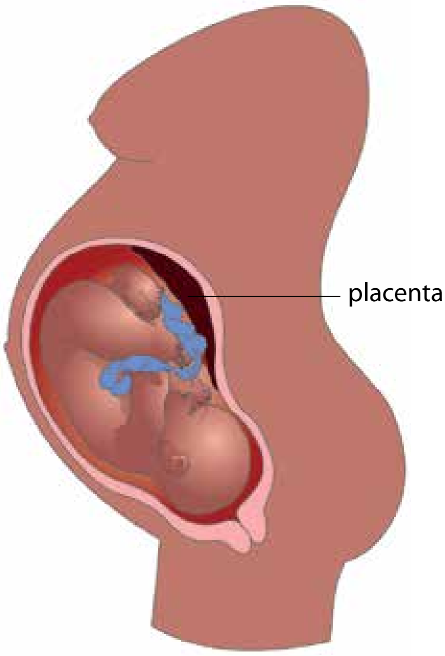


Figure 3.11: The foetus is attached to the placenta by the umbilical cord.

Birth

Towards the end of the pregnancy, the uterus starts to contract. This pushes the head of the foetus into the vagina (birth canal). After the head has appeared the rest of the body comes out quite quickly. The last to come out is the placenta.

Figure 3.10: A newborn baby

ACTIVITY Debate surrogacy

Many couples, for various reasons, are unable to fall pregnant. A surrogate mother can be impregnated with the couple's fertilised embryos and can therefore carry the couple's baby to full term. South African law allows only certain individuals to do this; it is not available to just anybody.

Instructions

1. Work in groups of 6.
2. Debate the issue of surrogacy in your group. Base your debate on the ethical concerns below or any others you may think of.
3. Appoint a spokesperson for the group.
4. Each of the groups' spokesperson must then share their group's points of view with the class.
5. Debate these issues in the class.

There are many ethical issues concerning surrogacy:

- In many cases, the surrogate mother is paid to grow the baby inside her body. Often women who are poor agree to be surrogate mothers and it is suggested that people who pay them to carry their babies are exploiting them. Should women be paid to be pregnant and deliver babies?
- In certain religions, surrogacy (including the donation of sperm and ova) is seen as 'highly immoral' because it involves the intrusion of a third person on a couple's relationship. Should religious institutions be allowed to prevent surrogacy in this way?

Write down some notes on any other points your group discusses.

Influences on the unborn baby

During pregnancy, what the mother eats, drinks and takes into her body has been shown to directly affect the unborn child. Other substances such as cigarettes, alcohol and drugs have a negative influence on an unborn baby.

The placenta transports nutrients and oxygen to the foetus, and removes metabolic waste products and carbon dioxide. However, it cannot differentiate between nutrients and harmful products, such as nicotine, alcohol or drugs. If the mother uses these substances during pregnancy, they will most likely pass through the placenta to the foetus causing great harm to the unborn child.

Pregnant mothers who drink alcohol during pregnancy may cause irreversible birth defects in their unborn babies. This is called Foetal Alcohol Syndrome (FAS).

Did you know?

Hearing is one of the first senses to develop in a foetus. So it is thought that unborn babies can hear and are affected by sounds while in the womb.

Prevention of pregnancy and contraceptives

Anyone who is sexually active and who wants to prevent an unwanted or unplanned pregnancy can take certain precautions.

There is a range of different contraceptives that can be used to prevent pregnancy. There are four different types of contraceptives:

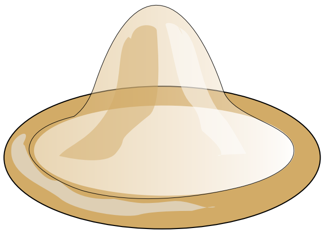
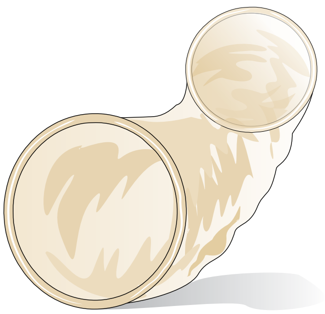
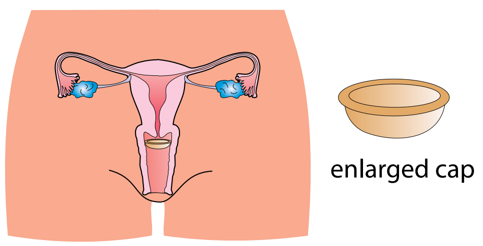
1. **barrier** – physically prevent sperm from reaching uterus
2. **hormonal** – prevent ovulation and fertilisation in the female using hormones
3. **intra-uterine devices** – prevent the embryo from implanting
4. **sterilisation** – by surgery in men and women . It is permanent and not reversible.

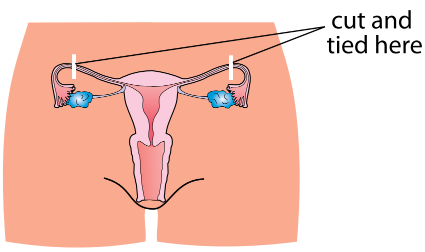
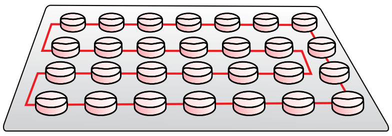
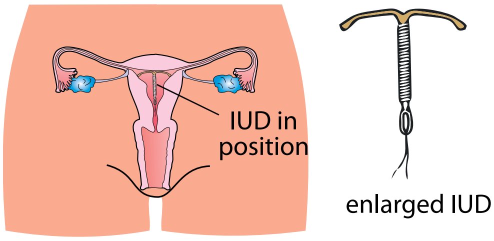
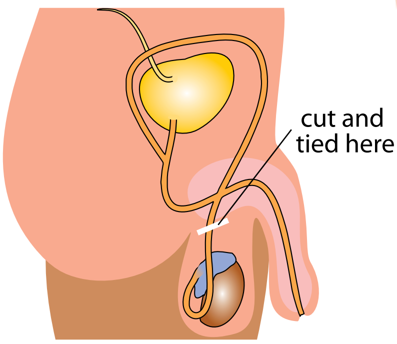
ACTIVITY Describing different contraceptives

Instructions

> **Take note**
>
> 'Contra-' means against, so 'contraception' means against conception.

1. In the following table, several different types of contraceptives have been listed, along with a picture and description.
2. You need to read the information, look at the images and classify the types of contraceptive as one of the following:
   a) barrier
   b) hormonal
   c) intra-uterine device
   d) sterilisation.

Contraceptive	Description	Classification
Male condoms 	These thin sheaths of rubber are placed over the erect penis before inserting it into the vagina. When the male ejaculates the sperm and seminal fluid is caught in the condom and cannot enter the cervix.	
Female condoms 	A female condom is a barrier method that a woman can use to help prevent pregnancy or sexually transmitted infections, including HIV. A thin polyurethane sheath, with two rings. It lines the vagina and the area outside and prevents sperm from entering the woman's vagina during sex.	
Diaphragm 	The diaphragm is a small rubber cap that is placed at the entrance to the uterus before sexual intercourse to create a seal and prevent sperm from entering the uterus	

Contraceptive	Description	Classification
Tubal ligation in women 	A surgical procedure in women in which the oviducts are cut and tied, which prevents mature eggs from reaching the uterus for fertilisation.	
Oral contraceptive pill 	Often referred to as 'the Pill', it is taken every day by mouth. It contains a combination of female hormones which prevents ovulation each month.	
Female intra-uterine device (IUD) 	A small 'T'-shaped device is inserted into the uterus to prevent fertilisation. It is long-acting, reversible contraception as the device may be removed again. It is not suitable for women who have not yet had a baby, and must be inserted by a doctor.	
Vasectomy 	A surgical procedure in males in which the *vas deferens* is cut and tied. Sperm are therefore prevented from becoming part of the ejaculate.	

Sexual intercourse with many different partners is very risky behaviour as there are many diseases that are transmitted through the fluids involved in the sexual act. We call these **Sexually Transmitted Diseases** (STDs). There are many different STDs, for example; HIV/AIDS, Herpes virus, Syphilis, Gonorrhoea and genital warts.

Being faithful to one partner limits your chances of contracting STDs. If you know that your partner has an STD he or she can either get medical treatment for this and/or you can take the necessary precautions to prevent contracting the disease. One of the most popular precautions to prevent the transmission of STDs is for the male partner to wear a condom. However, condoms can break and this can expose you to an STD, so you still have to be careful.

ACTIVITY Forum discussion

Hold a forum discussion regarding the choices women have when they do not want to be pregnant or raise a child. Before the discussion, do research and interviews with your parents, caregivers, or health professionals, or ask your Life Orientation teacher.

How to hold a forum discussion

In a forum discussion, experts are asked to sit on a panel and give their opinion about a particular topic. There are specific roles in a forum discussion:

- **Moderator**: This person keeps the discussion focused and on track.
- **Participants**: The experts. This will be you, the learners, after you have conducted your research.

1. Work in groups of 6.
2. Choose a moderator.
3. Discuss the different choices that women have regarding unwanted pregnancies using the information you obtained from the interviews you conducted.

Rules for a forum discussion

1. The speakers need to take turns to give their opinions.
2. Treat everyone with dignity and respect. Speak politely.
3. Use the correct scientific terminology.

Record your findings

Record the findings from the forum discussion in your exercise books explaining what choices women have when faced with an unwanted pregnancy.

Summary

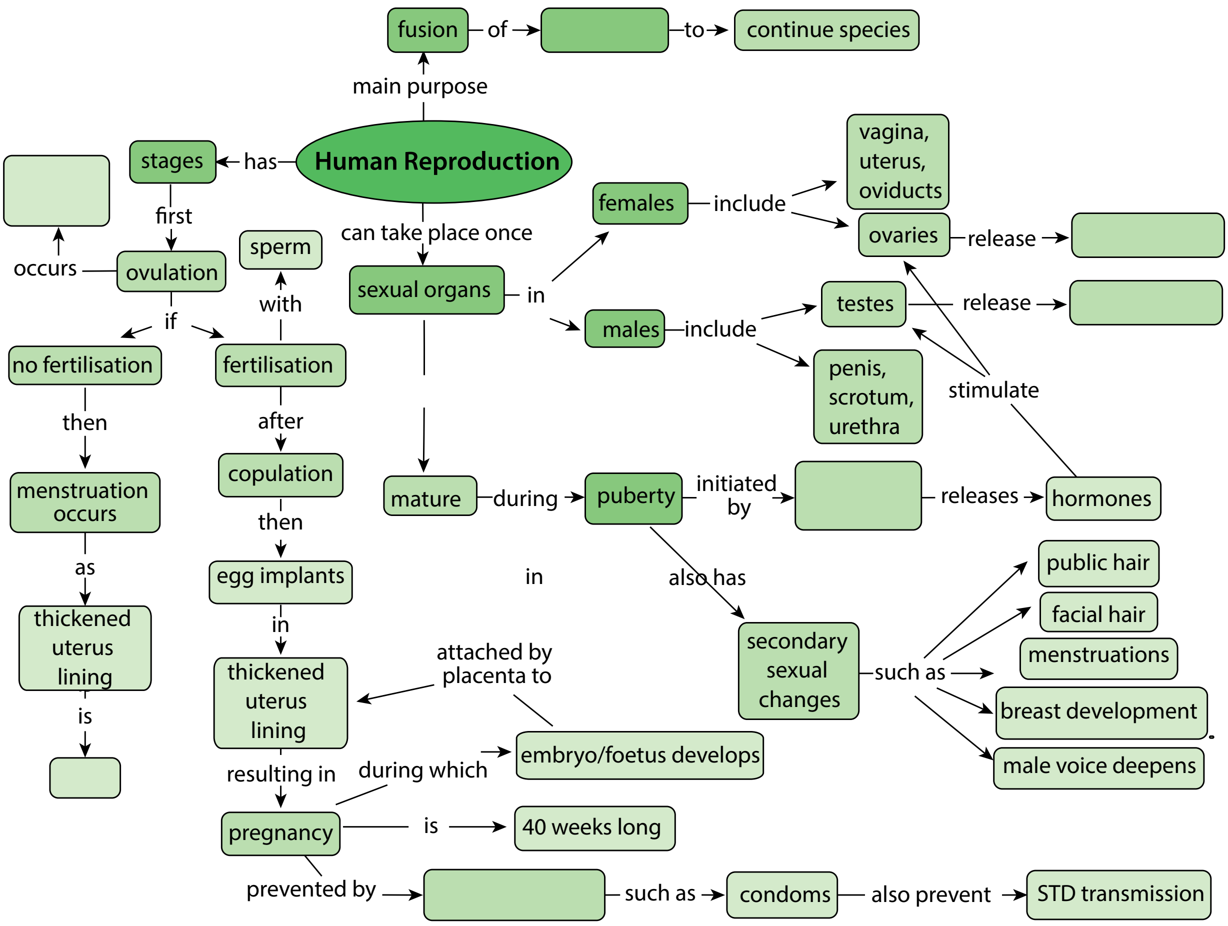
Key concepts

- The main aim of human reproduction is to produce babies to continue the species.
- In human reproduction, two gametes (the sperm and egg cell) fuse during conception to form a zygote that will eventually become a new baby.
- Puberty is the stage in the human life cycle when the sexual organs mature and prepare for reproduction.
- The pituitary gland below the brain releases hormones that stimulate the testes and ovaries to release hormones that will start the production of sperm in the male and the maturation of ova (egg cells) in the female.
  - In males, the hormone testosterone stimulates the testes to produce sperm.
  - In females, the hormone oestrogen stimulates the ovaries to produce mature ova.
- Testosterone and oestrogen cause different secondary changes in the body.
  - Females begin to menstruate, grow breasts and grow pubic and underarm hair, and may experience acne.
  - Males grow hair on the pubic area, on the face, chest and underarms, develop a deep voice and may develop acne.
- The male reproductive organs are: penis, sperm duct (*vas deferens*), testes, scrotum and urethra. Sperm is produced in the testes.
- The female reproductive organs are: vagina, cervix, uterus, oviducts (Fallopian tubes) and the ovaries. Ova are produced in the ovaries.
- Stages in the reproductive cycle include:
  - ovulation
  - copulation
  - fertilisation
  - embryo implants in uterus
  - results in pregnancy
  - gestation lasts 40 weeks
  - childbirth.
- Pregnancy can be prevented by using contraceptives. Condoms prevent the sperm from reaching the ovum and also prevent the spread of STDs.
- Pregnant women have various options if they do not want to keep their babies. Very early in the pregnancy they can undergo an abortion. They may also give the baby up for adoption.

Concept map

This concept map shows all that we have learned about reproduction in humans. Write the words to complete the blank spaces your exercise books. You may find this quite tricky, but you need to learn to 'read' a concept map by constructing sentences. For example, 'Human reproduction can take place once sexual organs are mature. They mature during puberty which is initiated by __________. The ____________ releases hormones which stimulate the ovaries and testes.'

What is this gland which initiates puberty and releases hormones, and which hormones do the ovaries and testes release?



Revision

1. Explain the changes that occur to the male and female body during puberty. [10]
2. Describe the hormonal control of the start of puberty. Name the organs involved and the hormones. [5]
3. At what stage of the reproductive cycle can one say that a woman is pregnant? [1]
4. There is an urban legend or myth that says that a girl cannot fall pregnant the first time she has sexual intercourse. Think carefully about everything you have learnt about conception and fertilisation, and discuss whether this myth is true or false. [2]
5. Explain why you think it is important for someone who considers becoming sexually active to know how reproduction occurs in humans. [1]
6. Imagine someone who has many sexual partners asks you for advice on which contraceptive to use. What advice would you give them? [3]
7. Some people have religious reasons for not using contraceptives. Decide whether you agree with them or not and why. Write a short letter to the editor of local newspaper expressing your concerns about contraceptives from this specific point of view. [6]
8. Do you think schools should teach learners about different contraceptives? Why do you say so? [3]
9. During pregnancy the pregnant mother needs to take care of herself in order to provide a healthy and safe environment for the unborn child. Your local clinic has asked you to produce a brochure that they can display in their waiting room for first-time mothers. Write a detailed list of instructions for a pregnant woman explaining what she needs to do to keep herself and her unborn baby healthy. You can chose how you want to do this - perhaps a list of 'Do's' and 'Dont's', or else provide some headings under which you can list some instructions such as 'Diet', 'Lifestyle', and so on. [6]

Total [37 marks]

4 Circulatory and respiratory systems

Key questions

- Why do we have to breathe?
- Are our lungs like big balloons in our chest, or what do they look like?
- How does the oxygen in the air that we breathe in pass from our lungs into our blood?
- How does blood move around our bodies and get to each cell to deliver oxygen?
- We know that carbon dioxide is produced as a waste product in cellular respiration, so how is it removed from our bodies?
- How are the circulatory and respiratory systems linked?

If we do not get oxygen for a few minutes, humans get permanent brain damage and may die. Cell respiration needs a constant oxygen supply to provide us with enough energy, so we constantly need to breath and keep blood circulation going to deliver this oxygen and remove the carbon dioxide. The respiratory and circulatory systems need to work together. Let's briefly revise the main components involved.

ACTIVITY Main components in the circulatory and respiratory systems

Instructions

1. Study the diagrams below.
2. In your exercise books, write down the different components that form part of the respiratory and circulatory system.

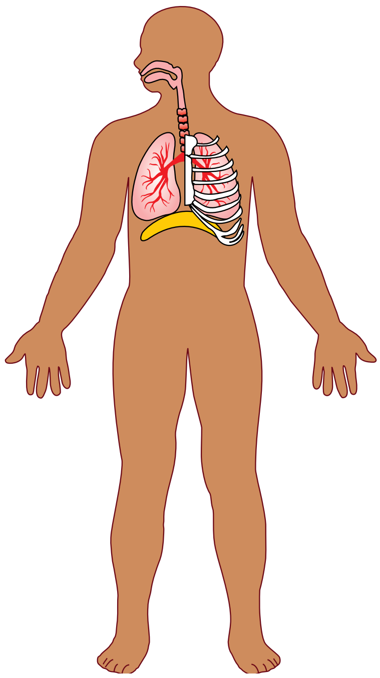


Figure 4.1: The respiratory system.

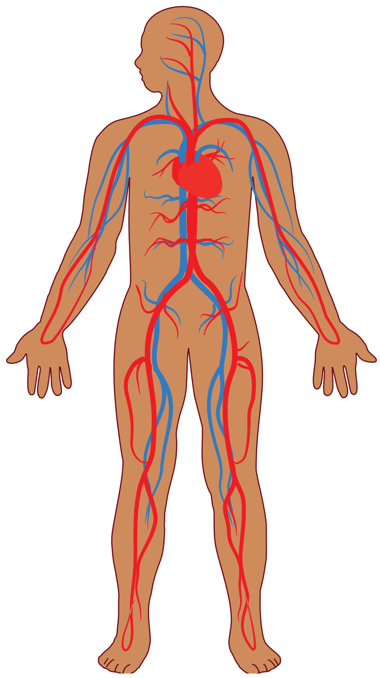


Figure 4.2: The circulatory system.

We will now look at these two systems under the following processes:

- breathing
- gaseous exchange
- circulation and respiration.

4.1 Breathing

We already learnt in Unit 2 that breathing consists of:

1. inhalation
2. exhalation.

When we inhale we take in air with a high concentration of oxygen and when we exhale we breathe out air that has more carbon dioxide in it. These processes take place in a continuous cycle.

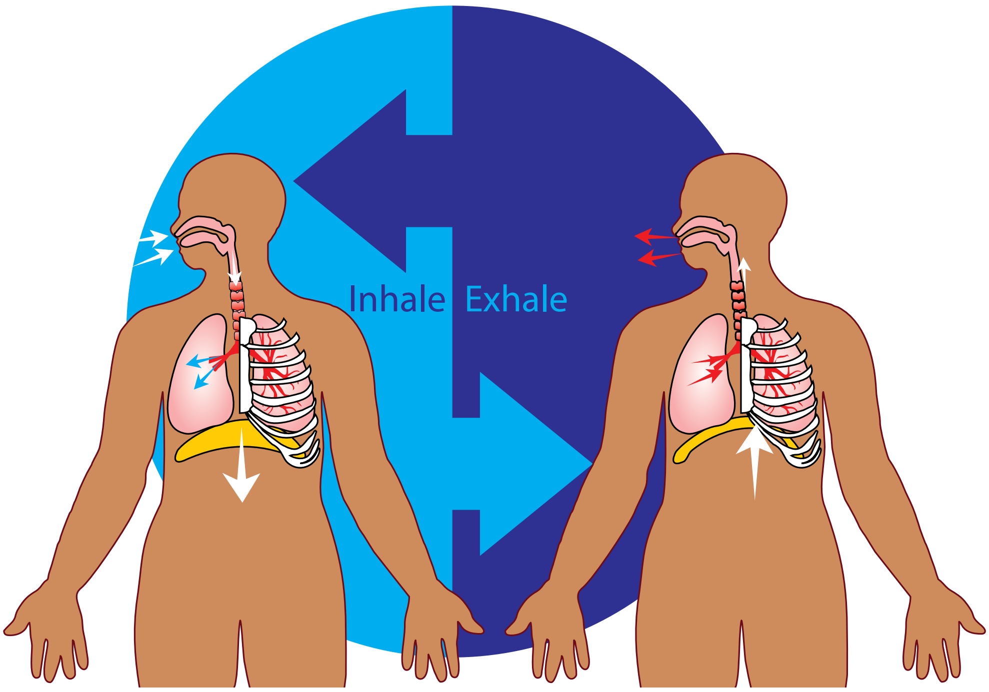


Figure 4.3: Breathing.

During **inhalation** the following takes place:

- The rib cage moves upwards and outwards.
- The diaphragm contracts and flattens causing it to move downwards.
- This causes the chest volume to increase and the pressure decreases.
- As a result the lungs are also forced to become bigger.
- This allows the air to be drawn into the extra space inside the lungs.

During **exhalation** the following takes place:

- The rib cage moves downwards and inwards.
- The diaphragm also relaxes, causing it to become more dome-shaped.
- This causes the chest volume to decrease and the pressure increases.
- As a result the lungs are squeezed smaller.
- This forces the air out of the lungs.

Keywords

- blood
- blood vessels
- bronchi
- bronchioles
- diaphragm
- diffuse
- exhale
- heart
- heart chamber
- inhale
- lungs
- pharynx
- pulse
- respiration
- trachea

Take note

The two tubes that branch from the trachea are called bronchi (plural) or a bronchus (singular).

ACTIVITY Summarise breathing using a flow chart

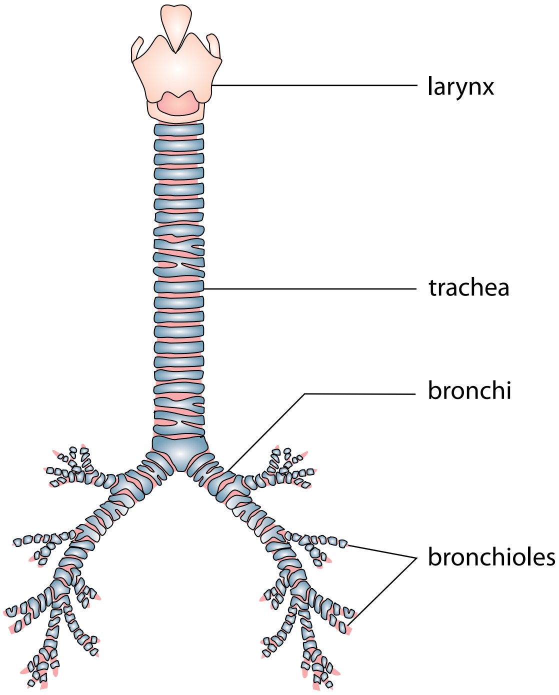


Figure 4.4: This image shows how the larynx joins the trachea, which branches into the bronchi within the lungs.

A flow chart allows us to write short summaries of processes that take place.

When you study for a test or exam you can picture the flow chart in your head, which often helps to trigger memories of what you learnt.

Use a flow chart to show how breathing (inhalation and exhalation) takes place.

You may choose your own design for the flow chart but it needs to show that inhalation and exhalation occur in a cycle.

During inhalation, air travels to the two bronchi – tubes that lead to each lung.

The bronchi are themselves branched (divided) into thousands of tiny bronchioles. During exhalation, the reverse takes place as air leaves the lungs and body.

What happens to the air within the lungs?

Keywords

- alveoli
- capillaries
- cartilage
- diffusion
- haemoglobin
- red blood cells

4.2 Gaseous exchange in the lungs

Gaseous exchange takes place in the lungs and in the cells of the body. The structure of the lung is adapted to fulfil the function of gaseous exchange.

Structure of the lung

Although the lungs inflate during inhalation and deflate during exhalation, they are not hollow. The lungs in a healthy individual are soft, pink and *spongy*.

Take note

The colours of the different bronchioles in the diagram indicate air travelling to different parts of the lungs.

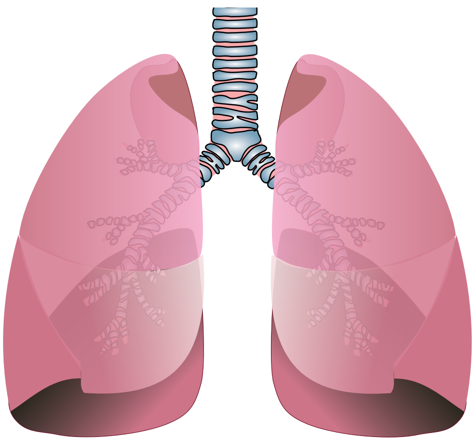


Figure 4.5: External structure of the lungs

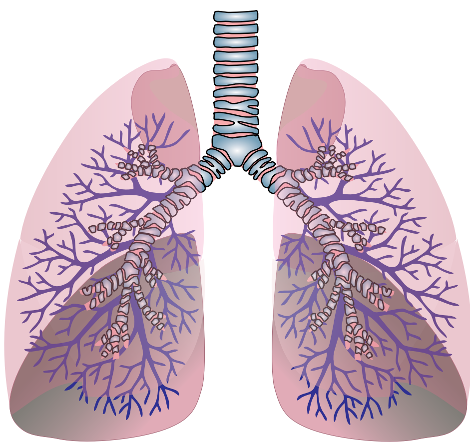


Figure 4.6: Internal structure of the lungs

The alveoli look like small grape-like structures made up of many individual air sacs. A big network of capillaries surrounds each alveolus. Have a look at the following image showing this.

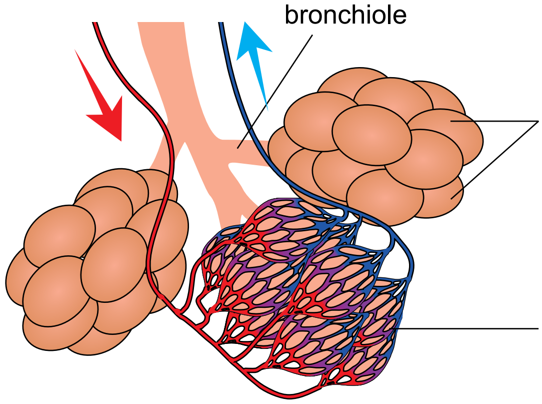


Figure 4.7: Alveoli and surrounding capillaries

An alveolus is one air sac but a group together (plural) is called alveoli.

ACTIVITY Lung dissection

If you are not able to do this in class, you can watch some of the videos showing a lung dissection.

Materials

- lung
- tray
- scalpel
- dissecting scissors
- rubber tubing (for example the Bunsen burner tubing) or hose pipe
- ruler
- beaker of water
- water and soap for washing hands
- disinfectant

Health and safety tips

1. The lungs may carry some bacteria. It is not necessary to wear gloves, as we do not wear gloves when preparing meat in a kitchen, but you must wash your hands thoroughly.
2. Clean all equipment and your work surface with disinfectant after the dissection.
3. Be careful when handling sharp equipment, such as the scalpel.
4. Decide how you are going to dispose of the lungs.

Instructions

Part 1: Preparation

1. Place the lung on the tray on your workbench.
2. Make sure all your dissecting instruments have been disinfected and are sharp. Lay them out next to your tray.
3. Make sure you have access to a first aid kit if necessary.

Part 2: External structure

1. Take note of the external structure of the lung. Look at the general shape, colour and texture.
2. If you have access to a scale, measure the mass of the lung.
3. Use your ruler to measure the length of the lung.
4. Identify the following parts of the lung.
   a) The **trachea** (windpipe) which is the main tube bringing air into and out of the lungs
   b) The **hard rings** in the trachea. What do you think these rings are for?
5. The **bronchi**. There are two bronchi that branch off from the trachea – one to each lung.
6. See if you can identify the first **bronchioles** branching off from the bronchi.
7. Are there any **blood vessels** visible that are attached to your lung? If so, feel these vessels and describe what you feel.
8. Use the rubber tubing or straw or hosepipe and insert this into the tube leading into the lung. Hold the trachea tightly closed around the pipe. Blow on the end of this tube to see if you can inflate the lung. Do not breathe the air back into your own lungs!

Part 3: Internal structure

1. Using the scalpel and dissecting scissors, cut down into the lung.
2. Observe the inner tissue of the lung and think how you would describe it.

Discuss this with your group.

3. Cut out a piece of the lung tissue and feel for tiny bronchioles (they feel like hard little lumps in the soft lung tissue). Place this piece of lung tissue into a beaker of water. Observe the piece of lung tissue. Does it float or sink?

Questions

1. Write a description of the look, feel and colour of the lung you observed. If you were able to measure the mass, write it down, and include the length of the lung in centimeters.
2. What structures made the trachea stay open, but still able to bend?
3. When you cut the lung open, was it like a hollow balloon or bag, or was it spongy inside? What else did you observe when you cut the lung open and observed the inside?
4. When you placed a piece of the lung tissue into water, why do you think it floated?
5. When you blew air into the lung, what did it look and feel like? Did you have to squeeze the lung to force the air out again?
6. In a human, what is responsible for pushing the air out of the lungs?

The process by which gaseous exchange occurs is called **diffusion**.

How does diffusion work?

The movement of particles from an area where there is a high concentration to where there is an area of low concentration is called **diffusion**.

In the lung, each alveolus is surrounded by a network of capillaries. The two gases which diffuse between the alveoli and the blood in the capillaries are oxygen and carbon dioxide.

- Oxygen diffuses into the cells of the alveolus and then into the blood in capillaries.
- Carbon dioxide diffuses out of the blood and into the cells of the alveoli, then into the air.

ACTIVITY Drawing gaseous exchange in the alveoli

Instructions

1. Draw a diagram to show alveoli surrounded by a capillary.
2. On this diagram, name the gases and indicate the direction in which the gases diffuse.
3. Indicate whether the blood is oxygenated or deoxygenated in the capillaries that travel towards and away from the alveolus.
4. Give your diagram a heading.

4.3 Circulation and respiration

Blood is continually circulated to support cell respiration. Let's have a look at how this takes place.

Blood circulation from the lungs to the heart

Keywords

- arteries
- atrium
- blood pressure
- veins
- ventricles

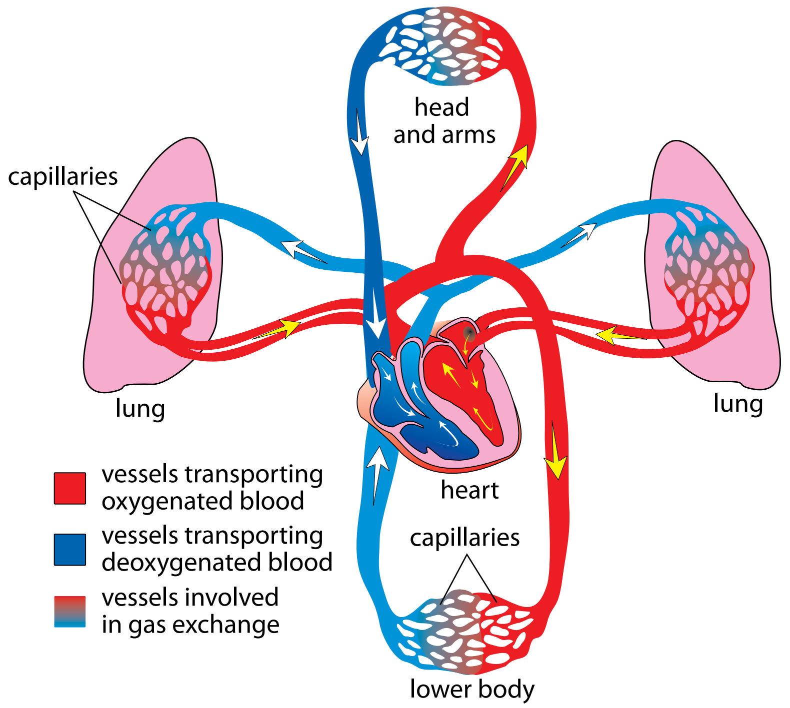


Figure 4.8: Blood circulation from the heart to the lungs

The heart pumps the blood around your body by rhythmic, repeated contractions. This is felt as your **heartbeat**.

The oxygenated blood flows from the lungs to the left side of the heart.

The left side then contracts to pump the blood out of the heart and into the aorta. The aorta is the main artery leaving the heart.

Have a look at the diagram on the previous page, which shows how the blood flows from the lungs to the heart and then to the rest of the body.

Let's take a closer look at the structure of this vital organ in the circulatory system.

ACTIVITY Heart dissection

Materials

- heart (sheep or pig)
- tray
- scalpel
- dissecting scissors
- rubber tubing (for example the Bunsen burner tubing) or straw
- ruler
- beaker of water
- water and soap for washing hands
- disinfectant

Figure 4.9: The heart of a sheep.

Health and safety tips

As with the lung dissection, the same health and safety tips apply to the heart dissection.

Instructions

Part 1: Preparation

1. Place the heart on the tray on your workbench.
2. Make sure all your dissecting instruments have been disinfected and are sharp. Lay them out next to your tray.
3. Make sure you have access to a first aid kit if necessary.

Part 2: External structure

1. Take note of the external structure of the heart. Look at the general shape, colour and texture.
2. If you have access to a scale, measure the mass of the heart.
3. Use your ruler to measure the length of the heart.
4. Identify the following parts of the heart:
   a) There are blood vessels entering and leaving the heart (**arteries and veins**). Arteries have much thicker, more rubbery walls than veins which have thin walls. See if you can identify the difference.

b) Place your fingers inside the blood vessels to feel their texture and strength. Look inside the main arteries and veins as well and describe to your group what you see. Push one finger down the aorta and see if you can feel any structures. The photo on the right shows the aorta opening.

5. Examine the **surface** of the heart for blood vessels. Why do you think the surface of the heart also has blood vessels attached to it?
6. Locate the **atria** and **ventricles**.
7. Locate which is the right- and which is the left-hand side of the heart.

Part 3: Internal structure

1. We are now going to cut into the heart to view the internal structure. Use the following diagrams to help you orientate the heart before cutting.
2. Make a cut down the aorta and then through the left ventricle to the tip of the heart. A tip is to cut through the aorta first using scissors, and then to cut through the left ventricle using the scalpel.
3. Once you have made the cut, pull the ventricle walls apart so that you can view the inside. Can you see the structures at the base of the aorta that you felt in Part 1 (step b)? What do you think these structures do?
4. Look at the following diagram, and make the second cut upwards into the left atrium.
5. Using your ruler, measure the thickness of the left atrium wall and the left ventricle wall. Write these measurements down.
6. You can now cut open the right side of the heart in the same way. Measure the thickness of the right ventricle wall. The

Figure 4.10: Can you see the large opening of the aorta?

Figure 4.11: Take note of the surface of the heart and the blood vessels attached to it.

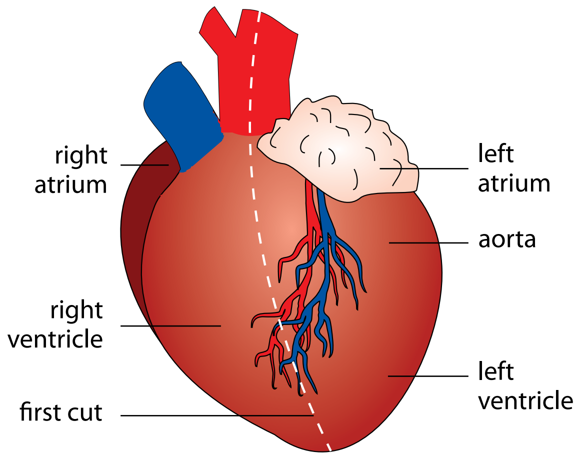


Figure 4.12: Structure of the heart.

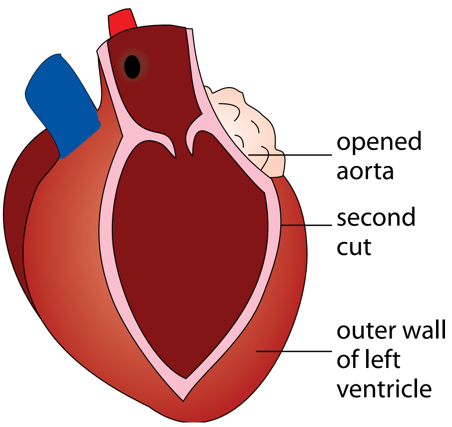


Figure 4.13: The heart showing the cuts.

Did you know?

The average human heart, beating at 72 beats per minute, will beat approximately 2,5 billion times during an average 66 year lifespan.

Take note

Remember that arteries carry blood AWAY FROM the heart and veins carry blood TOWARDS the heart.

following diagram provides a detailed overview of the internal structure of the heart. We have not discussed all of these structures and you are not required to know all of these. However, for this dissection, use this diagram to see how many of these parts you can identify in your dissected heart. If you are able to locate them in the actual heart, draw a ring around the label in the following diagram.

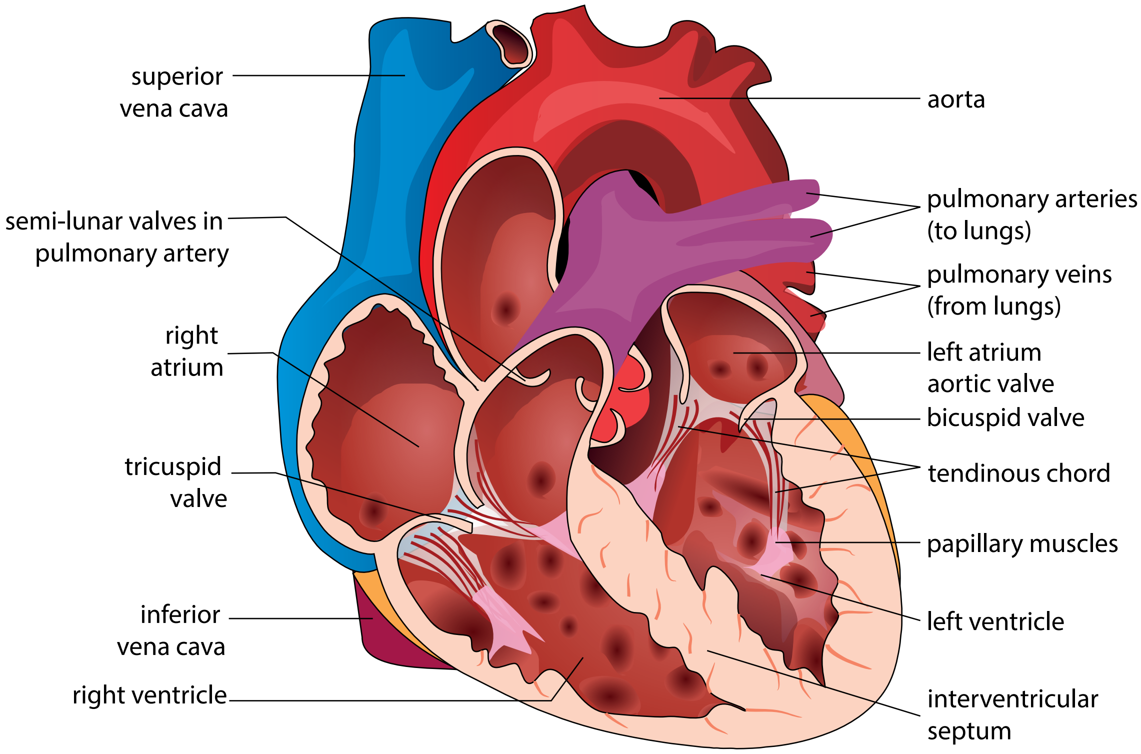


Figure 4.14: Diagram of the internal structure of the heart.

Questions:

1. Write a description of the look, feel and colour of the heart you have observed. If you were able to measure the mass, write this down, and include the length of the heart in centimetres.
2. Write down the thickness that you measured for the left ventricle and atrium walls. Why do you think there is a difference in the thickness of these walls? Hint: Think back to where the atria have to pump the blood and to where the ventricles have to pump blood.
3. Write down the thickness that you measured for the right ventricle wall. Mention possible reasons for the difference in thickness between the left and right ventricle walls. Once again, think about the area to which each ventricle is pumping blood.

Once the blood is pumped out of the heart, it enters the circulatory system in the body.

Blood circulation from the heart to the rest of the body

Once blood leaves the heart in the aorta, this main artery branches into smaller arteries which form a network throughout the body.

ACTIVITY Feel your blood rushing through your body!

Instructions

Figure 4.15: Measuring heart rate in the wrist.

1. Put your index (pointer) and middle fingers against your neck in the hollow between your trachea (windpipe) and the large neck muscles. Use your finger tips as these are more sensitive. You should feel the throbbing of your blood.
2. Can you find your pulse in your wrist?
   Place your middle and index fingers just below the creases in the skin of your wrist - on the side of your thumb.
   Press lightly until you feel the pulse which means the blood is pushing under your skin.
3. You can also try and find your pulse behind your knee, on the inside of your elbow or near the ankle joint.
4. Each throb of your pulse is when your heart pumps the blood from the left side of your heart into the arteries of your body, causing the pressure in the arteries to rise.

Questions

1. Count how many times your heart beats in one minute. Alternatively, count your heart beats for 30 seconds while a friend or your teacher times you, and write the number in your exercise books.
2. Now, calculate your heart rate in beats per minute and write your answer in your exercise books.

Take note

A rate always measures something over time. In this activity we are calculating heart rate as beats per minute, as this is the standard measurement used for heart rate. Can you think of some other units of measurement which indicate a rate?

Arteries subdivide to form capillaries. Capillaries are in close contact with the body cells. Capillaries are much smaller than arteries. They form a fine network throughout the body's cells to make sure that all cells get a supply of blood and oxygen. The capillaries leaving the cells with deoxygenated blood then combine to form veins. Veins from the body carry deoxygenated blood back to the heart.

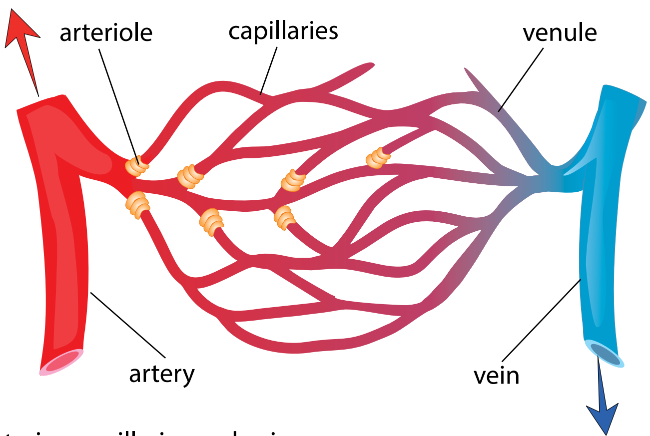


Figure 4.16: Arteries, capillaries and veins

Take note

An exception to the rule of arteries carrying oxygenated blood are the pulmonary arteries, which carry deoxygenated blood away from the heart to the lungs to become oxygenated.

Arteries

- Arteries transport blood **away** from the heart.
- Arteries transport oxygenated blood (except for the pulmonary arteries).
- Arteries need to have strong muscular walls because they carry blood away from the heart under high pressure.

Veins

- Veins transport blood **towards** the heart.
- Veins transport deoxygenated blood (except for the pulmonary veins).
- The blood is flowing back to the heart and therefore the blood pressure in the veins is much lower.

Capillaries

- Capillaries form webs or networks around each cell to ensure that all cells receive nutrients and oxygen.
- Capillaries are much smaller than veins and arteries.

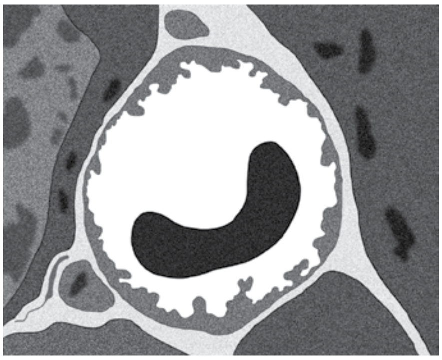


Figure 4.17: This transmission electron micrograph shows a cross-section through a capillary. The semi-circular black structure within the capillary is a red blood cell. This shows how small capillaries are. They are only just wider than a red blood cell.

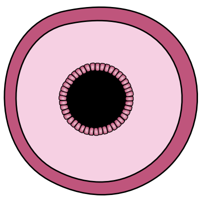
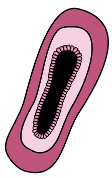
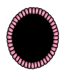
Did you know?

If you put TEN capillaries next to each other, together they would be as thick as only ONE human hair!

ACTIVITY Tabulating differences between the blood vessels

Instructions

1. Compare arteries, veins and capillaries.
2. Use the following table in which to do this comparison.

Blood vessel type	Artery	Vein	Capillary
Image			
Function			
Type of blood			
Exceptions			

Respiration within the cells

Within the cells, the mitochondria use oxygen to respire. This is called cellular respiration.

- The mitochondria combine oxygen with food particles, such as glucose.
- Energy from the food particles is released and can be used by the cell to perform various processes.
- During cellular respiration, carbon dioxide is released as a by-product.

The carbon dioxide diffuses from the cells back into the blood in the capillaries.

This blood therefore becomes deoxygenated as oxygen has been removed and carbon dioxide is added.

Blood circulation from the body back to the heart and lungs

The deoxygenated blood in the body then returns to the right side of the heart through the veins in the circulatory system.

The right side of the heart pumps the deoxygenated blood to the lungs through the pulmonary arteries.

ACTIVITY A circulation simulation!

We are going to create a simulation of our circulation!

Instructions

1. Imagine that you are a red blood cell and you will be carrying oxygen around the body.
2. Your teacher will help your class to lay out the huge body in your exercise books using A4 sheets with labels and hula hoops as in the diagram.
3. There are two colours of paper blocks at each organ or body part and in the lungs. One colour (preferably red) will represent oxygen and the other colour (preferably blue) will represent carbon dioxide.
4. Start off by standing in the lungs and pick up oxygen. You now represent oxygenated blood.
5. Walk to the left side of the heart.
6. The heart now pumps you out to the body in the circulatory system. Leave the left heart hula hoop and walk to the organ or body part you are going to supply with oxygen.
7. When you reach the body part, drop off your oxygen block into the container and now pick up a coloured block representing carbon dioxide. You now represent deoxygenated blood.
8. Walk to the right side of the heart.
9. From here, the heart pumps you to the lungs. Walk to the lungs.

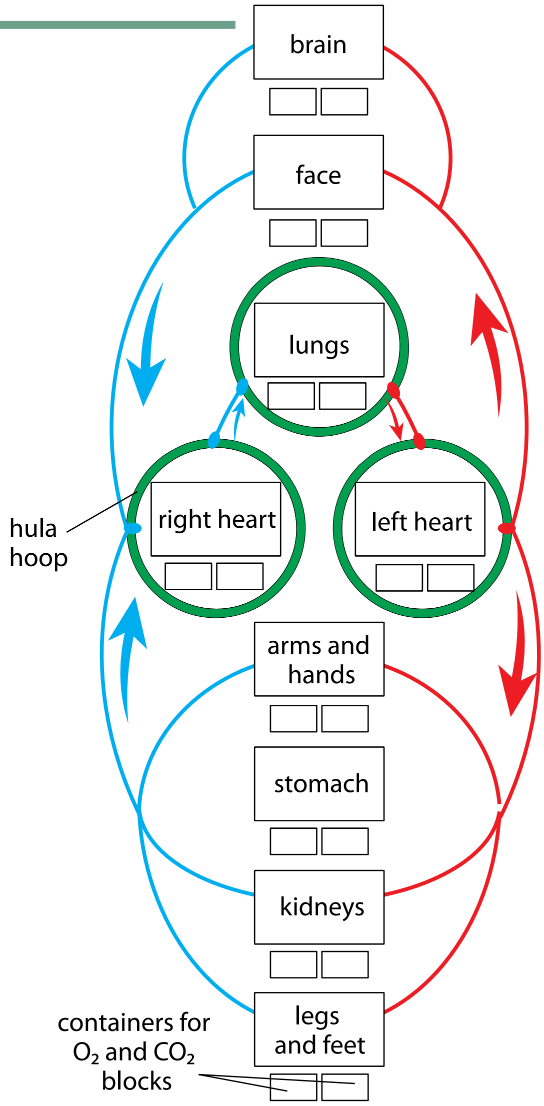


Figure 4.18: A circulation simulation

10. At the lungs, gaseous exchange takes place. Drop off the carbon dioxide you were carrying and pick up oxygen again.
11. You can now repeat the cycle and walk to a different body part.

Heart rate

Your resting heart rate is often used as an indicator of how fit you are or whether there are possible health concerns that you should pay attention to.

ACTIVITY Homework activity to measure your resting heart rate

Instructions

1. Take your resting heart rate first thing when you wake up in the morning. Record how many times per minute your heart beats.
2. Repeat this over 3 days to get an average – this is more reliable than a once-off reading.
3. Copy the table into your exercise books, then record your resting heart rates.

	Heart rate (beats per minute)
Day 1	
Day 2	
Day 3	
Average	

We have now had a look at our heart rate when we are resting. But what happens when we do some kind of physical activity? Will your heart rate increase or decrease? Do you think you could use your heart rate as a measure of how fit you are? Let's investigate!

Investigation Measuring and comparing heart rates before and after exercise

Instructions

Measure the heart rate of at least 10 learners in your class after they have done 2 minutes of skipping or running on the spot. Discuss in your group how you are going to do this and write down your method. Record your measurements and use a graph to display your findings. Make deductions about your class' fitness levels based on their heart rates after completing the graphs and discuss the benefits of exercise for the circulatory and respiratory system (also known as the cardiovascular system).

Aim

What is the aim of your investigation?

Hypothesis

What is your hypothesis for your investigation?

Variables

In any scientific investigation, it is crucial to identify the variables at the start.

- When you do an investigation you are going to change or vary one factor to answer your question. This is called the **independent variable**.
- The factor that you are measuring or observing is the **dependent variable**.
- Normally, you will have a third variable, the **control variable**. These are the factors that you want to keep the same (unvaried) during your test so they cannot affect your results.

What are the variables involved in this investigation?

Materials

Write a list of the materials you will need for this investigation.

Possible materials to be listed are:

- stopwatch
- skipping rope (if learners are to skip, otherwise they may just run on the spot)
- recording sheet and pen.

Method

Write down the method in your exercise books. The steps must be numbered.

Results

1. Design a table that will record the heart rate of the 10 learners when at rest and after 2 minutes of physical activity (skipping or jogging on the spot).
2. Remember to give your table a heading.

Analysis

In order to analyse your results, it is helpful to plot a graph, as this helps you to see the relationship between the dependent and independent variables and to make comparisons. Below is a description of different types of graphs and when they are used.

- **Line graph:** A line graph is used if the data you have is numerical and changes continuously, often over time. A line graph is useful for visualising a trend in the data over time.
- **Bar graph:** A bar graph is used to compare different categories or groups, normally when the categories are words. There are spaces between the bars in a bar graph.
  A *double* bar graph can compare two sets of data. In a double bar graph, two of the bars touch and are shown in different colours, and are separated by a space from the next two bars.

- **Histogram**: A histogram is used when the data for the independent variable is numerical and can be grouped into categories which are continuous. The bars in a histogram touch each other.
- **Pie graphs:** Pie graphs (or sector diagrams) are used to show the relative proportions or percentages of the categories when they make up a whole.

1. Which type of graph will you use to represent the data in this investigation? Give a reason for your answer.
2. How will you differentiate on your graph between the two sets of measurements for each learner (in other words, heart rate before and after exercise)?

Tips for drawing your graph

- Start by giving your graph a title, something that shows which dependent and independent variables you were studying.
- Use the appropriate axes for each variable: x-axis = independent variable (along the bottom of the graph) and y-axis = dependent variable (along the side).
- Label your x-axis and y-axis.
- Use an appropriate scale and use the graph paper that you have been given to draw the graph wisely.

Draw a graph of your results

1. Which learner in your group had the smallest increase in heart rate from before to after physical activity?
2. Which learner in your group had the largest increase in heart rate from before to after physical activity?
3. Rank the learners in your group from the smallest increase to the largest increase.
4. What deductions can you make about the fitness level of the learners in your group based on their heart rates before and after the physical activity? When you make deductions, ask yourself these questions:
   a) What do you see is happening?
   b) What do you notice that is different?
   c) What does this imply?

Discussion and evaluation

An important part of an investigation is to discuss your results and observations and evaluate them. At this point you get to talk about your results and explain them.

You also point out any shortcomings of the investigation. What could you have done to improve the investigation? You can also point out any unexpected results in your investigation and try to explain these using your science background. You should do some background research into the benefits of exercise for the cardiovascular system and write some points in your discussion.

Conclusion

Write a conclusion for your investigation. In a conclusion, you need to refer back to your hypothesis to see whether your results support or reject the hypothesis.

References

If you researched any additional information to support your discussion, you need to reference these sources in the following way:

- **Books:** Surname of author, Name of book, Year published, Name of publisher, Page numbers you used.
- **Internet:** Give the full URL for the website.
- **Person:** Personal communication with 'Name, Surname, Occupation.'

Take note

Simply listing Google or Wikipedia as your source is not recognised as a reference for your discussion.

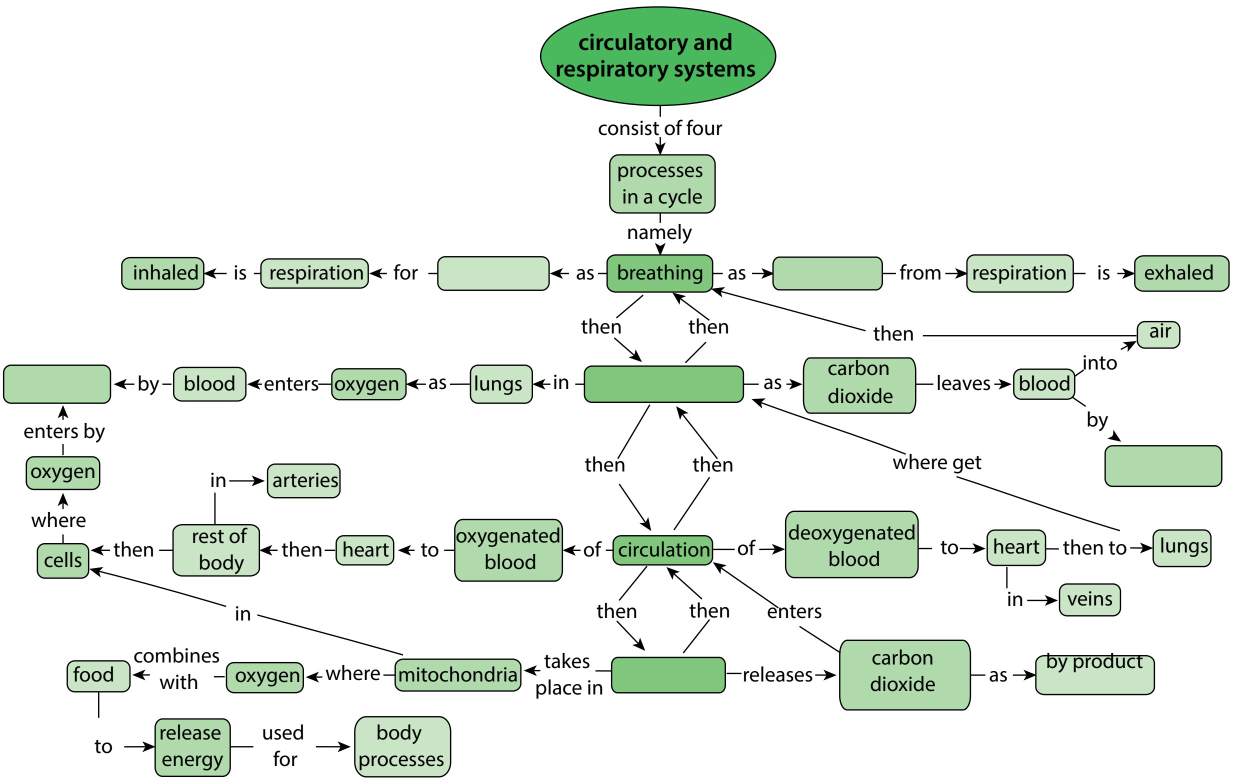
Summary

Key concepts

- During breathing oxygen is inhaled and carbon dioxide is exhaled.
- In the lungs, gaseous exchange occurs by diffusion.
- Oxygenated blood is transported in pulmonary veins from the lungs to the left side of the heart.
- The oxygenated blood is pumped through the aorta and arteries to the different parts of the body.
- Arteries divide into capillary networks between the cells, where oxygen and food diffuse from blood to cells.
- The cells carry out cell respiration, forming carbon dioxide, which diffuses back to the capillaries.
- Capillaries flow into veins that carry the deoxygenated blood to the right side of the heart.
- When the deoxygenated blood reaches the heart, it is transported to the lungs by the pulmonary artery, where gaseous exchange takes place once more.
- The carbon dioxide from cellular respiration diffuses out of the blood into the lungs and is exhaled.

Concept map

From what we have learnt in this unit, we can say that the circulatory and respiratory systems consist of 4 processes which occur in a cycle. Two of these processes are named in the concept map. Write the other two in your exercise books. During breathing, what is the gas which is inhaled for respiration, and which is the gas which is exhaled during respiration? Fill these in too. What is the name for the process by which these gases move across the cell membranes?



Revision

1. Draw a flow diagram to show how the different components of the respiratory and circulatory systems function in a cycle. [6]

2. Complete these sentences. Write just the word in your exercise books. [13]
   a) Oxygen diffuses into the blood from the air in the ______________
   b) The blood vessels that carry blood away from the heart are called ______________
   c) Tiny blood vessels called ______________come into close contact with ______________
   d) Carbon dioxide ______________ out of the cells into the ______________
   e) ______________ carry the ______________ blood to the heart from where it is sent to the ______________to be oxygenated.
   f) The chemical reaction that takes place in the ______________ of the cell when oxygen and glucose combine to release ______________ is called ______________

3. Copy and complete the table in your exercise books to describe what happens in the chest during breathing. [6]

	Inhaling	Exhaling
Chest volume		
Pressure on lungs		
Air movement		

4. Copy the table in your exercise books and match the word on the left to its correct meaning on the right. Write only the letter next to the word to indicate the correct meaning. Use each letter only once. [13]

Breathing	a	arteries, veins and capillaries
Diaphragm	b	the type of tissue that keeps the trachea open
Alveoli	c	small grape-like bunches at the ends of the bronchiole
Trachea	d	the movement of particles through a permeable membrane from a high to a low concentration
Heart	e	the tube that carries air to and from the mouth to the bronchi
Veins	f	blood vessels that transport blood away from the heart
Respiration	h	blood vessels that carry blood towards the heart
Cartilage	i	tubes leading from the trachea into the lungs
Bronchi	j	the process takes place in mitochondria to release energy for the cell to use
Capillaries	k	the organ responsible for pumping blood throughout the body
Types of blood vessels	l	inhaling and exhaling
Diffusion	m	these blood vessels surround alveoli to allow for gaseous exchange
Arteries	n	a large dome-shaped muscle across the bottom of the ribcage

5. The following image is an artist's drawing of one of the structures you learnt about in this unit. What does it represent? Give three reasons for your answer. [3]

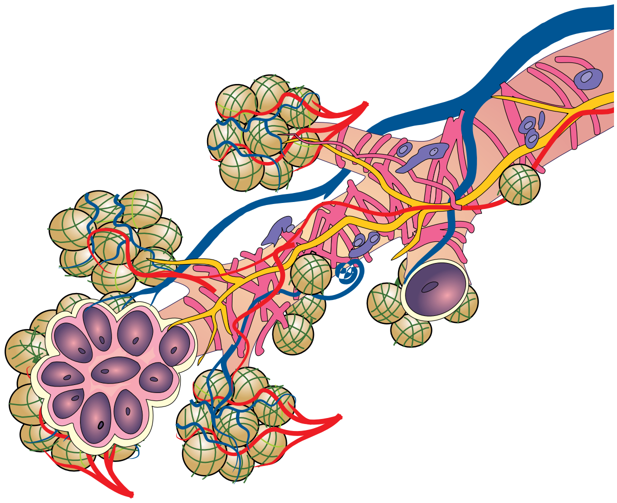


Figure 4.19

6. Describe how capillaries are suited to their function of allowing gaseous exchange within the lungs and at the cellular level in the body. [3]

Total [44 marks]

5 Digestive system

Key questions

- Why do we need to follow a healthy diet? What does a healthy diet consist of?
- What makes one type of food healthy and another type of food unhealthy?
- Is it possible to prevent things like diarrhoea or constipation? What about ulcers?
- Why do we need to digest food?
- How is food digested in our bodies?
- Where does the digested food go?

In this unit we are going to look more closely at the food we eat to see why certain foods are considered healthy and others unhealthy. We will then investigate how the food from our plates gets to our cells and why our digestive system is so well adapted for its job.

5.1 A healthy diet

Our human bodies are very active. Our bodies need a huge variety of different nutrients and substances in order to perform all these processes. We obtain these nutrients from the food we eat. The human body needs a balanced, healthy diet to keep functioning property.

ACTIVITY Comparing healthy and unhealthy foods

Instructions

1. Work with a partner.
2. We often know if a food is healthy or unhealthy. Make a list of at least 10 healthy and 10 unhealthy foods in your exercise books.

When you are done share your food list with the class and record the class's ideas of healthy and unhealthy foods on a large sheet of paper or on the board.

Display this in the class.

Study the list of healthy and unhealthy food.

1. What common characteristics can you identify in the food that the class listed as healthy?
2. What common characteristics can you identify in the food that the class listed as unhealthy?

Let's take a closer look at what makes up a healthy diet.

Keywords

- balanced diet
- carbohydrates
- dehydration
- diet
- enzymes
- fats
- fibre
- glucose
- iodine solution
- minerals
- nutrients
- protein
- starch
- sugars
- vitamins

The seven components of a healthy diet

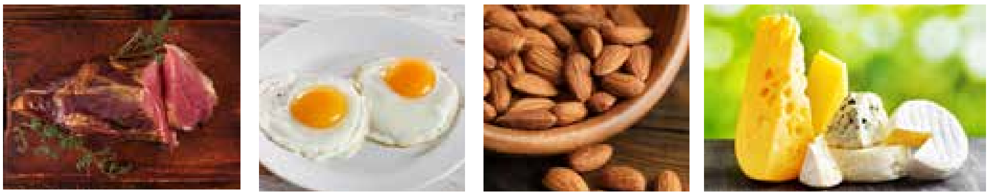
The foods that we eat can be divided into different groups:

- proteins
- carbohydrates
- fats and oils
- vitamins
- minerals
- fibre (non-digestible carbohydrates)
- water.

A healthy diet consists of foods from all of these groups.

Proteins

Proteins are our bodies' building blocks. They build and repair body cells and tissues. Foods rich in protein are: fish, meat, poultry, eggs, cheese and other food from animal sources. There are also many sources of protein from plants, for example, products made from soya beans, peas and beans, nuts and seeds.



Meat Eggs Almond nuts Cheese

Figure 5.1: Example of proteins

Carbohydrates

Carbohydrates are the main supply of energy for our bodies. They break down in our digestive system to form glucose (which is a sugar). Examples of foods that contain carbohydrates are: wholegrain bread, potatoes, pasta, rice, fruit, vegetables, maize and legumes.

Whole wheat bread Potatoes Rice Mealies (corn)

Figure 5.2: Example of carbohydrates

Unfortunately, many people eat too many carbohydrates, especially processed carbohydrates like sweets and biscuits, chips, pastries, soft drinks, and sweetened fruit juices.

Take note

Often, fruit juice is not the healthiest choice of drink as some fruit juices contain the same amount of sugar or more than the average soft drink. The best choice is water!

Fats and oils

Fats and oils are important for many body processes:

- Fat protects and insulates your organs
- They help maintain healthy hair and nails.
- Some vitamins can be absorbed and transported only when attached to fat molecules.
- Fats and oils also provide the body with energy.

However, some fats are better than others, and having too much of any type is not a good idea.

Figure 5.3: Olive oil and canola oil are both healthy oils.

Figure 5.4: Sardines are high in healthy fats.

> **Take note**
>
> Many people take large amounts of supplements to ensure they get enough vitamins and minerals. However, it is rather wasteful, as the body excretes any excess vitamins and minerals and stores only a few vitamins such as Vitamins A, D, E and K.

Vitamins

Figure 5.5: Our sources of vitamins are fruit and vegetables.

Vitamins help with the different chemical reactions in our bodies:

- **Vitamin A** helps strengthen our immune system and is good for eyesight in the dark.
- **B vitamins** help us process energy from food.
- **Vitamin C** helps to keep your skin and gums healthy and improves the immune system.
- **Vitamin D** helps to build strong bones and teeth.

Our main sources of vitamins are from fruit and vegetables, as shown in the diagram alongside.

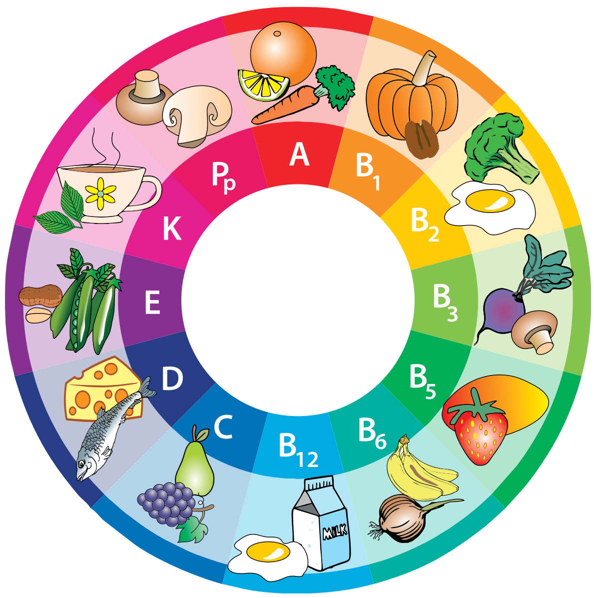


Figure 5.6: Food sources of different vitamins.

Minerals

Our bodies cannot produce minerals and we therefore need to include these in our diets. Some of the minerals we should include in our diets are:

- **calcium**, which is essential for strong bones and teeth
- **iron**, which is needed for healthy blood
- **magnesium**, which is used for building strong bones, teeth and muscles
- **sodium**, which is also needed for muscle and nerve function, and more importantly, it helps regulate the amount of water in the blood.

There are a variety of sources of minerals. For example, high levels of calcium are found in dairy products, meat is a high source of iron, and magnesium is found in lots of foods such as bananas, nuts, green leafy vegetables and milk.

The most common source of sodium is in sodium chloride, which is table salt.

Fibre

Fibre found in the skins of fruit and vegetables, and in wholegrain cereals, cannot be digested. It therefore travels through the alimentary canal. We need fibre in our diet as it helps us to have regular bowel movements and avoid constipation.

Beans are a good source of fibre.

High-fibre breakfast cereal.

Figure 5.7: Examples of sources of fibre.

Figure 5.8: You need to drink water daily.

Water

Our bodies are made up of more than 50 percent water. Water is necessary to help our blood carry nutrients and waste around the body and to help the chemical reactions that occur in our cells. Water forms most of sweat, saliva and tears.

ACTIVITY Comparing meals

Instructions

1. Below are photographs of different meals for breakfast, lunch and dinner.
2. One of the meals is healthier than the other.
3. Choose the healthier option and explain why. Write the answers in your exercise books.
   a) Breakfast:

Option 1: Fruit loops	Option 2: Fruit salad
	

b) Lunch:

Option 1: Hamburger	Option 2: Omelette with salad
	

c) Supper:

Option 1: Chicken pieces	Option 2: Beef, peas and rice
	

Different cultures and religions follow different diets. Some cultures will eat only certain types of food and will avoid other combinations. Some religions may restrict their followers to only certain foods, while others have no real dietary laws. Within South Africa, we have a very diverse population with people from many cultures, backgrounds and religions. This makes our country a truly unique, diverse and interesting place in which to live!

Testing food

There are various chemical tests which are used to easily identify the type of food molecules present in different foods.

One such test is the **starch test**. We can also test for the presences of fats and oils using the **emulsion test**.

Keywords

- emulsion

Investigation — Which foods contain starch or fats and oils?

In this investigation you will have to design the investigation and then write up your findings in an experimental report.

Materials

- various food items to test for starch: for example, pieces of bread, apple, tomato, boiled egg, cheese, cucumber, potato, yoghurt, ham (some substances must contain starch and some not)
- various food items to test for fats and oil: for example, the above food items can be used, and in addition, you could also provide peanut butter and butter
- petri dish per group or learner for the starch test
- bottle of iodine solution and dropper

Take note

Precaution
If you are allergic to iodine, rather observe this experiment and do not participate.

Take note

You may recall doing the starch test on plants in lower grades to see which leaves store starch from the glucose produced in photosynthesis.

Take note

If you are allergic to iodine, rather observe this experiment and do not participate.

- several test tubes for the fat emulsion test
- water
- glass rod (or any other suitable hard, round item) for crushing food substances for the fat emulsion test
- bottle of ethanol
- forceps

Instructions

1. Conduct an investigation to test whether the food substances you have been provided with contain starch or fats and oils or both.
2. A summary of each test is given below. You will need to design your investigation and conduct it.
3. Before starting, think about how you will record your results and write out your proposed method.

Starch iodine test

Iodine solution is an orange-brown colour. When iodine is added to a substance which has starch in it, the iodine reacts with the starch to produce a blue-black colour. The blue-black colour indicates the presence of starch.

Fat emulsion test

To conduct the test, crush a piece of the food (or liquid) in a small amount of ethanol. Pour some of the mixture onto paper. Once the ethanol has evaporated, oil stains on the paper will indicate the presence of fats or oils in the food.

Aim

What is the aim of your investigation?

Hypothesis

What is your hypothesis for this investigation?

Materials and apparatus

List the materials and apparatus you used in this investigation.

Method

Write down the method which you followed in this investigation.

Results and observations

Record your results and observations from this investigation in your exercise books.

Discussion

Discuss and evaluate your results and findings and the importance of food tests.

Conclusion

What do you conclude from this investigation?

Health problems relating to diet

In Unit 2 this term, we looked at some of the health issues relating to the digestive system, such as ulcers, diarrhoea and eating disorders. There are also health issues which arise directly from your diet. The following activity will introduce you to some of these health concerns.

ACTIVITY How does your diet affect your health in the short and long term?

Instructions

Use any available resource and research about the listed malnutrition

1. Below is a table with descriptions of several health issues relating to a poor diet.
2. You need to read the descriptions and use your knowledge of the food groups to classify what the diet of the person is deficient in, or what they have a surplus of in their diet.
3. For some conditions, there may be a variety of causes, but this activity is focusing on the causes related to diet.

Name of health issue	Description	What does this person have a deficiency or surplus of in their diet?
Osteoporosis	Osteoporosis is a disease, most common in older women, where the bones become fragile and are more likely to break. Usually the bones lose density and become porous.	
Anaemia	Anaemia is a condition of the blood where there are not enough healthy red blood cells. A patient feels tired and weak as the tissues and organs in the body are not able to get enough oxygen so respiration slows down.	
Marasmus	This is a severe form of malnutrition due to starvation. The person becomes extremely thin (emaciated).	
Constipation	A person has constipation when they have a bowel movement less than 3 times per week. The person may have hard stools and difficulty and pain when passing stools.	
Kwashiokoh	Kwashiorkor, also known as "edematous malnutrition" because of its association with edema (fluid retention), is a nutritional disorder most often seen in regions experiencing famine.	

5.2 Digestion and the alimentary canal

Keywords

- digestion
- dissolve
- enzymes
- faeces
- gastric
- peristalsis

What is digestion?

Digestion involves a variety of complex processes that turn the food that you eat into tiny molecules that can then be absorbed and transported to the cells of the body.

There are two types of digestion:

1. **Mechanical digestion** occurs when food is physically broken down – these include chewing, churning and mashing. Mechanical digestion takes place in your mouth and in your stomach.
2. **Chemical digestion** takes place when different digestive **enzymes** break down the bits of food into smaller molecules. Enzymes are special proteins that speed up certain chemical reactions in the body. Chemical digestion starts in the mouth, where enzymes in your saliva start to break down starch. Chemical digestion also takes place in the stomach and small intestine.

The alimentary canal

We already studied the alimentary canal in Unit 2, so we'll start by reviewing what we learnt there.

ACTIVITY The different organs in the digestive system

Instructions

1. Label the following diagram.

Labels to include

- large intestine
- anus
- oesophagus
- rectum
- stomach
- mouth
- small intestine
- liver
- gall bladder
- pancreas

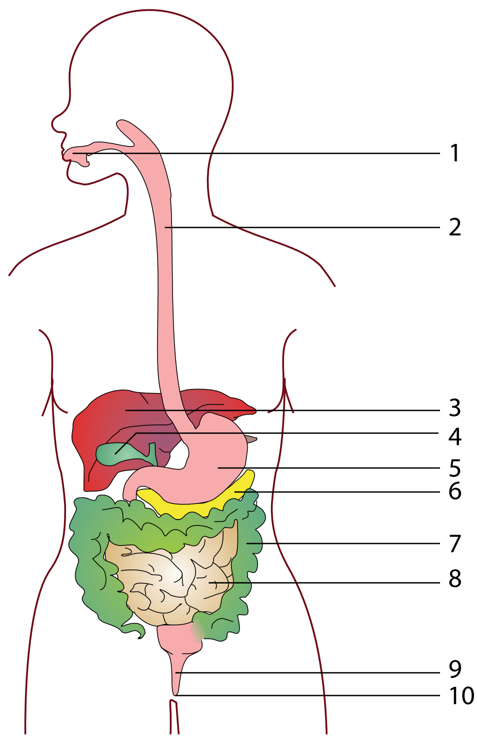


Figure 5.8

2. Let us make a model of the alimentary canal that can demonstrate mechanical and chemical digestion in the different parts. We will also learn about how the different parts are structurally adapted to suit their function.

ACTIVITY A digestion simulation

Materials

Each group will need the following:

- large dish to work over, or black bags and newspaper
- crackers, white bread or viennas
- mixing bowl
- scissors, pestle and potato masher
- water bottle that can squirt water through a small hole
- the inner tube of a kitchen paper towel roll or toilet paper roll
- a clear plastic Ziploc bag
- 30 – 40 ml of lemon juice, vinegar or a fizzy drink
- full-length stocking with the toe section cut open – it helps if one leg is put inside the other to form a double layer
- bicarbonate of soda dissolved in water in syringes (10ml)
- large bowl

Instructions

1. Work in groups and construct a model to demonstrate the different processes that food goes through in the different parts of the alimentary canal.
2. Make careful observations and describe in detail what happens at each stage.
3. Work over a large bowl or tray or sheets of newspaper and a black bag to contain the mess, which may be produced during this activity.

Stage 1 – The mouth

The function of the mouth is to ingest food and to start to digest the food. The mouth is specifically adapted for its function as follows:

- The lips keep the food in the mouth while chewing.
- Food is bitten off with the front teeth.
- Food is cut, torn and mashed into smaller parts by the different teeth in the mouth – this is mechanical digestion.
- The tongue moves the food around the mouth while it is being chewed. It also prepares the food for swallowing.
- Salivary glands secrete saliva. Saliva coats the food in the mouth making it easier to swallow. Saliva also contains enzymes which start to digest chemically.

1. Using the mixing bowl to represent the mouth and the scissors, pestle and potato masher to represent and simulate the digestion of your food type that occurs in the mouth.
2. Squirt some water onto the mixture as you are 'digesting' the food.
3. Describe what is happening to the food at this point.
4. Compare the model to the actual process in your mouth and what each part and action you are performing in the simulation represents.

Did you know?

Dr William Beaumont (1785 – 1853) discovered how food is digested in the stomach. He dropped food, attached to silk threads, into the stomach of a patient who had a gunshot wound in the stomach that would not close, and examined what happened.

Stage 2 – The oesophagus

- The pharynx (the throat) moves food from the mouth to the oesophagus. The oesophagus transports food from the pharynx to the stomach.
- A flap in the pharynx covers the trachea (windpipe) to prevent food from accidentally going into the trachea and causing the person to choke.
- The oesophagus is a muscular tube that moves the food by contracting in sections and relaxing in other sections. This is called peristalsis.
- A special circular muscle shuts the entrance of the stomach. It prevents the contents of the stomach from pushing back into the oesophagus, which may lead to vomiting.

Did you know?

Our bodies produce about 1.7 litres of saliva *each day*.

1. Roll the ball of food you created in the mouth down the cardboard tube and into the clear Ziploc bag.
2. Describe what is happening to the food at this point.
3. Compare the model to the actual process in your oesophagus. Can you think of a better way of simulating the action of moving the food from the mouth to the stomach?

Stage 3 – The stomach

The stomach is specifically adapted for its function as follows:

- The stomach has strong muscles which help churn the food to break it up further. This also mixes the pieces of food with the digestive gastric juices.
- Since the stomach has to store food and liquid, it has many folds and ridges in the wall that help to expand the stomach further.
- The lining of the stomach is replaced to prevent the stomach from digesting itself.
- The stomach secretes gastric juices when food is present. This helps the functioning of the enzymes in the chemical digestion of proteins.
- Cells in the stomach lining are adapted to absorb water.
- The lower end of the stomach has muscles which can control the emptying of the stomach contents.

1. The Ziploc bag represents the stomach. After the food has entered the stomach, pour one of the digestive juices (lemon juice, vinegar or Coca Cola) into the bag over the ball of food.
2. In your body, a special circular muscle closes and seals the stomach and digestive juices from the oesophagus. Seal the Ziploc as if you were sealing the actual upper end of the stomach.
3. Squeeze the bag to show the churning of food in the stomach.
4. Describe what is happening to the food at this point.
5. Compare the model to the actual process in your stomach.

Stage 4 – The small intestine

In the small intestine, the digestion of proteins, carbohydrates and fats is completed and the end-products of these digestion processes are absorbed. The small intestine is specifically adapted for its function as follows:

- Since most of the digestion and absorption process takes place in the small intestine, it is exceptionally long and folded to create an even bigger absorption area.
- The inner layer of the small intestine is lined with small finger-like structures called villi, which aid absorption and increase the area for absorption.
- The small intestine has a large network of capillaries surrounding it to transport the absorbed food away.
- The muscles of the small intestine control the direction in which the food flows by peristalsis.
- The stocking that you have been provided with represents the small intestine. Cut a small corner off the bottom of the Ziploc bag and insert this end into the stocking.

1. Work over a large dish or black plastic bags for this part. While one learner is holding the stocking, the other learner should squeeze the food mixture into the stocking.
2. Use the syringes with the dissolved bicarbonate of soda and squirt the bicarbonate of soda into the food as it enters the stocking.
3. Simulate the action that takes place in the small intestine to move the food mixture through.
4. Describe what is happening to the food at this point.
5. Compare the model to the actual process in your small intestine.

Stage 5 – The large intestine

The large intestine absorbs water and mineral salts, to make some vitamins, and to decay the undigested food materials and form faeces. The large intestine is specifically adapted for its function as follows:

- Undigested waste remains in the large intestine for up to 24 hours in order to maximise the absorption of water from this region.
- The muscles in the large intestine are able to turn the waste material into faeces preparing it for egestion.
- When it is time to egest waste, the muscles in the large intestine create strong peristaltic movements to force the faeces out of the body via the rectum and anus.
- Circular muscles in the anus control the emptying of the waste materials.

1. Was this a worthwhile activity for you? Explain what you learnt from this activity and whether you think it was a worthwhile activity or not, giving reasons for your opinion.

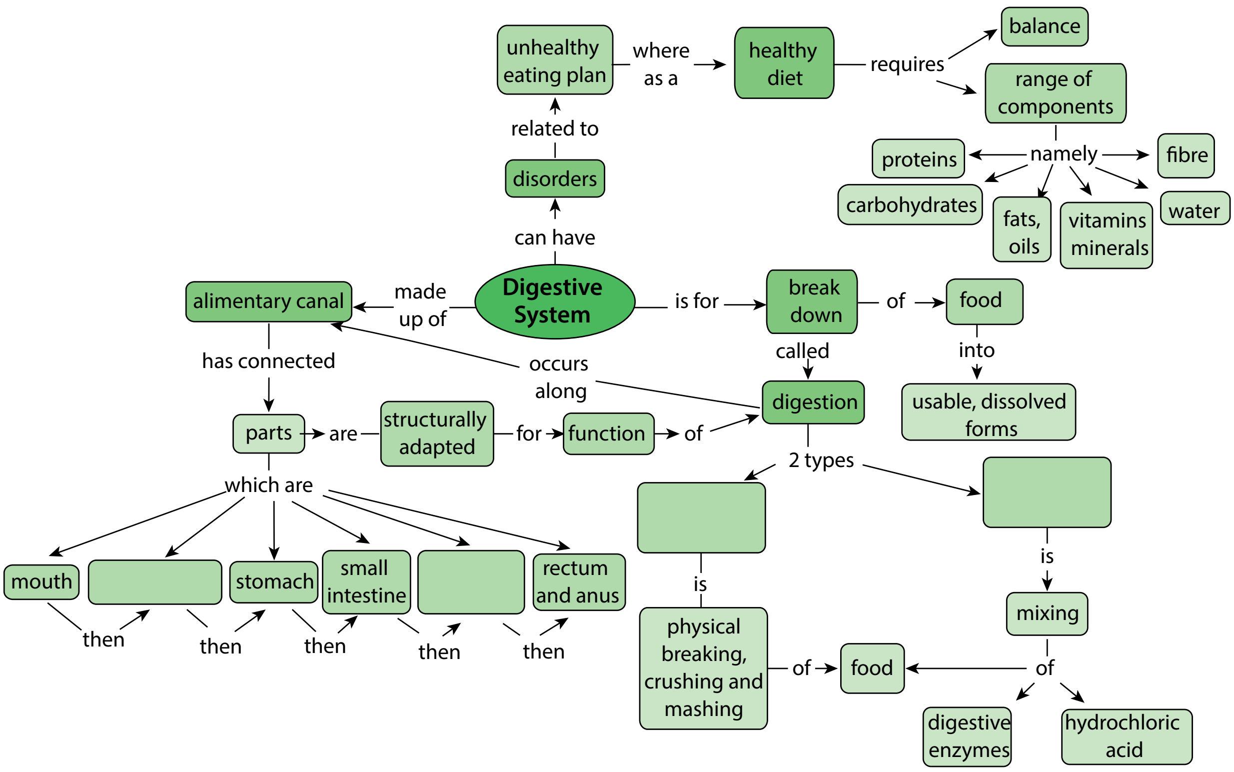
Summary

Key concepts

- There are seven buildings blocks in a healthy diet: proteins, carbohydrates, fats and oils, vitamins, minerals, fibre and water.
- A healthy diet includes the correct proportions of the seven building blocks.
- Problems in our digestive system can be related to an inappropriate diet that does not give our bodies the correct nutrients.
- Our alimentary canal is composed of the mouth, oesophagus, stomach, small intestine, large intestine, rectum and anus.
- Digestion is the breaking down of food into usable, dissolvable forms that can be absorbed.
- There are two types of digestion: mechanical (or physical) and chemical digestion.
- Each structure in the alimentary canal is specifically adapted to suit its purpose.

Concept map

The alimentary canal is made of several parts linked together – two of these parts are missing from the concept map. We also looked at two types of digestion in this unit. What are these? When filling them in on the concept map provided by your teacher, you need to decide which space to put them in by looking at the concepts, which come afterwards to describe each type.



Revision

1. Describe what you understand the term 'healthy diet' to mean. [2]

2. Copy the following table in your exercise books. For each of the following food items, classify what nutrients you can get from them (i.e. protein, carbohydrates, vitamins, etc). Some food items provide more than one class of nutrient. [10]

Food item	Nutrients	Food item	Nutrients
 Fried chips		 Strawberries	
 Chicken pieces		 Bran biscuits	
 Butternut		 Yoghurt	
 Assorted nuts		 Split peas and lentils	
 Green beans		 Margarine	

3. Which of the foods in Question 2 contain starch? How can you test if they contain starch? [9]

4. Why is it important to limit your intake of take-aways? [3]

5. Give at least 2 reasons why we should eat raw fruit and vegetables. [2]

6. Some food may take up to 24 to 36 hours to digest and be fully absorbed. Why do you think this process takes so long and why is this a good thing? [2]

Total [28 marks]

Glossary

absorption food molecules pass into the bloodstream

alimentary canal the tube that runs from your mouth to your anus where food is digested, nutrients are absorbed and solid waste is egested

alveoli clusters of tiny air sacs in the lung that together provide a very large surface area

antibiotic a medicine that kills bacteria

anus the opening at the lower alimentary canal through which waste is eliminated from the body

arteries blood vessels that carry blood away from the heart

atrium the upper left and right chambers of the heart

auditory of or relating to the sense of hearing

balanced diet a way of eating that includes adequate amounts of the necessary nutrients required for healthy growth and activity in their correct proportions

birth control the limitation or control of the number of children that a couple or a woman want to conceive by the planned use of contraceptive techniques

bladder the membranous, balloon-like sac in our bodies in which urine is collected for excretion

blood the red liquid in the blood vessels of the body that transports nutrients and oxygen to cells and removes waste and carbon dioxide from the cells

blood pressure the pressure of the blood in the circulatory system against the walls of the blood vessels

blood vessels tube-like structures that carry blood to and from tissues and organs

bowing bending

brain the organ in the skull made of soft nervous tissue that coordinates activities, senses and intelligence

breathing taking air into the body through the mouth, trachea, bronchi and lungs and releasing carbon dioxide-rich air from the lungs, trachea and mouth

brittle hard but easily broken or shattered

bronchi the two large air tubes going into each lung from the trachea

bronchioles smaller, branched air passages in the lungs

capillary /capillaries the smallest branching blood vessels that form a network between cells and join arteries to veins; diffusion between blood and cells occurs here

carbohydrates nutrients from plants, such as sugar and starch, that serve as a major source of energy in animals' diets

carbon dioxide a colourless, odourless gas that is released from the chemical breakdown of food during cellular respiration

cartilage firm, whitish, flexible connective tissue found in joints, outer ear, larynx, nose, and in rings around the trachea

cell the structural and functional unit of all living organisms; the smallest living part of plants and animals

cell membrane the selectively permeable membrane that surrounds the cytoplasm of a cell

cellular respiration process whereby organic substances (from food) combine with oxygen in order to release energy; carbon dioxide and water are by-products

cell wall tough, usually flexible layer that surrounds a plant's cell membrane; supports and protects the plant cell

cellulose a special type of carbohydrate made up of many glucose molecules that are packed very tightly together so that it doesn't dissolve in water; provides support in plants

cervix the neck of the uterus

chemical digestion breaking food into molecules that can dissolve in the blood and be transported to the cells using chemical agents (enzymes)

chloroplast a cell organelle found in plants that contains chlorophyll and can therefore photosynthesise

cilia small hair-like extensions in specialised cells in the lining of the nose and all breathing tubes that trap and remove dust and germs from the body

closed blood system blood never leaves the blood vessels

collagen a strong, flexible protein in connective tissue that cannot stretch

conception moment of fertilisation when the male sperm and the female ovum fuse together and a new individual is formed

conduct to carry impulses from one neuron the next

contraception any method that prevents pregnancy

contract to get smaller or shorter

contraction the shortening (tensing) of a muscle; term used to refer to the forceful tensing of the uterus muscles during childbirth

cover slip a small glass square which is placed over the specimen on a slide to view under a microscope

cytoplasm the gel-like material found within a plant or animal cell that is enclosed by the cell membrane but excluding the cell nucleus

degenerative a worsening in function over time

dehydration when the body loses too much water

deoxygenate to remove oxygen

deprived not given enough of something

diaphragm the dome-shaped muscle that separates the thorax from the abdomen; it plays a major role in breathing

diet what a person (or an animal) regularly eats or drinks

differentiation process by which a less specialised cell type becomes more structurally specialised to perform certain functions

diffuse move from an area of high concentration to an area of low concentration through a permeable membrane

diffusion the movement of a substance from an area of high concentration to a region of low concentration

digest break into pieces that are small enough to dissolve in the bloodstream and be absorbed into the cytoplasm

digestion breaking up food into small soluble parts that can be absorbed

dissolve when a solid breaks down into smaller and smaller particles until it mixes completely with a liquid (goes into solution)

DNA DeoxyriboNucleic Acid; molecule that stores information on how to make proteins and what characteristics the organism inherited from its parents

egestion passing out solid, undigested waste

ejaculation/ejaculate the release of sperm from the penis

embryo a very young, developing baby

emulsion a mixture of two liquids that normally do not mix together, such as oil and water

enzymes special proteins that help reactions to take place in the body of the organism

erection the enlarged state or condition of tissues around the penis

eukaryote an organism that has genetic material inside a nucleus

excrete to remove metabolic waste products and carbon dioxide from the body

excretion removing harmful wastes that were made in the body and need to be removed from the body

exhale letting air rich in carbon dioxide out of the body through the mouth or nose, breathing out

faeces the waste from your body formed from undigested food in the intestines and passed out through the anus

fallopian tube (oviduct) a tube extending from the ovary to the uterus to transport a mature ovum

fats nutrients that are very high in energy, don't mix with water, and are found in oils and greasy foods

fertilisation when a sperm fuses with an egg

fibre the cell walls of plant material that we eat that cannot be digested by humans
flaccid soft and hanging loosely
foreskin a layer of skin that covers and protects the head of the penis
fracture crack or break
frame structure a structure made by connecting beams and columns

gamete cells another name for 'sex cells' that fuse during fertilisation
gaseous exchange the process in the lungs when oxygen enters the bloodstream and carbon dioxide is removed; at cellular level, occurs when oxygen is removed from the bloodstream and enters the cells and carbon dioxide is removed from cells and enters the bloodstream
gastric of or relating to the stomach
gestation (pregnancy) the period (9 months in human beings) of development in the uterus from conception to birth
glucose a simple sugar molecule that is produced during photosynthesis and is the main source of energy for living organisms

haemoglobin a red iron-rich protein responsible for transporting oxygen in the blood
heart the organ responsible for pumping blood throughout the body
heart chamber any of the four spaces of the mammalian heart
hereditary characteristics that are transmitted from the parent to the offspring
hormone the body's chemical messengers that travel in the bloodstream to tissues and organs to affect many different reactions in the body

implantation the attachment of the fertilised egg into the wall of the uterus of the mother
impulse an electrical signal travelling along a nerve cell
infection when bacteria or viruses invade and multiply in the body's tissues and cells causing disease and illness
ingestion taking food into the mouth and body
inhale taking air rich in oxygen into the body through the mouth or nose; breathe in
inherited genetic characteristics received from the parent
integrate to make into a whole by bringing all the parts together; unify
iodine solution a brownish-orange liquid that is used as an antiseptic and dye; it changes colour in the presence of starch

jaundice yellowing of the eyes and skin, common in liver conditions
joint the place where two or more bones meet

kidney organ in the abdomen that filters the blood and produces urine

labour the process or effort of childbirth; the time during which this takes place
ligament a short band of tough, flexible, fibrous connective tissue that connects two bones or cartilage, or holds together a joint
locomotion movement or the ability to move from one place to another
lungs the organs used for breathing and gaseous exchange

medium a solution in which cells or organelles are suspended and in which reactions take place
membrane a thin flexible sheet or skin that acts as a boundary around a cell or cell organelle
menopause the changes that occur in an older female (around age 50) body when she is no longer able to reproduce
menstrual cycle a recurring series of bodily changes in women that occurs roughly every 28 days in which the lining of the uterus thickens in preparation for the possible implantation of a fertilised egg; when that doesn't happen the lining of the uterus breaks down and is discharged as menstrual blood
metabolic relating to the chemical processes and changes that happen within the cells of plants and animals
metabolic waste products any unwanted substance produced by the various body processes
metabolise any build-up or break-down process in the body

microscope an optical instrument used for viewing very small objects not often visible to the naked eye

microscopic so small that it can be seen only under a microscope

mineral salts chemical elements in food needed for growth and development, like, sodium, potassium, calcium, iron, phosphorous, and so on

minerals the elements (such as iron, sulfur and calcium) that are essential to animals and plants

mitochondria a cell organelle that uses oxygen and food molecules to release energy for the cell

mucus a slimy substance secreted by the mucous membranes and glands (in the nose for instance) for lubrication and protection

multicellular organisms organisms that have many cells

muscle a type of tissue in the body that can contract to produce movement

nerve a whitish bundle of neuron fibres that transmits impulses between the nerve centres in the brain and spinal cord and various parts of the body

network a structure that interconnects many different parts

neuron a specialised nerve cell that transmits nerve impulses

nuclear membrane a double-layered membrane that separates the content of the nucleus from the cytoplasm

nucleolus small dense round structure in the nucleus of a cell

nucleus structure with a membrane around it that contains the cell's hereditary information and controls the cell's growth and reproduction

nutrients components of food that provide the body with energy or supply the building blocks for growth and repair

oestrogen the female sex hormone that causes the development of many of the female secondary sex characteristics

optic of or relating to the eye or vision

organelle(s) specialised structures inside the cytoplasm of the cell that perform functions for the cell

organisms an individual animal, plant or single-celled life form

ovary the organ that produces the female ova (egg cells), as well as the female hormones oestrogen and progesterone

ovulation the process whereby a mature ovum or egg cell gets released from the ovaries

ovum the female egg cell produced in the ovaries of a woman

oxygen a colourless, odourless reactive gas is used in cell respiration of all organisms

oxygenate to supply with oxygen

penis one of the male sex organs

peristalsis the wave-like contraction and relaxation of the walls of the alimentary canal that helps move food forward

pharynx throat

population growth rate growth of a population over time seen as the change in the number of individuals (of any species) in a population per unit of time

prokaryote a type of organism that does not have a separate nucleus but has its hereditary material in the cytoplasm

protein group of biological molecules that provide structure and enable chemical reactions

puberty the time between childhood and adulthood when the sex organs mature with accompanying changes in the body that prepare the person's body for reproduction

pulse the rhythmical throbbing of the arteries as blood is pumped through them by the heart

red blood cells specialised cells in the bloodstream that contain haemoglobin and can therefore carry oxygen

reproduction any process by which organisms produce offspring

respiration the chemical process in cells that releases energy from food molecules by using oxygen and forming carbon dioxide as a waste product

rupture break or burst open

saliva the watery substance in the mouth that covers chewed food and moistens the mouth

scrotum the external sac of skin that encloses the testes in males

selectively permeable **a** feature and a function of the cell membrane that allows it to regulate the substances that enter and leave the cell

self-propulsion having the ability to move itself

semen the fluid that is produced in the male reproductive organs, containing sperm and other chemicals suspended in a liquid medium

sexual intercourse how the male sperm is introduced into a woman's body when the penis is placed inside the vagina

slide a small glass plate on which we mount specimens to examine under a microscope

small intestine the part of the alimentary canal between the stomach and large intestine where most of the digestion and absorption of nutrients takes place

specialised able to perform a particular function

species the most basic biological classification of organisms; organisms that are capable of mating with one another to produce FERTILE offspring

specimen a sample or small part of a larger organism that we want to examine or analyse; it can also mean an object or organism that was selected and presented as part of a collection or series

sperm the male sex cell produced by the testes

sperm duct (*vas deferens*) the tube that connects the testes to the ejaculation duct

starch a large storage molecule in plants that is made from many glucose molecules joined together

stem cell a special undifferentiated cell that can become any of the other cell types

stimulus any change that is detected inside or outside the body, to which we need to react

stomach the wider part after the oesophagus where food is stored for a short while; proteins are digested here

sugars group of sweet-tasting simple carbohydrates that are made by plants during photosynthesis

surrogacy when a person or animal acts as a substitute for another third person; when a woman carries and delivers a child for another couple or person

synthesis the process by which organic molecules are made inside organisms

temperature how much heat is present in an object, substance or body; the degree of internal heat of someone's body

tendons inelastic cords of strong fibres made of collagen tissue that attaches a muscle to a bone

testes male glands that produce sperm cells and male hormones

testosterone the male sex hormone that causes physical changes during puberty and controls the production of sperm

toxic poisonous

trachea (windpipe) the tube that carries air from the mouth and nose to the bronchial tubes in the lungs

transmit send out a message

transport move from one part of the body to another

turgid swollen or bulging outwards

ulcer an open sore in the alimentary canal

umbilical cord the cord or tube-like structure that connects the foetus at the abdomen with the placenta of the mother; it transports nourishment and oxygen to the foetus and removes waste

unicellular consisting of a single cell

urea a metabolic waste product that is formed when protein is broken down in the liver

ureter the duct (tube) that joins the kidney and bladder and allows urine to pass from the kidney to the bladder

urethra the thin tube that allows urine to flow from the bladder to the outside

urinate to excrete or pass urine out of the body

uterus the hollow muscular organ in the pelvic area of female mammals in which the fertilised egg implants and develops (also known as the *womb*)

vacuoles a fluid-filled bag in the cytoplasm of most plant cells

vagina an elastic muscular tube or canal that connects the neck of the uterus (cervix) with the external opening

variation a change or slight difference

veins blood vessels that carry blood back to the heart

ventricles the lower left and right chambers of the heart

vision the ability to see

vitamins organic substances essential to normal growth and development in the body and found naturally in plant and animal products

wet mount a specimen mounted on a slide using a drop of liquid

womb another non-technical term for uterus

zygote the result of two gametes that fuse; a fertilised ovum